NESTED SWEEPOUTS AND FREE BOUNDARY MIN-MAX THEORY IN COMPLETE RIEMANNIAN MANIFOLDS

BY

XINZE LI

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Graduate Department of Mathematics
University of Toronto

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ABSTRACT

Nested Sweepouts and Free Boundary Min-Max Theory in Complete Riemannian Manifolds
Xinze Li
Doctor of Philosophy
Graduate Department of Mathematics
University of Toronto
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We extend the min–max theory for free-boundary minimal hypersurfaces to complete Riemannian manifolds with boundary and sublinear volume growth.

To my parents,
for their unconditional support.

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PUBLICATIONS

Parts of this thesis have been disseminated in the following manuscripts:
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1 INTRODUCTION

The existence of minimal hypersurfaces is a central problem in differential geometry and the calculus of variations. The Almgren–Pitts min-max theory [Alm65, Pit81], together with the regularity results of Schoen–Simon [SS81a], establishes that every closed Riemannian manifold M^{n+1} with $2 \leq n \leq 6$ contains a smooth, closed, embedded minimal hypersurface. This foundational result has since inspired far-reaching developments, including the resolution of the Willmore conjecture by Marques–Neves [MN14], Yau's conjecture on infinitely many minimal hypersurfaces by Marques–Neves [MN17] (for positive Ricci curvature) and Song [Son23] (in full generality), and the generic equidistribution results of Irie–Marques–Neves [IMN18] and Marques–Neves–Song [MNS19].

Two natural directions of generalization arise. The first concerns manifolds *with boundary*: given a compact Riemannian manifold M with non-empty boundary ∂M, one seeks *free boundary minimal hypersurfaces* — hypersurfaces that are minimal in the interior and meet ∂M orthogonally along their boundary. Li–Zhou [LZ21] developed a min-max theory for free boundary minimal hypersurfaces and proved that every smooth compact Riemannian manifold M^{n+1} ($2 \leq n \leq 6$) with $\partial M \neq \varnothing$ contains such a hypersurface, without any assumption on the geometry or topology of M or ∂M.

The second direction concerns *non-compact* manifolds. For complete non-compact manifolds without boundary, Chambers–Liokumovich [CL20] introduced a localized min-max framework based on *good sets*: bounded open sets $U \subset M$ whose interior boundary $\partial_I U$ is small relative to the

width $W(U)$. Using good sets, they proved that every complete non-compact manifold of finite volume (more generally, of sublinear volume growth) contains a closed embedded minimal hypersurface.

This thesis combines both directions: we develop a min-max theory for free boundary minimal hypersurfaces on *complete, non-compact* Riemannian manifolds *with boundary*. Our main result is the following (see Figure 1.1).

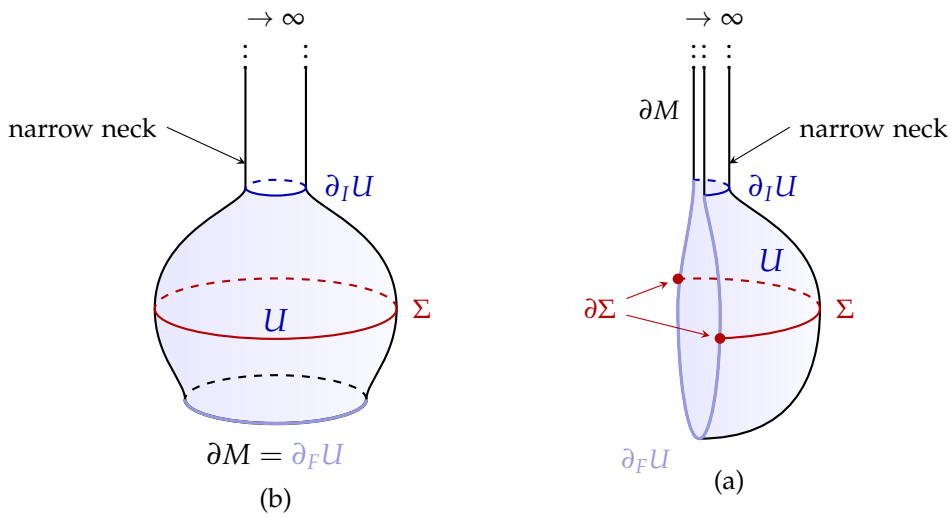

Figure 1.1: A good set U (shaded) inside a non-compact manifold M with boundary ∂M (bold). The *free boundary* $\partial_F U \subset \partial M$ is the portion of ∂M adjacent to U, and the *interior boundary* $\partial_I U$ sits at a narrow neck, so $\mathbf{M}(\partial_I U)$ is small relative to $W_\partial(U)$. (a) ∂M parallel to the neck, both extending to infinity; (b) ∂M transverse to the neck.

Theorem 1.0.1 (Main Theorem)**.** *Let (M^{n+1}, g) be a complete Riemannian manifold with non-empty smooth boundary ∂M, $2 \leq n \leq 6$. Let $U \subset M$ be a* good set *(Definition 2.3.1), i.e., a bounded open set of finite perimeter satisfying $\mathbf{M}(\partial_I U) \leq \frac{1}{10} W_\partial(U)$, with $\overline{U} \cap \partial M \neq \varnothing$. Then there exists a smooth, almost properly embedded minimal hypersurface $(\Sigma, \partial\Sigma) \subset (M, \partial M)$ with (possibly empty) boundary $\partial\Sigma \subset \partial M$ such that*

(i) *Σ is minimal in the interior and meets ∂M orthogonally along $\partial\Sigma$;*

(ii) (Area bound) $\mathcal{H}^n(\Sigma) \leq W_\partial(U) + \mathbf{M}(\partial_I U)$;

(iii) (Localization) $\Sigma \cap \overline{U} \neq \varnothing$.

The two conclusions in the theorem address issues specific to the non-compact setting. The area bound $\mathcal{H}^n(\Sigma) \leq W_\partial(U) + \mathbf{M}(\partial_I U)$ gives quantitative control on the size of the minimal hypersurface produced by the min-max process; combined with the good set condition $\mathbf{M}(\partial_I U) \leq \frac{1}{10} W_\partial(U)$, this yields $\mathcal{H}^n(\Sigma) \leq \frac{11}{10} W_\partial(U)$, so the area is controlled at the scale of the width. The localization condition $\Sigma \cap \overline{U} \neq \varnothing$ ensures that the limit hypersurface does not escape to infinity: it genuinely passes through the good set U. In the compact setting both conclusions are automatic, but on non-compact manifolds they require the good set machinery of Chambers–Liokumovich [CL20] together with the no-escape-to-infinity argument developed in Chapter 7.

2

SETTINGS AND PRELIMINARIES

This chapter establishes the foundational framework for the free boundary min-max theory in complete Riemannian manifolds with boundary.

Let (M^{n+1}, g) be a smooth, complete Riemannian manifold with non-empty smooth boundary ∂M. We assume $2 \leq n \leq 6$ throughout. We write $\nu_{\partial M}$ for the inward unit normal to ∂M. See Appendix A.2 for details on Fermi coordinates near ∂M.

We denote by $\mathcal{H}^k$ the k-dimensional Hausdorff measure and by $\mathbf{M}(T)$ the mass of a current T. We assume M is isometrically embedded in some $\mathbb{R}^L$. For vector fields, let $\mathfrak{X}_c(M)$ denote the space of *compactly supported* smooth vector fields on M that are tangent to M:

$$\mathfrak{X}_c(M) := \big\{X \in \mathfrak{X}(\mathbb{R}^L) : X \text{ is compactly supported}, \\ X(p) \in T_pM \text{ for all } p \in M\big\}.$$

The subspace of vector fields additionally tangent to the boundary is

$$\mathfrak{X}_{\tan}(M) := \{X \in \mathfrak{X}_c(M) : X(p) \in T_p(\partial M) \text{ for all } p \in \partial M\}.$$

Remark 2.0.1 (Role of the isometric embedding). The assumption that M is isometrically embedded in $\mathbb{R}^L$ (which exists by Nash's embedding theorem [Nas56]) is needed because integer rectifiable currents are defined as functionals on differential forms in an ambient Euclidean space (see Simon [Sim84, §26–27], Li–Zhou [LZ21, §3]). The embedding also guaran-

tees the isoperimetric inequalities (Lemmas 2.1.7 and 2.1.8) with constants depending only on the embedding.

For a relatively open subset $U \subset M$, we write $\mathrm{int}_M(U)$ for the relative interior of U in M. We use $A \Subset B$ to denote that $\overline{A}$ is compact and contained in B.

2.1 CURRENTS AND VARIFOLDS

We work with currents and relative cycles following Li–Zhou [LZ21]. We write $\mathcal{R}_k(M;\mathbb{Z}_2)$ for the space of mod 2 flat chains supported in M, with mass norm $\mathbf{M}$ and flat topology.

Remark 2.1.1 (Coefficient group). Throughout this thesis we work with $\mathbb{Z}_2$ coefficients. The reduced boundary $\partial^*\Omega$ of a Caccioppoli set carries no canonical orientation, only a distinction between interior and exterior, so $\mathbb{Z}_2$ is the natural coefficient group. Li–Zhou [LZ21] state their results for $\mathbb{Z}$ coefficients, but the key tools—the isoperimetric lemma, slicing theorem, and regularity theory—hold equally for $\mathbb{Z}_2$ (see [MN17, Alm62]).

Definition 2.1.2 (Relative cycles and equivalence classes)**.** For subsets $A \supset B$ of M, the space of *relative cycles* is

$$\mathcal{Z}_n(A,B;\mathbb{Z}_2) := \{T \in \mathcal{R}_n(M;\mathbb{Z}_2) : \mathrm{spt}(T) \subset A,\ \mathrm{spt}(\partial T) \subset B\},$$

equipped with the flat topology. The *space of equivalence classes of relative cycles* is the quotient

$$\mathbf{Z}_n(A,B;\mathbb{Z}_2) := \mathcal{Z}_n(A,B;\mathbb{Z}_2)/\mathcal{Z}_n(B,B;\mathbb{Z}_2).$$

Two cycles $T_1, T_2 \in \mathcal{Z}_n(A,B;\mathbb{Z}_2)$ represent the same class $\tau \in \mathbf{Z}_n(A,B;\mathbb{Z}_2)$ if and only if $T_1 - T_2 \in \mathcal{Z}_n(B,B;\mathbb{Z}_2)$, i.e., $T_1 - T_2$ is supported in B.

[**Xinze:** Proposition and proof rewritten; unified notation $\partial[\![\Omega]\!]$ throughout.]

Proposition 2.1.3 (Caccioppoli sets and mod 2 cycles)**.** *Let $\Omega \subset M$ be a measurable subset. Then Ω is a Caccioppoli set (i.e. $\mathrm{Per}(\Omega) < \infty$) if and only if the integration current $[\![\Omega]\!] \in \mathcal{R}_{n+1}(M; \mathbb{Z}_2)$ satisfies $\mathbf{M}(\partial[\![\Omega]\!]) < \infty$. In this case,*

$$\partial[\![\Omega]\!] \in \mathcal{Z}_n(M, \partial M; \mathbb{Z}_2)$$

is a mod 2 relative cycle with

$$\mathrm{Per}(\Omega) = \mathbf{M}(\partial[\![\Omega]\!]).$$

The current $\partial[\![\Omega]\!]$ is supported on the reduced boundary: $\partial^*\Omega \subset \mathbf{spt}\partial[\![\Omega]\!] \subset \overline{\partial^*\Omega}$, and the varifold $|\partial^*\Omega|$ induced by the reduced boundary agrees with the varifold $|\partial[\![\Omega]\!]|$ induced by the mod 2 current. Moreover, Ω and $M \setminus \Omega$ induce the same mod 2 cycle: $\partial[\![\Omega]\!] = \partial[\![M \setminus \Omega]\!]$.

Proof. The set Ω defines an integer rectifiable $(n+1)$-current $[\![\Omega]\!]$ by integration: $[\![\Omega]\!](\omega) = \int_\Omega \omega$ for any smooth $(n+1)$-form ω. Reducing modulo 2, we regard $[\![\Omega]\!] \in \mathcal{R}_{n+1}(M; \mathbb{Z}_2)$. By definition, Ω is a Caccioppoli set if and only if its distributional boundary μ_Ω is a finite Radon measure, i.e. $|\mu_\Omega|(M) < \infty$. Since both $|\mu_\Omega|(M)$ and $\mathbf{M}(\partial[\![\Omega]\!])$ express the total variation of the distributional derivative of $\mathbf{1}_\Omega$, we have $|\mu_\Omega|(M) = \mathbf{M}(\partial[\![\Omega]\!])$, and the equivalence $\mathrm{Per}(\Omega) < \infty \Leftrightarrow \mathbf{M}(\partial[\![\Omega]\!]) < \infty$ follows.

The chain complex axiom gives $\partial(\partial[\![\Omega]\!]) = \partial^2[\![\Omega]\!] = 0$, so $\mathrm{spt}(\partial(\partial[\![\Omega]\!])) = \emptyset \subset \partial M$ and $\partial[\![\Omega]\!] \in \mathcal{Z}_n(M, \partial M; \mathbb{Z}_2)$.

By De Giorgi's structure theorem (Maggi [Mag12, Theorem 15.9]), the Gauss–Green measure satisfies $|\mu_\Omega| = \mathcal{H}^n \llcorner \partial^*\Omega$, and the density $\Theta^n(\partial^*\Omega, x) = 1$ for every $x \in \partial^*\Omega$ (Maggi [Mag12, Corollary 15.8]). Therefore

$$\mathbf{M}(\partial[\![\Omega]\!]) = |\mu_\Omega|(M) = \mathcal{H}^n(\partial^*\Omega) = \mathrm{Per}(\Omega),$$

and $\partial^*\Omega \subset \mathbf{spt}\partial[\![\Omega]\!] \subset \overline{\partial^*\Omega}$. Note that the current $\partial[\![\Omega]\!]$ is defined by first forming the integration current and then applying ∂, requiring no a priori knowledge of the reduced boundary. One could alternatively define an

n-current $[\![\partial^*\Omega]\!]$ by integrating directly over $\partial^*\Omega$; De Giorgi's structure theorem identifies the two: $\partial[\![\Omega]\!] = [\![\partial^*\Omega]\!]$. Throughout this thesis we write $[\![\partial\Omega]\!] := \partial[\![\Omega]\!]$, with the understanding that $\partial\Omega$ refers to the reduced boundary $\partial^*\Omega$ in the context of currents.

Since $\partial[\![\Omega]\!]$ is carried by $\partial^*\Omega$ with multiplicity one, the associated varifold $|\partial[\![\Omega]\!]|$ acts on $f \in C_c(G_n(M))$ by

$$|\partial[\![\Omega]\!]|\,(f) = \int_{\partial^*\Omega} f(x, T_x(\partial^*\Omega))\, d\mathcal{H}^n(x) = |\partial^*\Omega|\,(f),$$

where $T_x(\partial^*\Omega)$ is the approximate tangent plane, which exists $\mathcal{H}^n$-a.e. by rectifiability.

Finally, in $\mathbb{Z}_2$ coefficients we have $[\![\Omega]\!] + [\![M \setminus \Omega]\!] = [\![M]\!]$. Since $\partial[\![M]\!]$ is supported on ∂M, it vanishes as a relative cycle in $\mathcal{Z}_n(M, \partial M; \mathbb{Z}_2)$, so $\partial[\![\Omega]\!] + \partial[\![M \setminus \Omega]\!] = 0$, which in $\mathbb{Z}_2$ gives $\partial[\![\Omega]\!] = \partial[\![M \setminus \Omega]\!]$. □

The min-max machinery of Li–Zhou [LZ21] operates on equivalence classes of relative cycles rather than on individual currents. We therefore need a well-defined notion of mass on these classes.

Definition 2.1.4 (Canonical representative)**.** For $\tau \in \mathbf{Z}_n(A, B; \mathbb{Z}_2)$, there exists a unique representative $T \in \tau$ satisfying $T \llcorner B = 0$. This T is called the *canonical representative* of τ. When no confusion arises, we identify τ with its canonical representative.

Definition 2.1.5 (Mass of an equivalence class)**.** For $\tau \in \mathbf{Z}_n(A, B; \mathbb{Z}_2)$, the *mass* of τ is defined as $\mathbf{M}(\tau) := \mathbf{M}(T)$, where T is the canonical representative of τ. By [LZ21, Lemma 3.3], $\mathbf{M}(\tau) = \inf_{S \in \tau} \mathbf{M}(S)$, so the canonical representative achieves the infimum of mass over all representatives.

[**Xinze:** Review this paragraph again later.] The flat norm and isoperimetric lemmas below are the analytic backbone of the min-max theory. The flat norm $\mathcal{F}$ metrises the weak topology on cycles: two cycles are $\mathcal{F}$-close when they differ by a thin filling plus a small boundary residue. The two isoperimetric lemmas serve complementary purposes. The $\mathcal{F}$-isoperimetric

lemma (Lemma 2.1.7) takes flat-norm proximity as input and produces a filling Q with $\mathbf{M}(Q) \leq C_M \mathcal{F}(\tau_2 - \tau_1)$; it is the tool for constructing continuous maps into the cycle space, and we use it in Proposition 2.2.4 to establish the $\mathcal{F}$-continuity of the cycle-valued map $t \mapsto \partial[\![\Omega_t]\!]$. The $\mathbf{M}$-isoperimetric lemma (Lemma 2.1.8) takes mass-norm proximity as input and decomposes the difference $T_2 - T_1 = \partial Q + R$ into a filling Q and a boundary remainder R supported on ∂M, with $\mathbf{M}(Q) + \mathbf{M}(R) \leq C_M \mathbf{M}(\tau_2 - \tau_1)$; it is used in the pull-tight procedure and almost-minimizing selection of Chapter 8, where one needs to compare nearby cycles at the level of area.

Definition 2.1.6 (Relative flat norm)**.** For a compact set $K \subset M$ and $\tau \in \mathbf{Z}_n(A, B; \mathbb{Z}_2)$, the *relative flat norm* is

$$\mathcal{F}^K(\tau) := \inf_{T \in \tau} \mathcal{F}^K(T),$$

where $\mathcal{F}^K(T) := \inf\{\mathbf{M}(T - \partial S) + \mathbf{M}(S) : S \in \mathcal{R}_{n+1}(M; \mathbb{Z}_2), \mathrm{spt}(S) \subset K\}$ is the flat norm of T relative to K.

Lemma 2.1.7 ($\mathcal{F}$-isoperimetric lemma [LZ21, Lemma 3.15])**.** *There exist $\epsilon_M > 0$ and $C_M \geq 1$ depending only on the isometric embedding $M \hookrightarrow \mathbb{R}^L$ such that for any $\tau_1, \tau_2 \in \mathbf{Z}_n(M, \partial M; \mathbb{Z}_2)$ with $\mathcal{F}(\tau_2 - \tau_1) < \epsilon_M$, there exists a mod 2 $(n+1)$-current Q (an* isoperimetric choice*) such that:*

- $\mathrm{spt}(Q) \subset M$;
- $\mathrm{spt}(T_2 - T_1 - \partial Q) \subset \partial M$;
- $\mathbf{M}(Q) \leq C_M \mathcal{F}(\tau_2 - \tau_1)$,

where T_1, T_2 are the canonical representatives of τ_1, τ_2.

Lemma 2.1.8 ($\mathbf{M}$-isoperimetric lemma [LZ21, Lemma 3.17])**.** *There exist $\epsilon_M > 0$ and $C_M \geq 1$ depending only on the isometric embedding $M \hookrightarrow \mathbb{R}^L$ such that for any $\tau_1, \tau_2 \in \mathbf{Z}_n(M, \partial M; \mathbb{Z}_2)$ with $\mathbf{M}(\tau_2 - \tau_1) < \epsilon_M$, there exist a mod 2 $(n+1)$-current Q and a mod 2 rectifiable n-current R such that:*

- $\mathrm{spt}(Q) \subset M$, $\mathrm{spt}(R) \subset \partial M$;
- $T_2 - T_1 = \partial Q + R$;
- $\mathbf{M}(Q) + \mathbf{M}(R) \leq C_M \mathbf{M}(\tau_2 - \tau_1)$,

where T_1, T_2 are the canonical representatives.

[**Xinze:** Check this paragraph against Li–Zhou and Ch.8.] The isoperimetric lemmas above are stated by Li–Zhou for compact manifolds. When M is non-compact, we apply them to each precompact subset $\overline{K} \subset M$, which is compact and therefore has its own constants $\epsilon_{\overline{K}}, C_{\overline{K}}$. The exhaustion argument of Chapter 8 only requires these constants to exist for each fixed compact piece, not to be uniform over the exhaustion.

Given a Caccioppoli set Ω, we refer to the $(n+1)$-current $[\![\Omega]\!]$ as the *filling* of the relative cycle $\partial[\![\Omega]\!]$. For $\tau \in \mathbf{Z}_n(M, \partial M; \mathbb{Z}_2)$ with canonical representative T, the mass restricted to an open set V is $\|\tau\|(V) := \mathbf{M}(T \llcorner V)$.

Definition 2.1.9 (Interior boundary and free boundary part)**.** For a measurable set $\Omega \subset M$ with $\mathbf{M}(\partial[\![\Omega]\!]) < \infty$, the boundary current decomposes as

$$\partial[\![\Omega]\!] = \partial_I \Omega + \partial_F \Omega,$$

where:

- $\partial_I \Omega := \partial[\![\Omega]\!] \llcorner \mathrm{int}(M)$ is the *interior boundary*, supported on $\overline{\partial^* \Omega} \cap \mathrm{int}(M)$;
- $\partial_F \Omega := \partial[\![\Omega]\!] \llcorner \partial M$ is the *free boundary part*, supported on $\overline{\partial^* \Omega} \cap \partial M$.

In Chapters 6 and 7, we localise a global sweepout to a good set $U \subset M$ in order to apply the min-max theory on a precompact domain. This localisation requires cutting the Caccioppoli sets Ω_t along the interior boundary $\partial_I U$, which may introduce extra area near $\partial_I U$. To control this, we slice the filling $[\![\Omega_t]\!]$ by a Lipschitz function (such as the distance to $\partial_I U$) and use the coarea inequality to find cut-off radii where the extra area

is small; see Corollary 2.1.11 below. The slicing theorem is also used in the proof of the existence of good sets (Lemma 2.3.3), where we slice $[\![M]\!]$ by the distance function to find a level set with small interior boundary. The underlying tool is the slicing theorem for mod 2 flat chains (Simon [Sim84, Sections 28–29]).

Theorem 2.1.10 (Slicing [Sim84, 28.4, 28.5])**.** *Let $T \in \mathcal{R}_{n+1}(M;\mathbb{Z}_2)$ and let $f : M \to \mathbb{R}$ be a Lipschitz function. Then for almost every $t \in \mathbb{R}$, the* slice

$$\langle T, f, t\rangle \in \mathcal{R}_n(M;\mathbb{Z}_2)$$

is a well-defined mod 2 rectifiable n-current satisfying:

(i) *(Boundary formula) $\partial\langle T, f, t\rangle = \langle \partial T, f, t\rangle$ wherever ∂T has locally finite mass;*

(ii) *(Coarea inequality) $\int_{\mathbb{R}} \mathbf{M}(\langle T, f, t\rangle)\, dt \le \int |\nabla f|\, d\|T\|$.*

Corollary 2.1.11 (Coarea formula for fillings)**.** *Let $Q = [\![\Omega]\!]$ for a bounded open set $\Omega \subset M$ of finite perimeter, and let $f : M \to \mathbb{R}$ be Lipschitz. Then for almost every t, the slice $\langle Q, f, t\rangle \in \mathcal{Z}_n(M, \partial M;\mathbb{Z}_2)$ is a relative cycle, and*

$$\int_{\mathbb{R}} \mathbf{M}(\langle Q, f, t\rangle)\, dt \le \int_{\Omega} |\nabla f|\, d\mathcal{H}^{n+1}.$$

The min-max procedure produces limit objects not as currents but as varifolds: the pull-tight argument of Chapter 8 yields a stationary varifold, and the regularity theory ([LZ21, Theorem 6.2]; see also Chapter 9) recovers a smooth minimal hypersurface from it. We recall the basic definitions; standard references include Pitts [Pit81], Simon [Sim84], Colding–De Lellis [CL03], and De Lellis–Tasnady [DLT13].

Definition 2.1.12 (Varifolds)**.** Let $\mathrm{Gr}_n(M)$ denote the Grassmannian bundle of (unoriented) n-planes over M. An *n-varifold* on M is a Radon measure on $\mathrm{Gr}_n(M)$. The space of all n-varifolds is denoted $\mathcal{V}_n(M)$, equipped with the weak-$*$ topology.

For $V \in \mathcal{V}_n(M)$, the *mass measure* $\|V\|$ is the pushforward of V under the projection $\mathrm{Gr}_n(M) \to M$. The *total mass* is $\|V\|(M)$, and the *support* is $\operatorname{spt}\|V\|$. For an open set $A \subset M$, the *restriction* $V \llcorner A$ is the varifold defined by $V \llcorner A(E) := V(E \cap \mathrm{Gr}_n(A))$.

Definition 2.1.13 (First variation)**.** The *first variation* of $V \in \mathcal{V}_n(M)$ is the linear functional on smooth vector fields defined by

$$\delta V(X) := \int_{\mathrm{Gr}_n(M)} \operatorname{div}_S X(x)\, dV(x,S),$$

where $\operatorname{div}_S X(x) := \sum_{i=1}^n \langle \nabla_{e_i} X, e_i \rangle$ for any orthonormal basis $\{e_i\}$ of S.

Definition 2.1.14 (Stationarity)**.** A varifold $V \in \mathcal{V}_n(M)$ is:

(i) *stationary in* $\operatorname{int}(M)$ if $\delta V(X) = 0$ for all $X \in \mathfrak{X}_c(\operatorname{int}(M))$;

(ii) *stationary with free boundary* if $\delta V(X) = 0$ for all $X \in \mathfrak{X}_{\tan}(M)$.

Definition 2.1.15 (Pushforward of varifolds)**.** Let $\varphi : M \to M$ be a diffeomorphism. The *pushforward* $\varphi_\sharp V \in \mathcal{V}_n(M)$ is defined by

$$\varphi_\sharp V(A) := \int_{\mathrm{Gr}_n(M)} \mathbf{1}_A(\varphi(x), d\varphi_x(S)) \cdot |J_S\varphi(x)|\, dV(x,S)$$

for Borel $A \subset \mathrm{Gr}_n(M)$, where $J_S\varphi(x) := \mathbf{det}(d\varphi_x|_S)$ is the n-dimensional Jacobian of φ restricted to S.

Remark 2.1.16. Condition (ii) implies condition (i) since $\mathfrak{X}_c(\operatorname{int}(M)) \subset \mathfrak{X}_{\tan}(M)$. The additional constraint in (ii) is that the first variation vanishes even for vector fields that are supported up to ∂M, as long as they are tangent to ∂M.

The pull-tight procedure (Chapter 8) requires a metric on the space of varifolds that metrises weak-$*$ convergence and behaves well under localisation to open subsets. This role is played by the **F**-metric.

Definition 2.1.17 (**F**-metric)**.** The **F**-*metric* on $\mathcal{V}_n(M)$ is defined by

$$\mathbf{F}(V,W) := \sup\left\{ \int f\,dV - \int f\,dW : f \in C_c(\mathrm{Gr}_n(M)),\ |f| \le 1,\ \mathrm{Lip}(f) \le 1 \right\}.$$

On sets of uniformly bounded mass, **F** metrizes the weak-$*$ topology. For an open set $U \subset M$, the *localized* **F**-*metric* is $\mathbf{F}_U(V,W) := \mathbf{F}(V \llcorner U, W \llcorner U)$. See [LZ21, Section 2].

2.2 SWEEPOUTS AND WIDTH

The cut-and-paste operations in our proof (splitting, extension, and nested sweepout construction in Chapters 5–7) are most naturally described using Caccioppoli sets. We therefore adopt a continuous sweepout framework based on Caccioppoli sets (sets of finite perimeter), following the approach of Chambers–Liokumovich [CL20] and De Lellis–Tasnady [DLT13]. In Proposition 2.2.4 below, we show that every Caccioppoli sweepout gives rise to a continuous map into the space of mod 2 relative cycles; this is the bridge that allows us to invoke the min-max machinery of Li–Zhou [LZ21] (pull-tight, almost minimizing, regularity) developed in the cycle-valued setting. For applications to non-compact manifolds, we consider sweepouts of bounded open sets.

Definition 2.2.1 (Sweepout of $(U, \partial_F U)$)**.** Let $U \subset M$ be a bounded relatively open set of finite perimeter. A *sweepout of* $(U, \partial_F U)$ is a family of Caccioppoli sets $\{\Omega_t\}_{t\in[0,1]}$ in M satisfying:

(sw1) **Boundary condition**: $\Omega_0 \cap U = \emptyset$ and $U \subset \Omega_1$;

(sw2) **Continuity**: the map $t \mapsto \mathbf{1}_{\Omega_t}$ is continuous in $L^1(M)$, i.e.,

$$\mathcal{H}^{n+1}(\Omega_t \triangle \Omega_s) \to 0 \quad \text{as } s \to t;$$

(sw3) **Finite mass**: $\sup_{t\in[0,1]} \mathrm{Per}(\Omega_t) < \infty$;

(sw4) **No mass concentration**: for every $x \in M$,

$$\lim_{r\to 0} \sup_{t\in[0,1]} \operatorname{Per}(\Omega_t, B_r(x)) = 0.$$

The *height* of the sweepout is $F(\{\Omega_t\}) := \sup_{t\in[0,1]} \operatorname{Per}(\Omega_t)$.

Definition 2.2.2 (Width)**.** The *width* of $(U, \partial_F U)$ is

$$W(U) := \inf\big\{F(\{\Omega_t\}) : \{\Omega_t\} \text{ is a sweepout of } (U, \partial_F U)\big\}.$$

Remark 2.2.3. The free boundary condition is built into the Caccioppoli framework: by Proposition 2.1.3, each reduced boundary $\partial^*\Omega_t$ induces a mod 2 relative cycle $\partial[\![\Omega_t]\!] \in \mathcal{Z}_n(M, \partial M; \mathbb{Z}_2)$, whose "boundary" (in the sense of currents) is automatically supported on ∂M. Note that the sweepout "fills" U in the sense that the fillings Ω_t transition from being disjoint from U to containing U. Throughout this thesis we assume $\partial_F U \neq \emptyset$, i.e. $\overline{U} \cap \partial M \neq \emptyset$: this is essential for the min-max limit to be a free boundary minimal hypersurface. When $\partial_F U = \emptyset$, one recovers a closed minimal hypersurface, which is already covered by the work of Chambers–Liokumovich [CL20].

The following proposition provides the bridge between the Caccioppoli sweepout framework and the cycle-valued maps used in Li–Zhou [LZ21].

Proposition 2.2.4 (From Caccioppoli sweepouts to cycle-valued maps)**.** *Let $\{\Omega_t\}_{t\in[0,1]}$ be a sweepout of $(U, \partial_F U)$. Then the map*

$$\Phi : [0,1] \to \mathcal{Z}_n(M, \partial M; \mathbb{Z}_2), \quad t \mapsto \partial[\![\Omega_t]\!]$$

is continuous in the flat topology $\mathcal{F}$, and satisfies $\mathbf{M}(\Phi(t)) = \operatorname{Per}(\Omega_t)$ for all t by Proposition 2.1.3. Moreover, Φ satisfies the no mass concentration condition

$$\lim_{r\to 0} \sup_{t\in[0,1]} \|\Phi(t)\|(B_r(x)) = 0 \quad \textit{for every } x \in M.$$

Proof. The mass identity and no concentration condition are immediate from Proposition 2.1.3 and (sw4). For $\mathcal{F}$-continuity, $[\![\Omega_t]\!] - [\![\Omega_s]\!]$ is a filling of $\Phi(t) - \Phi(s)$ modulo ∂M, so $\mathcal{F}(\Phi(t) - \Phi(s)) \leq \mathcal{H}^{n+1}(\Omega_t \triangle \Omega_s) \to 0$ by (sw2). □

Definition 2.2.5 (Good sweepout)**.** A sweepout $\{\Omega_t\}$ of $(U, \partial_F U)$ is a *good sweepout* if there exists $\varepsilon > 0$ such that

$$\mathrm{Per}(\Omega_t) \leq 5\,\mathbf{M}(\partial_I U) \quad \text{for all } t \in [0, \varepsilon) \cup (1 - \varepsilon, 1].$$

We write $W_g(U)$ for the infimum of heights over all good sweepouts.

Definition 2.2.6 (∂-transverse function)**.** A smooth function $f : M \to \mathbb{R}$ is *∂-transverse* if $f|_{\partial M}$ has no critical points, i.e. $\nabla_{\partial M}(f|_{\partial M}) \neq 0$ everywhere on ∂M. A *∂-transverse Morse function* is a Morse function on M that is also ∂-transverse.

Definition 2.2.7 (Nested sweepout)**.** A sweepout $\{\Omega_t\}$ of $(U, \partial_F U)$ is *nested* if $\Omega_s \subset \Omega_t$ for all $0 \leq s \leq t \leq 1$. Equivalently, $\{\Omega_t\}$ arises from a ∂-transverse Morse function $f : M \to \mathbb{R}$ via $\Omega_t = f^{-1}((-\infty, c_t))$ for some increasing function $t \mapsto c_t$. We denote the collection of nested sweepouts of U by $\mathcal{S}_n(U)$, and write $W_n(U) := \inf_{S \in \mathcal{S}_n(U)} F(S)$.

Definition 2.2.8 (Relative sweepout and relative width)**.** A *relative sweepout* of U is a sweepout $\{\Omega_t\}$ of $(U, \partial_F U)$ with the additional property that $\partial^* \Omega_t \subset \overline{U}$ for all $t \in [0, 1]$. We denote the collection of relative sweepouts by $\mathcal{S}_\partial(U)$, and define the *relative width* as $W_\partial(U) := \inf_{S \in \mathcal{S}_\partial(U)} F(S)$.

By Proposition 2.2.4, each of the above sweepout classes (good, nested, relative) gives rise to an $\mathcal{F}$-continuous map into $\mathcal{Z}_n(M, \partial M; \mathbb{Z}_2)$ satisfying the no mass concentration condition, so the min-max machinery of Li–Zhou [LZ21] applies to each corresponding width.

Proposition 2.2.9 (Width inequalities)**.** *Let $U \subset M$ be a bounded open set with smooth boundary. Then:*

(i) $W_\partial(U) \le W_n(U) \le W_\partial(U) + \mathbf{M}(\partial_I U)$;

(ii) $W_n(U) = W(U)$, *where* $W(U)$ *is the width over all sweepouts;*

(iii) *If* U *is a good set (Definition 2.3.1; see Section 2.3), then* $W_g(U) = W(U)$.

Proof. Part (i) follows by restricting a nested sweepout to $\overline{U}$ (giving the lower bound) and extending a relative sweepout by adding $\partial_I U$ at each time (giving the upper bound). Part (ii) is proved by approximating an arbitrary sweepout by nested ones using the coarea formula (Theorem 2.1.10); see [CL20, Lemma 4.2]. Part (iii) follows from (ii) by a cut-off argument using the good set condition. □

2.3 GOOD SETS

On a compact manifold, the min-max process automatically produces a minimal hypersurface because all mass stays in a bounded region. On a non-compact manifold, mass can escape to infinity during the min-max sequence, and the limit varifold may be trivial. To prevent this, we work inside bounded open sets U whose interior boundary $\partial_I U$ is small relative to the width: sweepouts of U must transfer mass from outside to inside through $\partial_I U$, and when $\mathbf{M}(\partial_I U) \ll W_\partial(U)$, at least one slice must carry substantial mass inside $\overline{U}$. This is the role of good sets, used in the no-escape argument of Chapter 7.

Definition 2.3.1 (Good set)**.** A bounded open subset $U \subset M$ of finite perimeter is a *good set* if

$$\mathbf{M}(\partial_I U) \le \frac{1}{10} W_\partial(U),$$

where $\partial_I U = \partial[\![U]\!] \llcorner \mathrm{int}(M)$ is the interior boundary of U.

Remark 2.3.2 (Closed vs. free boundary limits). The min-max theory of Li–Zhou [LZ21] produces a stationary varifold whose support is a minimal

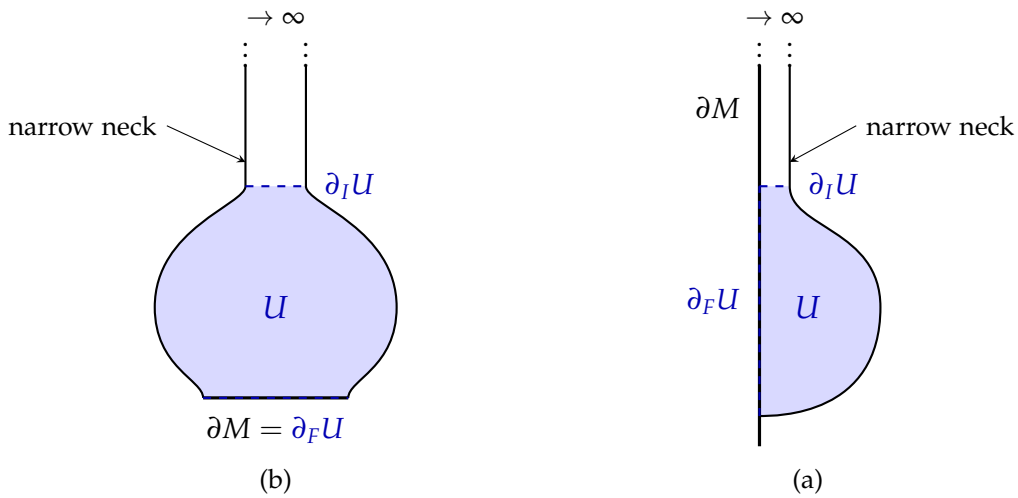

Figure 2.1: A good set U (shaded) inside a non-compact manifold M with boundary ∂M (bold). The free boundary $\partial_F U \subset \partial M$ is the portion of ∂M adjacent to U (dashed), and the interior boundary $\partial_I U$ (dashed) sits at a narrow neck, so $\mathbf{M}(\partial_I U)$ is small relative to $W_\partial(U)$. (a) ∂M parallel to the neck, both extending to infinity; (b) ∂M transverse to the neck. Note that in case (b), the min-max limit could in principle be a closed minimal hypersurface in the interior of M, away from ∂M; see Remark 2.3.2.

hypersurface, but does not by itself guarantee that $\partial\Sigma \neq \varnothing$: in general, the limit could be a closed minimal hypersurface in the interior of M (see [LZ21, Remark 1.3]). Following the standard convention, a closed minimal hypersurface is regarded as a (degenerate) free boundary minimal hypersurface with $\partial\Sigma = \varnothing$; the conditions $\partial\Sigma \subset \partial M$ and the orthogonal meeting condition in Theorem 1.0.1 are then vacuously satisfied. When M is a compact domain in $\mathbb{R}^{n+1}$, the closed case cannot occur since $\mathbb{R}^{n+1}$ admits no closed minimal hypersurface.

Lemma 2.3.3 (Existence of good sets)**.** *Let (M^{n+1}, g) be a complete Riemannian manifold with non-empty boundary and sublinear volume growth, i.e.*

$$\liminf_{R\to\infty} \frac{\mathcal{H}^{n+1}(B_R(x) \cap M)}{R} = 0 \qquad \textit{for some } x \in M.$$

Then there exists a good set $U \subset M$ with $0 < W_g(U) < \infty$.

Proof. Fix a point $x \in M$ and a radius $r > 0$ such that the local isoperimetric constant

$$C_I := \inf\{\mathbf{M}(\partial_I E) : E \subset B_r(x),\ \mathcal{H}^{n+1}(E) = \tfrac{1}{2}\,\mathcal{H}^{n+1}(B_r(x))\} > 0.$$

Let $d_x : M \to [0, \infty)$ denote the distance function from x. Applying the slicing theorem (Theorem 2.1.10) to the current $T = [\![M]\!]$ and the Lipschitz function $f = d_x$ with $|\nabla d_x| \leq 1$, we obtain

$$\int_0^\infty \mathbf{M}(\langle [\![M]\!], d_x, R\rangle)\, dR \leq \mathcal{H}^{n+1}(M \cap B_S(x))$$

for any $S > 0$. The sublinear growth assumption therefore allows us to pick a regular value $R > r$ such that

$$\mathbf{M}(\langle [\![M]\!], d_x, R\rangle) \leq \frac{C_I}{100}.$$

Set $U := B_R(x) \cap M$. Since $\partial_I U$ is supported on $\partial B_R(x) \cap \mathrm{int}(M)$, we have $\mathbf{M}(\partial_I U) \leq \mathbf{M}(\langle [\![M]\!], d_x, R\rangle) \leq C_I/100$. Any sweepout of U must bisect volume in $B_r(x)$ at some time, so $W_\partial(U) \geq C_I$. Since $\mathbf{M}(\partial_I U) \leq \frac{1}{10} W_\partial(U)$, the set U is good. Finiteness $W_g(U) < \infty$ follows from the existence of a sweepout by slices $\langle [\![M]\!], d_x, R'\rangle$ for $R' \in [0, R]$. $\square$

3

STRUCTURE OF THE PROOF

The proof of the main theorem requires overcoming two fundamental challenges: the **non-compactness** of the ambient manifold M, and the **presence of the boundary** ∂M. This chapter provides an overview of how these challenges shape the technical framework and guide the proof strategy.

3.1 CHALLENGES FROM NON-COMPACTNESS

In the compact setting, the classical Almgren–Pitts min-max theory produces minimal hypersurfaces by considering sweepouts of the entire manifold. When M is complete but non-compact, we must fundamentally rethink the setup.

Localizing the Sweepout

Since M is non-compact, the width of sweepouts of the entire manifold would typically be infinite or ill-defined. Following Chambers–Liokumovich [CL20], we instead consider *sweepouts of bounded open sets*: given a bounded open set $U \subset M$, a sweepout is a continuous family of relative cycles $\{\tau_t\}$ that starts "outside U" and ends having "filled U." The width $W(U)$ is defined as the infimum of the maximal slice mass over all such sweepouts.

This localization introduces two difficulties: **stationarity** (how to perform pull-tight on non-compact M) and **escape** (how to ensure mass doesn't escape from U).

Stationarity and Escape

These are solved by two independent mechanisms:

- **Good set condition solves escape.** We require $\mathbf{M}(\partial_I U) \leq \frac{1}{10} W_\partial(U)$. This makes $\partial_I U$ a "bottleneck" too small to transfer the filling alone, forcing every good sweepout to have critical slices with substantial mass inside U.
- **Exhaustion + pull-tight solves stationarity.** Since M is non-compact, we pull-tight layer by layer on an exhaustion $U_0 \Subset U_1 \Subset \cdots \Subset M$, then use a diagonal argument to obtain a globally stationary varifold.

The key point is that these two mechanisms must work together: we need the stationary slice to come from a good sweepout of a good set. Pull-tight preserves good sweepout structure (since endpoints have small mass and are not modified), so the extracted stationary slice inherits the non-escape property from the good set condition.

4

MORSE FOLIATIONS WITH CONTROLLED AREA OF FIBERS

The goal of this chapter is to develop technical tools for constructing *nested* sweepouts from arbitrary sweepouts. The central problem is the following: given a sweepout $\{\tau_t, Q_t\}$ that may not be nested (i.e., $Q_s \not\leq Q_t$ for $s < t$), how can we deform it into a nested sweepout while controlling the mass of the slices?

Our strategy is to work locally: we construct ∂-transverse Morse functions on regions of M whose level sets provide the desired nested structure. The key technical results are Lemma 4.1.1 and Lemma 4.1.2, which show that given any Morse function f and a target hypersurface Σ, we can perturb f to a new Morse function g having Σ as a level set, while preserving mass bounds on the slices. These local constructions are then glued together in Chapter 6 to produce global nested sweepouts.

4.1 GLUING MORSE FOLIATIONS

Lemma 4.1.1. *Let $N^{n+1} \subset M^{n+1}$ be a compact submanifold with non-empty boundary. Let $f : N \to [a, b]$ be a ∂-transverse Morse function. Let Σ^n be a closed submanifold of ∂N.*

For every $\varepsilon > 0$, there exists a Morse function $g : N \to [a, b]$, such that the following holds:

(1) $g^{-1}(b) = \Sigma$;

(2) $f^{-1}([a,t)) \subset N_{\varepsilon/2}(g^{-1}([a,t))) \subset N_\varepsilon(f^{-1}([a,t)))$;

(3) $\mathbf{M}(\langle [\![N]\!], g, t\rangle) \leq \mathbf{M}(\partial_I f^{-1}([a,t])) + \varepsilon$;

(4) *If* $\mathbf{dist}(x, f(\Sigma)) > \varepsilon$ *then* $f^{-1}(x) = g^{-1}(x)$.

We will also need a slightly different version of this lemma for a non-compact submanifold N.

Lemma 4.1.2. *Let $N \subset M$ be a not necessarily compact submanifold with non-empty boundary and $f : N \to (-\infty, b]$ be a proper ∂-transverse Morse function. Let Σ be a compact submanifold of ∂N.*

For every $\varepsilon > 0$ there exists a Morse function $g : N \to (-\infty, b]$, such that the following holds:

(1) $g^{-1}(b) = \Sigma$;

(2) $f^{-1}((-\infty,t)) \subset N_{\varepsilon/2}(g^{-1}((-\infty,t))) \subset N_\varepsilon(f^{-1}((-\infty,t)))$;

(3) $\mathbf{M}(\langle [\![N]\!], g, t\rangle) \leq \mathbf{M}(\partial_I f^{-1}((-\infty,t])) + \varepsilon$;

(4) *If* $\mathbf{dist}(x, f(\Sigma)) > \varepsilon$ *then* $f^{-1}(x) = g^{-1}(x)$.

4.2 GLUING MORSE FOLIATIONS ON A MANIFOLD SEPARATED BY A HYPERSURFACE TRANSVERSE TO THE BOUNDARY

Lemma 4.2.1. *Let $N^{n+1} \subset M^{n+1}$ be a manifold with compact boundary ∂N and Σ be a hypersurface with $\partial\Sigma \subset \partial N$ and such that Σ intersects ∂N transversally and separates N into two disjoint regions $N \backslash \Sigma = V_1 \sqcup V_2$.*

For every $\varepsilon > 0$, there exist open sets with smooth boundary Ω_1 and Ω_2 and a Morse function $g : \mathrm{cl}(N \backslash (\Omega_1 \cup \Omega_2)) \to [0,1]$, such that the following holds:

(1) *Ω_1 is a relative inward ε-perturbation of V_1; Ω_2 is a relative inward ε-perturbation of V_2.*

(2) $g^{-1}(0) = \partial\Omega_1 \cup \partial\Omega_2$ *and* $g^{-1}(1) = \partial N$*;*

(3) $\mathbf{M}(\langle [\![N]\!], g, t\rangle) \leq \mathbf{M}(\partial_I N) + 2\,\mathcal{H}^n(\Sigma) + \varepsilon$.

Proof. Let $f : M \to \mathbb{R}$ be a ∂-transverse Morse function such that there exists some sufficiently small $0 < \delta_1 < \varepsilon$,

$$f(x) = \operatorname{sdist}(x, \partial N)$$

for $x \in N_{\delta_1}(\partial N)$. The sign is chosen so that $f(x) \geq 0$ when $x \in N$. Without any loss of generality, we may assume the following three conditions hold:

(i) f has no critical points in the ε-neighborhood of Σ: a generic Morse function has no critical points on any given compact set, so a small perturbation of f (supported away from $N_{\delta_1}(\partial N)$) achieves this.

(ii) The restriction $f|_\Sigma$ is also Morse: by Sard's theorem, $f|_\Sigma$ is Morse for a generic choice of f.

(iii) The mass bound

$$\mathbf{M}(\langle [\![N]\!], f, t\rangle) \leq \mathbf{M}(\partial_I N) + \varepsilon/10$$

holds for $t \in [0, \delta_1]$: since $f = \operatorname{sdist}(\cdot, \partial N)$ in $N_{\delta_1}(\partial N)$, the level set $f^{-1}(t)$ converges to ∂N in the C^1 sense as $t \to 0$, so $\mathcal{H}^n(f^{-1}(t) \cap N)$ converges to $\mathcal{H}^n(\partial_I N)$, and we can choose δ_1 small enough for this bound to hold.

Pick $\delta_2 \in (0, \delta_1)$ so that for $t \in (0, \delta_2)$,

$$\mathcal{H}^n(f^{-1}(t) \cap N_{\delta_2}(\Sigma)) < \varepsilon/10$$

and

$$\mathcal{H}^n(\partial N_{\delta_2}(\Sigma)) \leq 2\,\mathcal{H}^n(\Sigma) + \varepsilon/10.$$

Let Ω_i denote a relative inward $\delta_2/2$-perturbation of V_i, i.e.

(a) $V_i \backslash N_{\delta_2/2}(\partial V_i) \subset \Omega_i \subsetneq V_i$;

(b) there exists a nested family of open sets $\{\Xi_t^i\}_{t\in[0,1]}$ and corresponding smooth isotopies $\partial\Xi_t^i$, such that $\partial\Xi_1^i$ is a smoothing of ∂V_i and

$$\mathcal{H}^n(\partial\Xi_t^i) \leq \mathcal{H}^n(\partial V_i) + \delta_2/2$$

for all $t \in [0,1]$.

We can make an arbitrarily small perturbation of f by an ambient diffeomorphism so that f is a Morse function on $\partial\Omega_1 \cup \partial\Omega_2$. For $t \in [0,\delta_2]$, since $f = \mathrm{sdist}(\cdot, \partial N)$ in $N_{\delta_1}(\partial N)$, the level set $f^{-1}(t)$ is C^1-close to ∂N. In particular, each leaf $\Sigma_t = (f^{-1}(t) \cap N)\backslash(\Omega_1 \cup \Omega_2)$ is a normal graph over some region $U_t \subset \partial N$, with

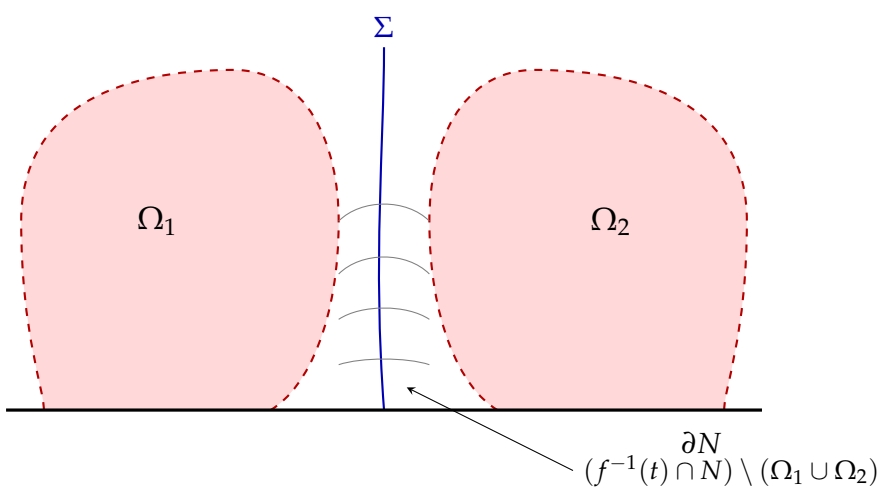

Figure 4.1: Gluing two submanifolds using a Morse function. The hypersurface Σ (blue) separates N into two regions; Ω_1 and Ω_2 (shaded) are inward perturbations of these regions. The foliation leaves (gray) in the gap $N \setminus (\Omega_1 \cup \Omega_2)$ connect $\partial\Omega_1 \cup \partial\Omega_2$ to ∂N.

$$\mathcal{H}^n(\Sigma_t) \leq \mathcal{H}^n(U_t) + \varepsilon/5$$

It follows that

$$\mathbf{M}(\langle [\![N \setminus (\Omega_1 \cup \Omega_2)]\!], f, t\rangle) \leq 2\,\mathcal{H}^n(\Sigma) + \mathbf{M}(\partial_I N) + \varepsilon/2.$$

We now verify that the hypotheses of Lemma 4.1.2 are satisfied for the restriction $f|_{N\setminus(\Omega_1\cup\Omega_2)}$:

- The domain $N' = \mathrm{cl}(N \setminus (\Omega_1 \cup \Omega_2))$ is a (non-compact) manifold with boundary $\partial N' = \partial\Omega_1 \cup \partial\Omega_2 \cup \partial N$.
- $f|_{N'}$ is proper: preimages of compact sets in $[0, \delta_1]$ are compact since the non-compact ends (i.e. the interiors of V_1 and V_2) have been removed by excising Ω_1 and Ω_2.
- $f|_{N'}$ is ∂-transverse on N', as this is inherited from the ∂-transversality of f on N.
- $\partial\Omega_1 \cup \partial\Omega_2$ plays the role of Σ in Lemma 4.1.2, i.e. the compact submanifold of $\partial N'$ sent to $b = \delta_1$.

Applying Lemma 4.1.2, we obtain the desired function g and the corresponding foliation, i.e.

$$\mathbf{M}(\langle [\![N]\!], g, t\rangle) \leq 2\,\mathcal{H}^n(\Sigma) + \mathbf{M}(\partial_I N) + \varepsilon.$$

□

5

SPLITTING AND EXTENSION LEMMAS

This chapter develops two fundamental technical tools needed for the construction of nested sweepouts: the *extension lemma* and the *splitting lemma*. The extension lemma allows us to modify a nested family of hypersurfaces to accommodate a new endpoint hypersurface while controlling the mass increase. The splitting lemma, based on Falconer's projection theorem, enables us to construct a nested family connecting two nested regions with controlled slice masses.

5.1 THE EXTENSION LEMMA

The extension lemma is a key ingredient in the monotonisation algorithm. It allows us to insert or remove a hypersurface at one endpoint of a nested family while keeping the mass bounded.

Lemma 5.1.1. *Suppose that $f : M \to [-1, \infty)$ is a ∂-transverse Morse function and $\{\Gamma_t\} = \{f^{-1}(t)\}_{t\in[0,1]}$ is a nested family of hypersurfaces such that $\mathcal{H}^n(\Gamma_t) \leq A$ for all $t \in [0,1]$ with associated open sets $\{\Omega_t\} = \{f^{-1}((-\infty, t))\}$.*

I. *Suppose Ω is a bounded open set with boundary Γ a smooth embedded manifold such that*

   (1) $\Omega \subset \Omega_1$;

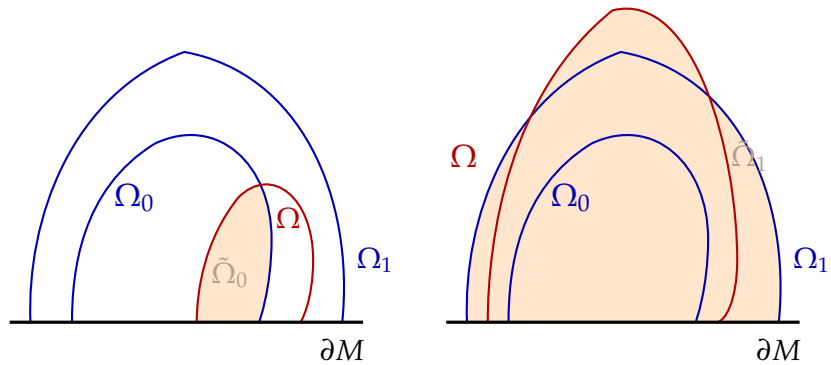

Figure 5.1: Illustration of Lemma 5.1.1. **Left** (Case I, $\Omega \neq \Omega_0$): Ω_0 and Ω partially overlap, with $\Omega \subset \Omega_1$. The shaded region $\tilde{\Omega}_0 = \Omega \cap \Omega_0$ serves as the new starting set for the modified nested family. **Right** (Case II, $\Omega \neq \Omega_1$): $\Omega_0 \subset \Omega$, and Ω partially overlaps Ω_1. The shaded region $\tilde{\Omega}_1 = \Omega \cup \Omega_1$ serves as the new ending set for the modified nested family.

(2) *There is an $\varepsilon > 0$ such that for every Ω' with $\Omega \subset \Omega' \subset \Omega_1$ we have*

$$\mathcal{H}^n(\Gamma) < \mathbf{M}(\partial_I \Omega') + \frac{\varepsilon}{4}.$$

Then, we can find a nested family $\{\tilde{\Gamma}_t\}$ and an associated family of open sets $\{\tilde{\Omega}_t\}$ such that $\tilde{\Omega}_0 \subset \Omega_0$, $\tilde{\Gamma}_1 = \Gamma$ and for all $t \in [0,1]$

$$\mathcal{H}^n(\tilde{\Gamma}_t) \leq A + \varepsilon.$$

Furthermore, if $\Omega_0 \subset \Omega$, then $\tilde{\Gamma}_0 = \Gamma_0$.

II. *Suppose that instead of properties (1) and (2) above, the following are true:*

(1)′ $\Omega_0 \subset \Omega$;

(2)′ *There is an $\varepsilon > 0$ such that for every Ω' with $\Omega_0 \subset \Omega' \subset \Omega$ we have*

$$\mathcal{H}^n(\Gamma) < \mathbf{M}(\partial_I \Omega') + \frac{\varepsilon}{4}.$$

Then, we can find a nested family $\{\tilde{\Gamma}_t\}$ *and an associated family of open sets* $\{\tilde{\Omega}_t\}$ *such that* $\Omega_1 \subset \tilde{\Omega}_1$, $\tilde{\Omega}_0 = \Gamma$, *and for all* $t \in [0,1]$

$$\mathcal{H}^n(\tilde{\Gamma}_t) \leq A + \varepsilon.$$

Furthermore, if $\Omega \subset \Omega_1$, *then* $\tilde{\Gamma}_1 = \Gamma_1$.

Proof. We begin the proof with the first half of this lemma. The desired family will be obtained by regularization of the collection of hypersurfaces $\{\partial(\Omega_0 \cap \Omega)\}$. As in the left side of Figure 5.1.

- **Case I(a)** $\Omega \subset \Omega_0$**:**
  For a sufficiently small $\delta > 0$ the function $g : \mathrm{cl}(N_\delta(\Gamma) \cap \Omega) \to [0,1]$ given by $g(x) = \frac{1}{\delta}\mathbf{dist}(x, \Gamma)$ is a smooth function without critical points and $\tilde{\Gamma}_t = g^{-1}(t)$ would satisfy

  $$\mathcal{H}^n(\tilde{\Gamma}_t) \leq \mathcal{H}^n(\Gamma) + \frac{\varepsilon}{2}.$$

  By I(2) and $\Omega \subset \Omega_0 \subset \Omega_1$, we know that

  $$\mathcal{H}^n(\Gamma) \leq \mathbf{M}(\partial_I \Omega_0) + \frac{\varepsilon}{4}$$

  thus

  $$\mathcal{H}^n(\tilde{\Gamma}_t) \leq A + \varepsilon.$$

  We extend g to a ∂-transverse Morse function on M in an arbitrary way and $\{\tilde{\Gamma}_t\}$ is a nested family satisfying the conditions of the theorem.

- **Case I(b)** $\Omega \backslash \Omega_0 \neq \varnothing$**:**
  Make a small perturbation to the hypersurface $\Gamma = \partial\Omega$ so that $f|_{\partial\Omega}$ is a Morse and (1) and (2) are still satisfied, possibly replacing $\varepsilon/4$ in (2) by $\varepsilon/2$.

Consider f restricted to Ω. We apply Lemma 4.1.1 with $N = \Omega$ and $\Sigma = \Gamma$ to obtain a Morse function $g : \Omega \to [-1, 1]$ for $\varepsilon/2$ such that $g^{-1}(-1)$ is a point in Ω and

1. $g^{-1}(1) = \Gamma$;
2. $f^{-1}([-1, t)) \subset N_{\varepsilon/4}(g^{-1}([-1, t))) \subset N_{\varepsilon/2}(f^{-1}([-1, t)))$;
3. $\mathbf{M}(\langle [\![\Omega]\!], g, t\rangle) \leq \mathbf{M}(\partial_I f^{-1}([-1, t])) + \varepsilon/2$;
4. If $\mathbf{dist}(x, f(\Gamma)) \geq \varepsilon/2$ then $f^{-1}(x) = g^{-1}(x)$.

It follows from 3. that

$$\begin{aligned}\mathcal{H}^n(g^{-1}(t)) \leq & \mathcal{H}^n(f^{-1}(t) \cap \Omega) \\ & + \mathcal{H}^n(f^{-1}([-1, t]) \cap \Gamma) + \frac{\varepsilon}{2}\end{aligned}$$

Furthermore, we have

$$g^{-1}([-1, 0)) \subset N_\varepsilon(\Omega \cap \Omega_0).$$

After a small perturbation of the function g we may assume that $g^{-1}([-1, 0)) \subset (\Omega \cap \Omega_0)$. We can extend g to a Morse function in an arbitrary way. We claim that $\tilde{\Gamma}_t = g^{-1}(t)$ for $t \in [0, 1]$ is the desired nested family and the only thing left is to prove the upper bound for the areas of $\tilde{\Gamma}_t$.

For any smooth hypersurface Σ_t obtained by a small perturbation of $\partial(\Omega \cup \Omega_t)$ since $\Omega \subset (\Omega \cup \Omega_t) \subset \Omega_1$. It follows that

$$\mathcal{H}^n(\Gamma) \leq \mathcal{H}^n(\Sigma_t) + \frac{\varepsilon}{4}$$

Furthermore, it follows that

$$\mathcal{H}^n(\Gamma) \leq \mathbf{M}(\partial_I(\Omega \cup \Omega_t)) + \frac{\varepsilon}{2}$$

Since $\partial(\Omega \cup \Omega_t) = (\Gamma_t \backslash \Omega) \cup (\Gamma \backslash \Omega_t)$ we have

$$\begin{aligned} &\mathcal{H}^n(\Gamma \cap \Omega_t) + \mathcal{H}^n(\Gamma \backslash \Omega_t) \\ &\leq \mathcal{H}^n(\Gamma_t \backslash \Omega) + \mathcal{H}^n(\Gamma \backslash \Omega_t) + \varepsilon/2 \\ \Longrightarrow\ &\mathcal{H}^n(\Gamma \cap \Omega_t) \leq \mathcal{H}^n(\Gamma_t \backslash \Omega) + \varepsilon/2 \end{aligned}$$

By Lemma 4.1.1, we have

$$\begin{aligned} \mathcal{H}^n(\tilde{\Gamma}_t) \leq &\mathcal{H}^n(\Gamma_t \cap \Omega) + \mathcal{H}^n(\Gamma \cap \Omega_t) + \frac{\varepsilon}{2} \\ \leq &\mathcal{H}^n(\Gamma_t \cap \Omega) + \mathcal{H}^n(\Gamma_t \backslash \Omega) + \varepsilon \\ \leq &\mathcal{H}^n(\Gamma_t) + \varepsilon \leq A + \varepsilon \end{aligned}$$

If $\Omega_0 \subset \Omega$, then by choosing sufficiently small $\varepsilon > 0$ and applying Lemma 4.1.1(4), we have

$$\tilde{\Gamma}_0 = f^{-1}(0) = \Gamma_0.$$

The proof of the second half is similar. The desired family will be a regularization of $\{\partial(\Omega_t \cup \Omega)\}$.

- **Case II(a) $\Omega_1 \subset \Omega$:**
  We define the desired nested family $\{\tilde{\Gamma}_t\}$ in a small tubular neighbourhood of Γ.

- **Case II(b) $\Omega_1 \backslash \Omega \neq \varnothing$:** We define $\tilde{f}(x) = -f(x)$. We apply Lemma 4.1.2 to $N = M \backslash \Omega$ and $\Sigma = \Gamma$ and the restriction $\tilde{f} : M \backslash \Omega \to (-\infty, 0]$. It follows that given $\varepsilon/2$, there exists a Morse function $\tilde{g} : M \backslash \Omega \to (-\infty, 0]$, such that
  1. $\tilde{g}^{-1}(0) = \Gamma$;
  2. $\tilde{f}^{-1}((-\infty, t)) \subset N_{\varepsilon/4}(\tilde{g}^{-1}((-\infty, t))) \subset N_{\varepsilon/2}(\tilde{f}^{-1}((-\infty, t)))$;
  3. $\mathbf{M}(\langle [\![M \backslash \Omega]\!], \tilde{g}, t \rangle) \leq \mathbf{M}(\partial_I \tilde{f}^{-1}((-\infty, t])) + \frac{\varepsilon}{2}$;
  4. If $\mathbf{dist}(x, \tilde{f}(\Sigma)) > \varepsilon$ then $\tilde{f}^{-1}(x) = \tilde{g}^{-1}(x)$.

We define $g(x) = -\tilde{g}(x)$ for $x \in M \backslash \Omega$ and extend it to a ∂-transverse Morse function from M to $[-1, \infty)$ in an arbitrary way. By property (2) of Lemma 4.1.2 we take that (possibly after a small perturbation)

$$\tilde{\Omega}_1 := g^{-1}([-1, 1]) \supset \Omega_1$$

For any smooth hypersurface Σ_t obtained by a small perturbation of $\partial(\Omega \cap \Omega_t)$. By (2)′, we have

$$\mathcal{H}^n(\Gamma) \leq \mathcal{H}^n(\Sigma_t) + \frac{\varepsilon}{4}$$

Furthermore, it follows that

$$\mathcal{H}^n(\Gamma) \leq \mathbf{M}(\partial_I(\Omega \cap \Omega_t)) + \frac{\varepsilon}{2}$$

Since $\partial(\Omega \cap \Omega_t) = (\Gamma_t \cap \Omega) \cup (\Gamma \cap \Omega_t)$ we have

$$\begin{aligned} &\mathcal{H}^n(\Gamma \cap \Omega_t) + \mathcal{H}^n(\Gamma \backslash \Omega_t) \\ &\leq \mathcal{H}^n(\Gamma_t \cap \Omega) + \mathcal{H}^n(\Gamma \cap \Omega_t) + \varepsilon/2 \\ \Longrightarrow\ &\mathcal{H}^n(\Gamma \backslash \Omega_t) \leq \mathcal{H}^n(\Gamma_t \cap \Omega) + \varepsilon/2 \end{aligned}$$

By Lemma 4.1.2, we have

$$\begin{aligned} \mathcal{H}^n(\tilde{\Gamma}_t) \leq &\mathcal{H}^n(\Gamma_t \backslash \Omega) + \mathcal{H}^n(\Gamma \backslash \Omega_t) + \frac{\varepsilon}{2} \\ \leq &\mathcal{H}^n(\Gamma_t \backslash \Omega) + \mathcal{H}^n(\Gamma_t \cap \Omega) + \varepsilon \\ \leq &\mathcal{H}^n(\Gamma_t) + \varepsilon \leq A + \varepsilon \end{aligned}$$

If $\Omega \subset \Omega_1$ then by property (4) of Lemma 4.1.2 we may assume that $\tilde{\Gamma}_1 = g^{-1}(1) = \Gamma_1$. □

5.2 THE SPLITTING LEMMA

The second lemma in this chapter deals with constructing a nested family between two nested open sets.

Theorem 5.2.1 (Falconer)**.** *There exists a constant $C(n)$ so that the following is true. Let $U \subset \mathbb{R}^{n+1}$ be an open set with smooth boundary. There exists a line $l \in \mathbb{R}^{n+1}$ so that the projection p_l onto l satisfies*

$$\mathbf{Vol}_n(U \cap p_l^{-1}(t)) < C(n)\mathbf{Vol}_{n+1}(U)^{\frac{n}{n+1}}$$

for all $t \in l$. Moreover, we can assume that p_l restricted to ∂U is a Morse function.

Lemma 5.2.2. *Let $\varepsilon > 0$, $L > 0$. Suppose $\Omega_0 \subset \Omega_1$ be bounded open sets with smooth boundary and $\Omega_1 \backslash \Omega_0 \subset B$ where B is $(1+L)$-bilipschitz diffeomorphic to a relative open subset U of $\mathbb{R}^{n+1}_+$. There exists a constant $C(n)$ and a nested family $\{\Gamma'_t\}$ with a family of corresponding open sets $\{\Omega'_t\}$, such that*

(1) the mass of the new sweepout can be estimated as

$$\begin{aligned}\mathcal{H}^n(\Gamma'_t) \leq & \mathbf{M}(\partial_I \Omega_0) + \mathbf{M}(\partial_I(\Omega_1 \backslash \Omega_0)) \\ & + C(n)(1+L)^n \mathcal{H}^{n+1}(\Omega_1 \backslash \Omega_0)^{\frac{n}{n+1}} + \varepsilon;\end{aligned}$$

(2) Ω'_0 is a relative inward ε-perturbation of Ω_0 and $\Omega'_1 = \Omega_1$;

(2′) Ω'_1 is a relative outward ε-perturbation of Ω_1 and $\Omega'_0 = \Omega_0$;

Proof. Let Ω' be a relative inward $\varepsilon/8$-perturbation of $\Omega = \Omega_1 \backslash \Omega_0$. By definition that means

(a) $\Omega \backslash N_{\frac{\varepsilon}{8}}(\partial\Omega) \subset \Omega' \subsetneq \Omega$;

(b) There exists a nested family of open sets $\{\Xi_t\}_{t \in [0,1]}$ and a smooth isotopy $\Sigma_t = \partial \Xi_t$, such that Σ_0 is a smoothing of $\partial\Omega$ and $\Xi_1 = \Omega'$ such that

$$\mathcal{H}^n(\Sigma_t) \leq \mathbf{M}(\partial_I \Omega) + \frac{\varepsilon}{8}$$

for all $t \in [0,1]$.

By Theorem 5.2.1, there exists a Morse function $f : \Omega' \to [0,1]$ such that

$$\mathcal{H}^n(f^{-1}(t)) \leq C(n)(1+L)^n \mathcal{H}^{n+1}(\Omega_1 \backslash \Omega_0)^{\frac{n}{n+1}} + \varepsilon/4.$$

We denote the nested sweepout of Ω' from g as $\{\Sigma_t^a\}$ with the corresponding family of open sets $\{\Xi_t^a\}$, such that $\Omega' = \Xi_1^a$. By Lemma 4.1.1, given $\varepsilon/4 > 0$, we can find $g : \Omega' \to [0,1]$ such that

$$\begin{aligned} \mathcal{H}^n(\Sigma_t^a) &\leq \mathbf{M}(\partial_I f^{-1}([0,t])) + \varepsilon/4 \\ &\leq \mathbf{M}(\partial_I(\Omega_1 \backslash \Omega_0)) + C(n)(1+L)^n \mathcal{H}^{n+1}(\Omega_1 \backslash \Omega_0)^{\frac{n}{n+1}} + \varepsilon/2. \end{aligned} \tag{5.1}$$

Let $\{\Sigma_t^b\}$ be a nested family with a corresponding family of open sets $\{\Xi_t^b\}$, such that $\{\Xi_0^b\}$ is a relative inward $\varepsilon/2$-perturbation of Ω_0, Ξ_1^b is a relative inward $\varepsilon/8$-perturbation of Ω_0. For all $t \in [0,1]$, the mass of the slices would satisfy

$$\mathcal{H}^n(\Sigma_t^b) \leq \mathbf{M}(\partial_I \Omega_0) + \varepsilon/2. \tag{5.2}$$

By Lemma 4.2.1, there exists a nested family $\{\Sigma_t^c\}$ with the corresponding family of open sets $\{\Xi_t^c\}$, such that $\Xi_1^c = \Omega_1$, $\Xi_0^c = \Xi^1 \sqcup \Xi^2$, where Ξ^1 is a relative inward $\varepsilon/8$-perturbation of Ω_0 and Ξ^2 is a relative inward $\varepsilon/8$-perturbation of $\Omega_1 \backslash \Omega_0$. It follows from the properties of the perturbations that, without any loss of generality, we may assume $\Xi^1 = \Xi_1^b$ and $\Xi^2 = \Omega'$. By (3) of Lemma 4.2.1, we have

$$\begin{aligned} \mathcal{H}^n(\Sigma_t^c) \leq& \mathbf{M}(\partial_I \Omega_1) + 2\,\mathcal{H}^n(\partial\Omega_0 \backslash \partial\Omega_1) + \varepsilon \\ \leq& \mathbf{M}(\partial_I(\Omega_1 \backslash \Omega_0)) + \mathbf{M}(\partial_I \Omega_0) + \varepsilon \end{aligned}$$

We define $\Gamma'_t = \Sigma^a_{2t} \cup \Sigma^b_{2t}$ for $t \in [0, 1/2)$ and $\Gamma'_t = \Sigma^c_{2t-1}$, $t \in [1/2, 1]$ with the open sets defined correspondingly. Then for $t \in [0, 1/2)$ combining 5.1 and 5.2, we obtain

$$\begin{aligned}\mathcal{H}^n(\Gamma'_t) \leq & \mathcal{H}^n(\Sigma^a_{2t}) + \mathcal{H}^n(\Sigma^b_{2t}) \\ \leq & \mathbf{M}(\partial_I \Omega_0) + \mathbf{M}(\partial_I(\Omega_1 \backslash \Omega_0)) \\ & + C(n)(1+L)^n \mathcal{H}^{n+1}(\Omega_1 \backslash \Omega_0)^{\frac{n}{n+1}} + \varepsilon\end{aligned}$$

□

6

NESTED SWEEPOUT

The construction of nested sweepouts is crucial for our work. As discussed in [CL20, §6], nested families allow us to glue local deformations while keeping precise control over the maximal mass of the sweepout. A family $\{\Gamma_t\}_{t\in[0,1]}$ is *nested* if it is given by the level sets of a ∂-transverse Morse function, so that the associated regions $\Omega_t = f^{-1}((-\infty, t))$ vary by elementary topological moves (handle attachments or deletions) as t increases.

The main result of this chapter, Proposition 6.0.1, shows that any sweepout of a bounded relative open set U can be approximated by a nested sweepout with arbitrarily small mass increase. The proof follows the strategy of [CL20, Lemma 6.2], adapted to the free-boundary setting, and proceeds in three main steps:

1. **Step 1 (Preliminary Modification):** Subdivide the original sweepout $\{\Gamma_t\}_{t\in[0,1]}$ into a finite number of subintervals $[t_i, t_{i+1}]$ such that on each subinterval, the corresponding regions Ω_t are either monotone increasing or monotone decreasing.

2. **Step 2 (Local Monotonization):** On each subinterval $[t_i, t_{i+1}]$, construct a nested family connecting slightly modified versions of Ω_{t_i} and $\Omega_{t_{i+1}}$. This uses the splitting and extension lemmas from Chapter 5 to build explicit Morse functions whose level sets have controlled mass.

3. **Step 3 (Gluing):** Glue the nested families from adjacent intervals into a single global nested sweepout. The construction ensures that

adjacent families satisfy the compatibility condition $\Omega_0^{i+1} \subset \Omega_1^i$, which allows for repeated application of a gluing proposition.

Proposition 6.0.1. *Let $U \subset M$ be a bounded relative open subset with smooth relative boundary. Then for any $\varepsilon > 0$, and any $\{\Gamma_t\} \in \mathcal{S}(U)$ with the corresponding open sets $\{\Omega_t\}$ with $\mathcal{F}(\{\Gamma_t\}) \leq A$, there exists a nested family $\{\tilde{\Gamma}_t\}$ with the corresponding family of open sets $\{\tilde{\Omega}_t\}$ such that $\tilde{\Omega}_0 \subset \Omega_0$, $\Omega_1 \subset \tilde{\Omega}_1$ with $\mathcal{F}(\{\tilde{\Gamma}_t\}) \leq A + \varepsilon$. In particular, this implies $W(U) = W_n(U)$.*

The proof of this proposition follows the strategy of [CL20, Lemma 6.2], with the following modifications for the free-boundary setting:

(i) *Boundary balls*: When a ball B_i in the covering meets ∂M, we work inside a fixed collar $\Phi : \partial M \times [0, \rho) \to M$ and regard B_i as bilipschitz to a half-ball in $\mathbb{R}^{n+1}_+$. All local deformations are generated by vector fields in $\mathfrak{X}_{\tan}(M)$, preserving the free-boundary condition.

(ii) *Relative cycles*: The sweepout takes values in $\mathbf{Z}_n(M, \partial M; \mathbb{Z}_2)$ (see Remark 2.1.1), and we use the relative coarea formula (Corollary 2.1.11) to control boundary contributions.

(iii) *∂-transverse Morse functions*: The Morse functions constructed in the preliminary modification are required to be ∂-transverse (Definition 2.2.6), ensuring that level sets meet ∂M transversally.

6.1 STEP 1: PRELIMINARY MODIFICATION

Fix $\{\Gamma_t\} \in \mathcal{S}(U)$ with the corresponding open sets $\{\Omega_t\}$ with $\mathcal{F}(\{\Gamma_t\}) \leq A$. We apply the preliminary modification as in [CL20, Lemma 6.2] to $\{\Gamma_t\}$.

Lemma 6.1.1 (Preliminary Modification)**.** *For any $\varepsilon > 0$ there exists a partition $0 = t_0 < \cdots < t_N = 1$ of $[0, 1]$ and a family $\{\Gamma'_t\}$ with the corresponding family of open sets $\{\Omega'_t\}$, such that the following holds:*

(1.1) $\Omega'_0 \subset \Omega_0$ *and* $\Omega_1 \subset \Omega'_1$*;*

(1.2) $\mathcal{F}(\{\Gamma'_t\}) \leq \mathcal{F}(\{\Gamma_t\}) + \varepsilon$*;*

(1.3) For each $i = 0, \dots, N-1$ *we have one of the two possibilities:*

(A) $\Omega'_{t_i} \subset \Omega'_{t_{i+1}}$ *and there exists a Morse function* $g_i : \mathrm{cl}(\Omega_{t_{i+1}} \backslash \Omega_{t_i}) \to [t_i, t_{i+1}]$*, such that* $\Gamma'_t = g_i^{-1}(t)$ *and* $\Omega'_t = \Omega'_{t_i} \cup g_i^{-1}((-\infty, t))$ *for* $t \in [t_i, t_{i+1}]$*.*

(B) $\Omega'_{t_{i+1}} \subset \Omega'_{t_i}$ *and there exists a Morse function* $g_i : \mathrm{cl}(\Omega_{t_i} \backslash \Omega_{t_{i+1}}) \to [t_i, t_{i+1}]$*, such that* $\Gamma'_t = g_i^{-1}(t)$ *and* $\Omega'_t = \Omega'_{t_i} \cup g_i^{-1}((-\infty, t])$ *for* $t \in [t_i, t_{i+1}]$*.*

Proof. Let M' be a compact subset of M that contains the closure of Ω_t for all $t \in [0, 1]$.

Set $L_{\text{ball}} := (1 + \frac{\varepsilon}{100W})^{1/n} - 1$ so that $(1 + L_{\text{ball}})^n \leq 1 + \frac{\varepsilon}{100W}$.

Choose r sufficiently small so that for every ball B of radius less than or equal to r in M' the following holds:

(i) B is $(1 + L_{\text{ball}})$-bilipschitz diffeomorphic to the Euclidean ball of the same radius (or to a half-ball in $\mathbb{R}^{n+1}_+$ if B meets ∂M);

(ii) $\mathcal{H}^n(B \cap \Gamma_t) < \frac{\varepsilon}{20}$ for $t \in [0, 1]$.

Condition (ii) can be realized because of the no concentration of mass property (sw4).

Boundary balls. If a ball B_i meets ∂M, we work inside a fixed collar $\Phi : \partial M \times [0, \rho) \to M$ and regard B_i as bilipschitz to a half-ball in $\mathbb{R}^{n+1}_+$. All local deformations in such B_i are generated by compactly supported vector fields in $\mathfrak{X}_{\tan}(M)$. This preserves the free-boundary condition and keeps the same bilipschitz constants as in (i).

Let $\{B_i\}$ be a collection of k balls of radius r covering M', such that balls of half the radius cover M'. We choose a partition $0 = s_0 < \cdots < s_{N'} = 1$, such that

(iii)

$$\begin{aligned}&\mathcal{H}^{n+1}(B_i \cap (\Omega_{s_j} \backslash \Omega_{s_{j+1}})) + \mathcal{H}^{n+1}(B_i \cap (\Omega_{s_{j+1}} \backslash \Omega_{s_j}))\\ &< \min\left\{\frac{r\varepsilon}{10k}, \left(\frac{\varepsilon}{10}\right)^{\frac{n+1}{n}}\right\};\end{aligned}$$

(iv) $\left|\mathcal{H}^n(B_i \cap \Gamma_{s_j}) - \mathcal{H}^n(B_i \cap \Gamma_{s_{j+1}})\right| \leq \frac{\varepsilon}{10k}$

for each $j = 0, \ldots, N'$ and $i = 1, \ldots, k$.

We define the new family $\{\Gamma'_t\}$ as follows. For $t = s_j$ we set $\Omega'_t = \Omega_t$ and $\Gamma'_t = \partial\Omega_t$ unless Γ_t is a finite collection of points in which case we set $\Gamma'_t = \Gamma_t$ and $\Omega'_t = \varnothing$.

Define a subdivision of $[s_j, s_{j+1}]$ into $2k$ subintervals, $s_j = s_j^0 < \cdots < s_j^{2k} = s_{j+1}$. Let $\{B'_i\}$ be a collection of k balls concentric with B_i of radius between $r/2$ and r and such that $\partial B'_i$ intersects Γ_{s_j} and $\Gamma_{s_{j+1}}$ transversally. Set

$$U_j^1 = \Omega_{s_j} \backslash \Omega_{s_{j+1}} \quad \text{and} \quad U_j^2 = \Omega_{s_{j+1}} \backslash \Omega_{s_j}.$$

By the slicing theorem (Corollary 2.1.11) and (iii), we may assume B'_i satisfies

$$\mathcal{H}^n(\partial B'_i \cap (U_j^1 \cup U_j^2)) \leq \frac{\varepsilon}{4k}. \tag{6.1}$$

Since the balls of radius $r/2$ already cover M', and each B'_i has radius in $[r/2, r]$ and is concentric with B_i, the family $\{B'_i\}_{i=1}^k$ still covers M'. Inductively we define

$$\begin{aligned}\Omega'_{s_j^0} &= \Omega'_{s_j}\\ \Omega'_{s_j^{2i-1}} &= \Omega'_{s_j^{2i-2}} \backslash (B'_i \cap U_j^1)\\ \Omega'_{s_j^{2i}} &= \Omega'_{s_j^{2i-1}} \cup (B'_i \cap U_j^2)\end{aligned} \tag{6.2}$$

for $i = 1, \ldots, k$.

Surfaces $\partial\Omega'_{s_j^l}$ may not be smooth, but there exists an arbitrarily small perturbation so that the boundaries are smooth. We perform these perturbations in the inward direction for $\Omega'_{s_j^{2i-1}}$ and in the outward direction for $\Omega'_{s_j^{2i}}$. To simplify notation we do not rename the sets after the perturbations; since the perturbations are arbitrarily small all the estimates for masses and volumes remain valid.

The following properties follow from the definition 6.2 and (i)-(ii):

(a) $\left|\mathbf{M}(\partial_I\Omega'_{s_j^l}) - \mathcal{H}^n(\Gamma_{s_j})\right| < \varepsilon/2$;

(b) $\Omega'_{s_j^{2i-1}} \subset \Omega'_{s_j^{2i}}$ and $\Omega'_{s_j 2i-1} \subset \Omega'_{s_j^{2i-2}}$.

We define $\Gamma'_{s_j^l} = \partial\Omega'_{s_j^l}$, unless $\Omega'_{s_j^l}$ is empty.

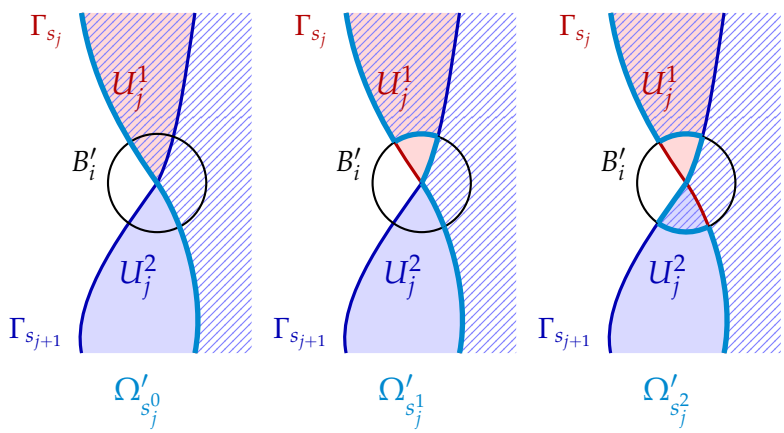

Figure 6.1: Construction of $\Omega'_{s_j^{2i-2}}$, $\Omega'_{s_j^{2i-1}}$, $\Omega'_{s_j^{2i}}$ for $i = 1$. The two curves Γ_{s_j} (red) and $\Gamma_{s_{j+1}}$ (blue) cross near B'_i, creating $U_j^1 = \Omega_{s_j} \setminus \Omega_{s_{j+1}}$ (red, hatched) and $U_j^2 = \Omega_{s_{j+1}} \setminus \Omega_{s_j}$ (blue). **Left**: $\Omega'_{s_j^0} = \Omega'_{s_j}$. **Middle**: $\Omega'_{s_j^1} = \Omega'_{s_j^0} \setminus (B'_i \cap U_j^1)$; the hatched part inside B'_i is removed. **Right**: $\Omega'_{s_j^2} = \Omega'_{s_j^1} \cup (B'_i \cap U_j^2)$; the blue part inside B'_i is added.

If $\Omega'_{s_j^l}$ is empty we set $\Gamma'_{s_j^l}$ to be a point inside $\Omega'_{s_j^{l-1}}$. By properties (iii) and (iv) surface $\Gamma'_{s_j^l}$ will satisfy the desired upper bound on the mass.

To complete our construction we need to show the existence of two types of nested families:

- a nested family that starts on $\Gamma'_{s_j^{2i-1}}$ and ends on $\Gamma'_{s_j^{2i-2}}$;
- a nested family that starts on $\Gamma'_{s_j^{2i-1}}$ and ends on $\Gamma'_{s_j^{2i}}$.

In both cases we want the homotopies to satisfy the desired upper bound on the masses.

Consider the set $\Omega'_{s_j^{2i-2}} \backslash \Omega'_{s_j^{2i-1}} = B'_i \cap U_j^1$. After smoothing the corner we call this set U. We map B'_i to $\mathbb{R}^{n+1}$ by a $(1 + L_{\text{ball}})$-bilipschitz diffeomorphism. Existence of the desired nested families follows by Lemma 5.2.2. The upper bound for the mass follows from the upper bound at the endpoints and properties (i) and (ii).

$$\begin{aligned}
\mathcal{H}^n(\Gamma'_t) \leq & \mathbf{M}(\partial_I \Omega'_{s_j^{2i-1}}) + \mathbf{M}(\partial_I(\Omega'_{s_j^{2i-2}} \backslash \Omega'_{s_j^{2i-1}})) \\
& + C(n)(1 + L_{\text{ball}})^n \mathcal{H}^{n+1}(\Omega'_{s_j^{2i-2}} \backslash \Omega'_{s_j^{2i-1}}) + \varepsilon \\
\leq & \mathcal{H}^n(\Gamma_{s_j}) + \varepsilon/2 + \varepsilon/10 + \varepsilon/4k \\
& + C(n)(1 + L_{\text{ball}})^n \min\left\{ \frac{r\varepsilon}{10k}, (\frac{\varepsilon}{10})^{\frac{n+1}{n}} \right\} \\
\leq & \mathcal{H}^n(\Gamma_{s_j}) + \varepsilon
\end{aligned}$$

□

6.2 STEP 2: LOCAL MONOTONIZATION

Assume that family $\{\Gamma_t\}$ satisfies conclusions of Lemma 6.1.1 for the subdivision $0 = t_0 < \cdots < t_N = 1$.

For every $\varepsilon > 0$ and each $i = 0, \ldots, N-1$, we will inductively define sets Ω_0^i and Ω_1^i such that the following holds:

(2.1) $\Omega_0^i \subset \Omega_1^i$;

(2.2) $\max\left\{\mathbf{M}(\partial_I \Omega_0^i), \mathbf{M}(\partial_I \Omega_1^i)\right\}$
$\leq \max\left\{\mathcal{H}^n(\Gamma_{t_i}), \mathcal{H}^n(\Gamma_{t_{i+1}})\right\} + \varepsilon$;

(2.3) $\Omega_{t_{i+1}} \subset \Omega_1^i$ and $\Omega_0^i \subset \Omega_{t_i}$;

(2.4) There exists a nested family of hypersurfaces $\left\{\Gamma_t^i\right\}$, $0 \leq t \leq 1$, with the corresponding family of nested open sets Ω_t^i, such that

$$\mathcal{H}^n(\Gamma_t^i) \leq \max\left\{\mathcal{H}^n(\Gamma_{t_i}), \mathcal{H}^n(\Gamma_{t_{i+1}})\right\} + \varepsilon.$$

Assume (2.1) - (2.4) holds for all Ω_0^j and Ω_1^j for $j < i$. To define Ω_0^i and Ω_1^i, we only need to consider the following two cases:

- **Case (A) of Lemma 6.1.1 (1.3):**
  In this case, $\Omega_{t_i} \subset \Omega_{t_{i+1}}$. We take $\Omega_0^i = \Omega_{t_i}$ and $\Omega_1^i = \Omega_{t_{i+1}}$ so that (2.1)-(2.3) would be automatically satisfied. (2.4) would follow from case (A) of Lemma 6.1.1 (1.3).

- **Case (B) of Lemma 6.1.1 (1.3):**
  In this case, $\Omega_{t_{i+1}} \subset \Omega_{t_i}$. We define $\Omega_1^i = \Omega_{t_{i+1}}$ and

  $$\Omega_0^i = \Omega_{t_{i+1}} \setminus \mathrm{cl}(N_\delta(\partial\Omega_{t_{i+1}}))$$

  for sufficiently small $\delta > 0$ so that $N_\delta(\partial\Omega_{t_{i+1}})$ is diffeomorphic to $\partial\Omega_{t_{i+1}} \times [-\delta, \delta]$ and hypersurfaces equidistant from $\partial\Omega_{t_{i+1}}$ in this neighbourhood and for each $s \in [-\delta, \delta]$, we have

  $$\mathcal{H}^n(\partial\Omega_{t_{i+1}} \times \{s\}) \leq \mathbf{M}(\partial_I \Omega_{t_{i+1}}) + \varepsilon/2.$$

  Thus, $\Omega_0^i \subset \Omega_{t_i}$ and we will set $\Omega_1^i = \Omega_{t_{i+1}}$.

Consider Ω_0^{i+1}. In case (A), it is defined as $\Omega_{t_{i+1}}$ thus

$$\Omega_0^{i+1} = \Omega_{t_{i+1}} = \Omega_1^i$$

In case (B), it is defined as

$$\Omega_0^{i+1} \subset \Omega_{t_{i+2}} \subset \Omega_{t_{i+1}} = \Omega_1^i.$$

Therefore, we can conclude that

(2.5) $\Omega_0^{i+1} \subset \Omega_1^i$.

Roughly speaking, the reason why (2.5) holds is because to construct Ω_0^{i+1} we push $\Omega_{t_{i+1}}$ inward (or not at all) and to construct Ω_1^i we push $\Omega_{t_{i+1}}$ outwards (or not at all).

6.3 STEP 3: GLUING NESTED FAMILIES

The final step is to glue together the nested families constructed on each interval $[t_i, t_{i+1}]$ into a single global nested sweepout of U. The key observation from Step 2 is that adjacent intervals satisfy the compatibility condition $\Omega_0^{i+1} \subset \Omega_1^i$, which allows us to apply the gluing proposition below repeatedly.

In this section, we prove the following:

Proposition 6.3.1. *Suppose $\{\Gamma_t^a\}$ and $\{\Gamma_t^b\}$ are two nested families (with corresponding families of open sets $\{\Omega_t^a\}$ and $\{\Omega_t^b\}$ respectively) and $\mathcal{H}^n(\Gamma_t^i) \leq W$. Suppose moreover that $\Omega_0^b \subset \Omega_1^a$. For any $\varepsilon > 0$ there exists a nested family $\{\Gamma_t\}$ and a corresponding family of open sets $\{\Omega_t\}$ such that*

$$\mathcal{H}^n(\Gamma_t) \leq W + \varepsilon,$$

$\Omega_1^b \subset \Omega_1$ *and* $\Omega_0 \subset \Omega_0^a$.

Proof. Let $\mathcal{S}$ denote the collection of all open sets Ω' such that $\Omega_0^b \subset \Omega' \subset \Omega_1^a$ and $\partial\Omega'$ is smooth. Let

$$A = \inf_{\Omega' \in \mathcal{S}} \mathbf{M}(\partial_I \Omega')$$

and choose $\Omega \in \mathcal{S}$ with

$$\mathbf{M}(\partial_I \Omega) < A + \frac{\varepsilon}{4}.$$

We set $\alpha = \partial\Omega$. We claim that Ω and α satisfy properties (1) and (2) from Lemma 5.1.1I for $\Omega_t = \Omega_t^a$. Indeed if Ω' satisfies $\Omega \subset \Omega' \subset \Omega_1$, we have

$$\begin{aligned}\mathcal{H}^n(\alpha) <& A + \frac{\varepsilon}{4} \\ \leq& \mathbf{M}(\partial_I \Omega') + \frac{\varepsilon}{4}\end{aligned}$$

Thus, we know that there exists a nested family $\{\tilde{\Gamma}_t^a\}$ with the corresponding family of open sets $\{\tilde{\Omega}_t^a\}$, such that $\tilde{\Omega}_0^a \subset \Omega_0^a$, $\tilde{\Gamma}_1^a = \alpha$ and

$$\mathcal{H}^n(\tilde{\Gamma}_t^a) \leq W + \varepsilon.$$

On the other hand, Ω and α also satisfy properties $(1)'$ and $(2)'$ from Lemma 5.1.1II for $\Omega_t = \Omega_t^b$. Indeed, for any open set Ω' with $\Omega_0^b \subset \Omega' \subset \Omega$ then again we have $\Omega_0^b \subset \Omega$ and

$$\mathcal{H}^n(\alpha) < \mathbf{M}(\partial_I \Omega') + \frac{\varepsilon}{4}$$

Therefore, there exists a nested family $\{\tilde{\Gamma}_t^b\}$ with the corresponding family of open sets $\{\tilde{\Omega}_t^b\}$, such that $\Omega_1^b \subset \tilde{\Omega}_1^b$, $\tilde{\Gamma}_0^b = \alpha$ and

$$\mathcal{H}^n(\tilde{\Gamma}_t^b) \leq W + \varepsilon.$$

We define the desired nested family Γ_t simply by concatenating these two nested families. □

Now we complete the proof of Proposition 6.0.1. We apply local monotonization 6.2 to define families $\{\Gamma_t^i\}$ for $i = 1, \ldots, N-1$.

By (2.5), we have

$$\Omega_0^2 \subset \Omega_1^1$$

Hence, we can apply Proposition 6.3.1 to the nested family $\{\Gamma_t^1\}$ and $\{\Gamma_t^2\}$. We obtain a new nested family $\left\{\Gamma_t^{1,2}\right\}$ with the corresponding family of open sets $\left\{\Omega_t^{1,2}\right\}$. Moreover, by (2.5) and Proposition 6.3.1, we have

$$\Omega_0^{1,2} \subset \Omega_0^1 \subset \Omega_0 \quad \text{and} \quad \Omega_1 \subset \Omega_1^2 \subset \Omega_1^{1,2}$$

Using (2.5) again, we have

$$\Omega_0^3 \subset \Omega_1^2 \subset \Omega_1^{1,2}$$

Hence, we can apply Proposition 6.3.1 to the nested family $\{\Gamma_t^3\}$ and $\left\{\Gamma_t^{1,2}\right\}$. We obtain a new nested family $\left\{\Gamma_t^{1,2,3}\right\}$ with the corresponding family of open sets $\left\{\Omega_t^{1,2,3}\right\}$.

We iterate this procedure inductively. At the i-th step, we can apply the Proposition 6.3.1 to families $\left\{\Gamma_t^{1,\dots,i}\right\}$ and $\left\{\Gamma_t^{i+1}\right\}$ to construct a new nested family $\left\{\Gamma_t^{1,\dots,i,i+1}\right\}$ with $\Omega_0^{1,\dots,i}$ and $\{\Omega_1\} \subset \Omega_1^{1,\dots,i}$, which would again guarantee that $\Omega_0^{i+2} \subset \Omega_1^{1,\dots,i}$, so that we can go to the next step.

After iteratively performing this operation N times we obtain the desired nested family. This finishes the proof of Proposition 6.0.1.

7

NO ESCAPE TO INFINITY

A key difficulty in min–max theory on *complete* manifolds (with or without boundary) is ensuring that the min–max sequence does not "escape to infinity"—that is, that slices with large area do not concentrate all their mass arbitrarily far from the good set U. As observed in [CL20], this is fundamentally a problem caused by the *completeness* of the manifold: in a compact manifold, such escape is topologically impossible, but in a complete non-compact setting, one must prove that the min–max construction forces slices to remain localized near U.

This chapter establishes a quantitative non-escape result: any good sweepout must have at least one slice carrying a definite amount of area inside U. The free-boundary setting introduces additional technical complications in the proof (particularly in handling boundary collars and relative isoperimetric estimates), but the fundamental difficulty remains the completeness of the ambient manifold.

The proof proceeds by contradiction. We assume such a sweepout exists and use the nested sweepout structure from Chapter 6 to construct a competing sweepout with strictly smaller width, contradicting the definition of $W_g(U)$.

7.1 STATEMENT OF THE MAIN RESULT

The main result of this chapter is the following proposition, which provides a quantitative lower bound on the area that must remain inside U.

Proposition 7.1.1. *For every good set U, there exists a positive constant $\varepsilon(U)$ which depends only on U such that the following holds: For every good sweepout $\{\Gamma_t\}$ of U with associated family of open sets $\{\Omega_t\}$, there exists $t' \in [0,1]$ such that*

$$\mathcal{H}^n(\Gamma_{t'}) \geq W_g(U),$$

and such that $\mathcal{H}^n(\Gamma_{t'} \cap cl(U)) \geq \varepsilon(U)$.

The proof is by contradiction. We assume that Proposition 7.1.1 does not hold, i.e. we can construct a good sweepout with volume of hypersurfaces strictly less than $W_g(U)$. The main tool in the proof is the structure of the nested sweepout from Proposition 6.0.1.

7.2 PUSHING SETS OUTSIDE GOOD SETS

The first technical ingredient is a lemma that allows us to deform a set with small intersection with U so that it avoids U entirely, while controlling the area increase.

Let U be a good set.

Lemma 7.2.1. *There exist $\varepsilon(U) > 0$, $\varepsilon_0(U) > 0$ and $\varepsilon_1(U) > 0$ such that for any open sets Ω' the following holds:*

1. $\max\{\varepsilon, \varepsilon_1\} < \mathbf{M}(\partial_I U)/10$;
2. *If $\varepsilon_0 < \mathcal{H}^{n+1}(\Omega' \cap U) < \mathcal{H}^{n+1}(U) - \varepsilon_0$, then $\mathbf{M}(\partial_I \Omega' \cap U) > 2\varepsilon$;*
3. *(A) If $\mathcal{H}^{n+1}(\Omega' \cap U) < 2\varepsilon_0$, then there exists a family of open sets $\{\Xi_t\}$ with*

$$\begin{aligned}
&\Xi_0 = \Omega';\\
&\Xi_t \backslash N_{\varepsilon_1}(U) = \Omega' \backslash N_{\varepsilon_1}(U);\\
&\Xi_1 \cap U = \varnothing\\
&\mathbf{M}(\partial_I \Xi_t) < \mathbf{M}(\partial_I \Omega') + \mathbf{M}(\partial_I U) + \varepsilon_1.
\end{aligned}$$

(B) *If* $\mathcal{H}^{n+1}(\Omega' \cap U) > \mathcal{H}^{n+1}(U) - 2\varepsilon_0$, *then there exists a family of open sets* $\{\Xi_t\}$ *with*

$$\begin{aligned}
&\Xi_0 = \Omega';\\
&\Xi_t \backslash N_{\varepsilon_1}(U) = \Omega' \backslash N_{\varepsilon_1}(U);\\
&\Xi_1 \cap U = U\\
&\mathbf{M}(\partial_I \Xi_t) < \mathbf{M}(\partial_I \Omega') + \mathbf{M}(\partial_I U) + \varepsilon_1.
\end{aligned}$$

Proof. Pick some $\varepsilon_1 \in (0, \mathcal{H}^n(\partial_I U)/10)$.

We will show that for all sufficiently small ε_0 (with the choice of ε_0 depending on ε_1) statement (3) holds;

We will show that for all sufficiently small ε (with the choice of ε depending on ε_0) statement (2) holds; the argument relies on the convexity properties of the isoperimetric profile of $\overline{U}$.

We will prove Statement(3)(A). And the statement (3)(B) would be similar.

Let $r_0 > 0$ be sufficiently small, so that every ball B of radius $r \in (0, r_0]$ centered at a point in $cl(U)$ is 2-biLipschitz diffeomorphic to a ball of the same radius in the Euclidean space.

Choose a covering $\{B_i\}$ of $cl(U)$ by N balls of radius r_0, so that concentric balls of radius $\frac{r_0}{2}$, denoted by $\frac{1}{2}B_i$, still cover $cl(U)$. Set

$$\varepsilon_0 := \min\left\{\frac{\varepsilon_1 r_0}{20N}, \left(\frac{\varepsilon_1}{2^{n+2}C(n)N}\right)^{\frac{n+1}{n}}\right\}, \tag{7.1}$$

where $C(n)$ is the constant from Lemma 4.2.1. Using the relative coarea formula (Corollary 2.1.11), we may choose a covering $\{B_i'\}$ of U by N balls of radius $r_i \in (r_0/2, r_0)$, so that

$$\mathcal{H}^n((\partial B_i') \cap (\Omega' \cap U)) \leq \frac{4\varepsilon_0}{r_0} \leq \frac{\varepsilon_1}{5N}. \tag{7.2}$$

Note that the second inequality follows from the first term in the definition of ε_0 in (7.1): since $\varepsilon_0 \leq \frac{\varepsilon_1 r_0}{20N}$, we have $\frac{4\varepsilon_0}{r_0} \leq \frac{4\varepsilon_1 r_0}{20Nr_0} = \frac{\varepsilon_1}{5N}$.

We are going to inductively push Ω' outside of $B_1' \cap U, B_2' \cap U, \ldots, B_N' \cap U$:

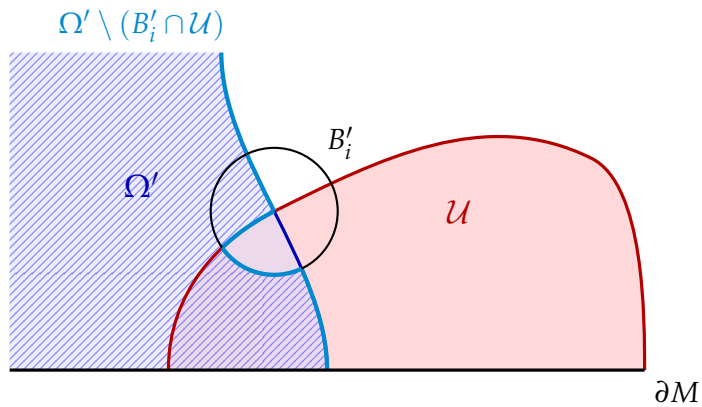

Figure 7.1: Pushing Ω' outside of $B_1' \cap U$. The region Ω' (blue) overlaps with $\mathcal{U}$ (red) near the ball B_i'. The set $\Omega' \setminus (B_i' \cap \mathcal{U})$ is obtained by removing the portion of Ω' inside $B_i' \cap \mathcal{U}$.

We can start with pushing Ω' outside of $B_1' \cap U$ as in Figure 7.1. By Lemma 4.2.1, for any $\varepsilon > 0$ sufficiently small (smaller than $\frac{\varepsilon_1}{20N}$ in fact), we can construct a nested family $\{\Xi_t^1\}$ that starts on Ω' and ends on a smoothing of $\Omega' \setminus (B_1' \cap U)$. We can choose the smoothing so that the set does not intersect $B_1' \cap U$. And in this step,

$$\begin{aligned}
\mathbf{M}(\partial_I \Xi_t^1) \leq & \mathbf{M}(\partial_I \Omega') + \mathbf{M}(\partial_I (B_1' \cap U \cap \Omega')) \\
& + 2^n C(n) \mathcal{H}^{n+1}(B_1' \cap U \cap \Omega') + \varepsilon \\
\leq & \mathbf{M}(\partial_I \Omega') + \mathcal{H}^n(B_1' \cap \partial_I U) + \mathcal{H}^n((\partial B_1') \cap (\Omega' \cap U)) \\
& + 2^n C(n) \varepsilon_0^{\frac{n}{n+1}} + \varepsilon \quad \text{By 7.1} \\
\leq & \mathbf{M}(\partial_I \Omega') + \mathcal{H}^n(B_1' \cap \partial_I U) + \frac{\varepsilon_1}{5N} + \frac{\varepsilon_1}{4N} + \varepsilon \quad \text{by 7.2} \\
\leq & \mathbf{M}(\partial_I \Omega') + \mathcal{H}^n(B_1' \cap \partial_I U) + \frac{\varepsilon_1}{2N}
\end{aligned}$$

We can iterate this procedure for each ball B_i'. During the i-th step, $1 \leq i \leq N$, we have that

$$\mathbf{M}(\partial_I \Xi_t^i) \leq \mathbf{M}(\partial_I \Omega') + \mathcal{H}^n(\partial_I U) + \frac{i\varepsilon_1}{2N}$$

That means we can conclude (3)(A) after gluing these together via Proposition 6.3.1. □

7.3 PROOF OF THE MAIN RESULT BY CONTRADICTION

We are now ready to prove Proposition 7.1.1 by contradiction. Suppose there exists a good sweepout $\{\Gamma_t\}_{t\in[0,1]}$ such that whenever

$$\mathcal{H}^n(\Gamma_t) \geq W_g(U)$$

we have $\mathcal{H}^n(\Gamma_t \cap U) < \varepsilon(U)$. Our goal is to construct a good sweepout entirely inside U whose maximal slice area is *strictly less* than $W_g(U)$, contradicting the definition of $W_g(U)$ as the infimum of maximal areas over all good sweepouts of U.

Continuity Regularization

Let $f(t) = \mathcal{H}^n(\Gamma_t \cap U)$. Note that $f(t)$ may not be continuous. However, it is easy to see that one can perturb the family $\{\Gamma_t\}$ so that it is δ-continuous in the following sense.

Definition 7.3.1. Function $f(t)$ is **δ-continuous** if the oscillation

$$\omega_f(t) := \lim_{a\to 0}[\sup_{s\in[t-a,t+a]} f(s) - \inf_{s\in[t-a,t+a]} f(s)]$$

satisfies $\omega_f(t) < \delta$ for every t.

Lemma 7.3.2 (δ-continuous perturbation)**.** *Let U be a bounded relative open subset to M with smooth relative boundary and $\{\Gamma_t\}$ be a good sweepout of U. For every $\delta > 0$ there exists a good sweepout $\{\Gamma'_t\}$ of U, such that*

- *$f(t) = \mathcal{H}^n(\Gamma'_t \cap U)$ is δ-continuous;*
- $\sup_t \mathcal{H}^n(\Gamma'_t) \leq \sup_t \mathcal{H}^n(\Gamma_t) + \delta$;
- $\sup_t \mathcal{H}^n(\Gamma'_t \cap U) \leq \sup_t \mathcal{H}^n(\Gamma_t \cap U) + \delta$.

Proof. Fix $\delta > 0$ and set $\varepsilon := \delta/6$. By Lemma 6.1.1 with parameter ε, from $\{\Gamma_t\}$ we obtain a piecewise nested family $\{\widetilde{\Gamma}_t\}$ and a partition $0 = t_0 < \cdots < t_N = 1$ such that

$$\sup_t \mathcal{H}^n(\widetilde{\Gamma}_t) \leq \sup_t \mathcal{H}^n(\Gamma_t) + \varepsilon,$$
$$\sup_t \mathcal{H}^n(\widetilde{\Gamma}_t \cap U) \leq \sup_t \mathcal{H}^n(\Gamma_t \cap U) + \varepsilon.$$

Indeed, the preliminary modification acts only inside finitely many thin annuli and endpoint collars; by (iii)–(iv) of Lemma 6.1.1 the M–area increase is $\leq \varepsilon$, and restricting to U with the relative coarea formula (Corollary 2.1.11) yields the same ε–bound for the U–area.

For each subinterval $[t_i, t_{i+1}]$ given by the preliminary modification, we may represent $\left\{\widetilde{\Gamma}_t\right\}_{t\in[t_i,t_{i+1}]}$ as level sets of an admissible Morse function. Let $\mathcal{S}$ denote the finite set consisting of the partition points $\{t_1, \ldots, t_{N-1}\}$ together with all critical values of these Morse functions. Choose $\eta > 0$ so small that the open intervals

$$J_s := (s - \eta, s + \eta) \subset (0, 1), \qquad s \in \mathcal{S}$$

are pairwise disjoint. We keep the family unchanged outside the buffers and set

$$\Gamma'_t := \widetilde{\Gamma}_t \qquad \text{for all } t \notin \bigcup_{s\in\mathcal{S}} J_s.$$

All subsequent modifications will be supported inside the union of the buffers $\bigcup_{s\in\mathcal{S}} J_s$.

Fix $s \in \mathcal{S}$ and set

$$\Omega^- := \widetilde{\Omega}_{s-\eta} \subset \Omega^+ := \widetilde{\Omega}_{s+\eta}, \qquad D_s := \Omega^+ \setminus \Omega^-.$$

Shrinking $\eta > 0$ if necessary and using the relative coarea formula (Corollary 2.1.11) and the small-volume choice from Lemma 6.1.1, we may assume

$$\mathcal{H}^n(\partial D_s \cap U) + (\mathcal{H}^{n+1}(D_s))^{\frac{n}{n+1}} < \varepsilon_s,$$

with $\sum_{s\in\mathcal{S}} \varepsilon_s \leq \varepsilon$. Cover $\overline{D_s}$ by the bilipschitz balls used in the preliminary modification (interior balls mapped to balls in $\mathbb{R}^{n+1}$; boundary balls, via a fixed collar, to half-balls in $\mathbb{R}^{n+1}_+$, with vector fields tangent to ∂M).

Applying Lemma 5.2.2 and gluing, we obtain a nested family $\{\Gamma'_t\}_{t\in J_s}$ connecting $\partial\Omega^-$ to $\partial\Omega^+$ such that, for all $t \in J_s$,

$$\begin{aligned}
\mathcal{H}^n(\Gamma'_t) \leq& \mathbf{M}(\partial_I \Omega^-) + \mathbf{M}(\partial_I D_s) \\
&+ C(n)(1 + L_{\text{ball}})^n \big(\mathcal{H}^{n+1}(D_s)\big)^{\frac{n}{n+1}} + \varepsilon_s, \\
\mathcal{H}^n(\Gamma'_t \cap U) \leq& \mathcal{H}^n(\partial\Omega^- \cap U) + \mathcal{H}^n(\partial D_s \cap U) \\
&+ C(n)(1 + L_{\text{ball}})^n \big(\mathcal{H}^{n+1}(D_s)\big)^{\frac{n}{n+1}} + \varepsilon_s,
\end{aligned}$$

where $L_{\text{ball}} := (1 + \frac{\varepsilon}{100W})^{1/n} - 1$ so that $(1 + L_{\text{ball}})^n \leq 1 + \frac{\varepsilon}{100W}$.

By shrinking $\eta > 0$ if necessary, choose numbers $\{\varepsilon_s\}_{s\in\mathcal{S}}$ so that the estimates in the previous paragraph hold and

$$\sum_{s\in\mathcal{S}} \varepsilon_s \leq \varepsilon.$$

Define the final family by

$$\Gamma'_t := \begin{cases} \widetilde{\Gamma}_t, & t \notin \bigcup_{s\in\mathcal{S}} J_s, \\ \text{the bridged slice constructed on } J_s, & t \in J_s \text{ for some } s \in \mathcal{S}. \end{cases}$$

Using the sup–bounds from the preliminary modification and the bridge estimates on each J_s, we obtain

$$\sup_t \mathcal{H}^n(\Gamma'_t) \le \sup_t \mathcal{H}^n(\Gamma_t) + \varepsilon + \sum_{s\in\mathcal{S}} \varepsilon_s \le \sup_t \mathcal{H}^n(\Gamma_t) + 2\varepsilon,$$
$$\sup_t \mathcal{H}^n(\Gamma'_t \cap U) \le \sup_t \mathcal{H}^n(\Gamma_t \cap U) + \varepsilon + \sum_{s\in\mathcal{S}} \varepsilon_s \le \sup_t \mathcal{H}^n(\Gamma_t \cap U) + 2\varepsilon.$$

(Any endpoint adjustment inside a fixed collar of ∂U preserves the "good" endpoints and adds at most ε to the above sup–bounds.) □

Let $g : [0,1] \to [0, \mathcal{H}^{n+1}(U)]$ be defined by

$$g(t) := \mathcal{H}^{n+1}(U \cap \Omega_t),$$

which is continuous. By Lemma 7.2.1 (2), if I' is a connected component of $g^{-1}([\varepsilon_0,\ \mathcal{H}^{n+1}(U) - \varepsilon_0])$, then for every $t \in I'$ we have

$$f(t) := \mathcal{H}^n(\Gamma_t \cap U) \ge 2\varepsilon.$$

Applying Lemma 7.3.2 with $\delta < \varepsilon/10$ we obtain a good sweepout $\{\Gamma'_t\}$ such that $f'(t) := \mathcal{H}^n(\Gamma'_t \cap U)$ is δ-continuous. By continuity of g and the sweepout property, there exists a closed interval $I = [t_0, t_1]$ with $I' \subset I \subset [0,1]$ such that

$$g(t_0) \le \varepsilon_0, \qquad g(t_1) \ge \mathcal{H}^{n+1}(U) - \varepsilon_0, \tag{7.3}$$

and (possibly shrinking I slightly) we have for all $t \in I$

$$f'(t) \geq \frac{3}{2}\varepsilon.$$

Moreover, by the choice of endpoints and the δ–continuity (with $\delta < \varepsilon/10$) together with the contrapositive of Lemma 7.2.1 (2), we may assume

$$\varepsilon \leq f'(t_i) \leq 2\varepsilon, \qquad i = 0, 1.$$

For the upper bound, note that the contrapositive of Lemma 7.2.1 (2) says: if $f'(t_i) > 2\varepsilon$ then

$$g(t_i) \in \big(\varepsilon_0,\ \mathcal{H}^{n+1}(U) - \varepsilon_0\big),$$

which contradicts our choice of the endpoints (see (7.3) for the bounds on $g(t_i)$).

For the lower bound, since f' is δ-continuous with $\delta < \varepsilon/10$ and on the interior interval we have $f' \geq \frac{3}{2}\varepsilon$, choosing t_i sufficiently close to the boundary of I' yields

$$f'(t_i) \geq \tfrac{3}{2}\varepsilon - \delta > \varepsilon.$$

Remark 7.3.3 (Why enlarge I' to a closed interval I). Enlarging a connected component $I' \subset B$ to $I = [t_0, t_1]$ with $g(t_0) \leq \varepsilon_0$ and $g(t_1) \geq \mathcal{H}^{n+1}(U) - \varepsilon_0$ is essential: (i) it yields a thin error set P (hence the inward perturbation $\overline{U}$ with controlled relative boundary); (ii) by δ–continuity it provides endpoint anchors $\varepsilon \leq f(t_i) \leq 2\varepsilon$ needed for the cut–cap–insert step; (iii) on the compact interval I we obtain a strict gap $\sup_{t\in I} \mathcal{H}^n(\Gamma_t) < W_g(U)$, which survives the nested replacement and underwrites the final contradiction.

Relabelling f' back to f and Γ'_t back to Γ_t, define

$$A := \left\{t \in [0,1] :\ \mathcal{H}^n(\Gamma_t) \geq W_g(U)\right\},$$
$$B := \left\{t \in [0,1] :\ g(t) \in (\varepsilon_0,\ \mathcal{H}^{n+1}(U) - \varepsilon_0)\right\}.$$

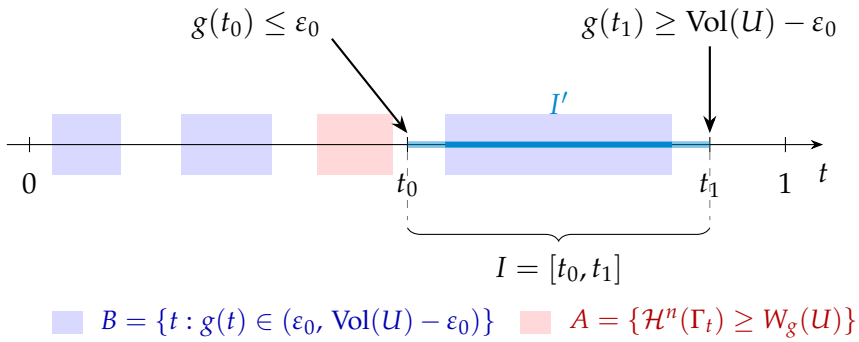

Figure 7.2: Time–axis schematic. The blue bands show $B = \{t : g(t) \in (\varepsilon_0, \mathrm{Vol}(U) - \varepsilon_0)\}$; the red band shows $A = \{t : \mathcal{H}^n(\Gamma_t \cap M) \ge W_g(U)\}$ with $A \cap B = \varnothing$. A connected component $I' \subset B$ is enlarged to $I = [t_0, t_1]$ with $g(t_0) \le \varepsilon_0$, $g(t_1) \ge \mathrm{Vol}(U) - \varepsilon_0$.

By Lemma 7.2.1 (2), $t \in B \Rightarrow f(t) \ge 2\varepsilon$, hence $A \cap B = \emptyset$. Since $I \subset B$, we have

$$\mathcal{H}^n(\Gamma_t) \;<\; W_g(U) \qquad \text{for all } t \in I.$$

By piecewise continuity of $t \mapsto \mathcal{H}^n(\Gamma_t)$ on the compact interval I, there exists

$$m := \sup_{t \in I} \mathcal{H}^n(\Gamma_t) < W_g(U), \qquad \delta_1 := W_g(U) - m > 0.$$

Apply the nested–replacement proposition from Section 6 (Proposition 6.0.1) to $\{\Gamma_t\}_{t \in I}$. We obtain a nested family $\{\overline{\Gamma}_t\}_{t \in [0,1]}$ with associated open sets $\{\overline{\Omega}_t\}$ such that

$$\sup_{t \in [0,1]} \mathcal{H}^n(\overline{\Gamma}_t) \;\le\; m - \tfrac{\delta_1}{2} \;<\; W_g(U) - \tfrac{\delta_1}{2},$$

and

$$\overline{\Omega}_0 \subset \Omega_{t_0}, \qquad \Omega_{t_1} \subset \overline{\Omega}_1.$$

Define the thin error set

$$P \;:=\; (\Omega_{t_0} \cap U) \;\cup\; \big(U \setminus \mathrm{cl}(\Omega_{t_1})\big).$$

By the endpoint volume bounds $g(t_0) \le \varepsilon_0$ and $g(t_1) \ge \mathcal{H}^{n+1}(U) - \varepsilon_0$, both $\Omega_{t_0} \cap U$ and $U \setminus \mathrm{cl}(\Omega_{t_1})$ have measure $\le \varepsilon_0$, hence $\mathcal{H}^{n+1}(P) \le 2\varepsilon_0$. Remove P from U and smooth inward inside a fixed collar of ∂U to obtain a slightly shrunken open set $\overline{U} \subset U$ (an relative inward δ-perturbation of $U \setminus \mathrm{cl}(P)$). Then, as in [CL20, p. 26], we may assume:

(i) $\{\overline{\Gamma}_t\}$ *restricts to a nested sweepout of* $\overline{U}$. Since $\overline{\Omega}_0 \subset \Omega_{t_0}$ and $\Omega_{t_1} \subset \overline{\Omega}_1$, removing the very thin set P does not break nestedness; reparametrising the family and performing corner-smoothing inside the collar of ∂U yields a nested sweepout of $\overline{U}$.

(ii) *Relative boundary control:*

$$\mathbf{M}(\partial_I \overline{U}) \;\le\; \mathbf{M}(\partial_I U) \;+\; C\,\varepsilon_0^{\frac{n}{n+1}} \;\le\; 2\,\mathbf{M}(\partial_I U)$$

for $\varepsilon_0 = \varepsilon_0(U)$ chosen sufficiently small. Here the extra term comes from the new boundary created by excising P and the inward smoothing; it is controlled by the relative isoperimetric/coarea inequalities applied to $\mathcal{H}^{n+1}(P) \le 2\varepsilon_0$.

(iii) *Push-out with controlled area:* there exists a homotopy supported in the collar of ∂U, generated by vector fields tangent to ∂M, that pushes $\partial_I \overline{U}$ outside U through slices of area at most $3\mathbf{M}(\partial_I U)$, after absorbing the $O(\varepsilon_0^{\frac{n}{n+1}})$ error into the constant (again by choosing $\varepsilon_0(U)$ small). This is the relative version of the short "push across the collar" used in [CL20].

Controlling Slices Outside the Good Set

Before proceeding, we isolate a slice of the nested family whose portion *outside* the shrunken set $\overline{U}$ is uniformly controlled. This is the key bridge between the global width bound and a sweepout confined to $\overline{U}$: once such a slice is found, we can (i) cut the family at that time, (ii) cap the outside part by using only boundary data of U, and (iii) build a sweepout of $\overline{U}$ whose maximal area is strictly below $W_g(U)$. The next lemma provides precisely this slice-level estimate and makes the subsequent "cut–and–insert" step possible.

Lemma 7.3.4 (relative version of Lemma 7.4 in [CL20])**.** *There exists* $t_0 \in [0,1]$ *such that*

$$\mathcal{H}^n(\overline{\Gamma}_{t_0} \setminus \overline{U}) \leq 2\mathbf{M}(\partial_I U).$$

Proof. Let $L := \sup_{t\in[0,1]} \mathcal{H}^n(\overline{\Gamma}_t \cap \overline{U})$. By the Morse–foliation construction on $\overline{U}$ and the relative boundary control in (ii), we can sweep out $\overline{U}$ by slices of area at most

$$L + \mathbf{M}(\partial_I \overline{U}) + \delta \leq L + 2\,\mathbf{M}(\partial_I U) + \delta.$$

Using (iii), we extend this to a good sweepout of U (the homotopy is supported in a fixed collar of ∂U and generated by vector fields tangent to ∂M) whose maximal slice area is at most $L + 2\,\mathbf{M}(\partial_I U) + \delta$. Hence, by the definition of $W_g(U)$,

$$W_g(U) \leq L + 2\mathbf{M}(\partial_I U) + \delta.$$

If, for every $t \in [0,1]$, we had $\mathcal{H}^n(\overline{\Gamma}_t \setminus \overline{U}) \;>\; 2\,\mathbf{M}(\partial_I U)$, then

$$\begin{aligned}
\mathcal{H}^n(\overline{\Gamma}_t) &= \mathcal{H}^n(\overline{\Gamma}_t \cap \overline{U}) + \mathcal{H}^n((\overline{\Gamma}_t \cap M) \setminus \overline{U}) \\
&> L + 2\mathbf{M}(\partial_I U) \\
&\geq W_g(U) - \delta
\end{aligned}$$

contradicting $\sup_t \mathcal{H}^n(\overline{\Gamma}_t) \leq W_g(U) - \delta_1/2$ for $\delta > 0$ chosen smaller than $\delta_1/2$. Therefore such t_0 exists. $\square$

Let

$$\mathcal{U}_0 := \left\{\Omega : \overline{\Omega}_0 \subset \Omega \subset \overline{\Omega}_{t_0} \setminus \overline{U}\right\};$$
$$\mathcal{U}_1 := \left\{\Omega : \overline{\Omega}_{t_0} \cup \overline{U} \subset \Omega \subset \overline{\Omega}_1\right\}$$

and define

$$A_i := \inf\left\{\mathbf{M}(\partial_I \Omega) : \Omega \in \mathcal{U}_i\right\}, \qquad i = 0, 1.$$

By Lemma 7.3.4 and property (ii) for $\overline{U}$, we have

$$A_i \;\leq\; 3\,\mathbf{M}(\partial_I U).$$

Choose $\Sigma_i = \partial \Xi_i$ with $\Xi_i \in \mathcal{U}_i$ such that

$$\mathcal{H}^n(\Sigma_i) \;\leq\; A_i + \delta/4.$$

Applying the relative nested–insertion lemmas (Lemma 5.1.1(I),(II) in our Section 5) *inside* $\overline{U}$ and concatenating, we obtain a nested sweepout of $\overline{U}$ from Σ_0 to Σ_1 whose maximal slice area (measured in M) satisfies

$$\sup_t \; \mathcal{H}^n(\Gamma_t^{\overline{U}}) \;\leq\; W_g(U) - \delta/4.$$

Using Lemma 7.2.1 (3) (relative push across a ∂U–collar), we upgrade this to a good sweepout of U with

$$\sup_t \; \mathcal{H}^n(\Gamma_t^{U}) \;\leq\; W_g(U) - \delta/4,$$

contradicting the definition of $W_g(U)$. This completes the proof of Proposition 7.1.1.

8

CONVERGENCE OF A MIN–MAX SEQUENCE TO A MINIMAL HYPERSURFACE

This chapter completes the proof of the main theorem by showing that the min-max sequence constructed in the previous chapters converges to a smooth free-boundary minimal hypersurface. The argument combines three key ingredients: the pull-tight procedure adapted to the complete setting, the almost minimizing property in relative annuli, and the free-boundary regularity theory of Li–Zhou [LZ21].

Throughout this chapter we use vector fields in $\mathfrak{X}_{\tan}(M)$ (tangent to ∂M). Varifold convergence is understood in M, so "stationary" means stationary with free boundary.

Starting from a good sweepout of the good set constructed in Chapter 6, we perform a pull–tight deformation on an exhaustion of M, select slices that are almost minimizing in small relative annuli, and pass to the limit. The free boundary regularity theory of Li–Zhou [LZ21], together with the almost minimizing property, then yields a smooth almost properly embedded free boundary minimal hypersurface.

8.1 PULLED–TIGHT SEQUENCES AND MIN–MAX VARIFOLDS

Fix a good set $U \subset M$. Proposition 6.0.1 produces good sweepouts with maximal area arbitrarily close to $W_g(U)$. To prevent loss of mass at infinity we perform the pull–tight deformation on a bounded exhaustion of M.

Let $U_0 \Subset U_1 \Subset \cdots \Subset M$ be bounded relative open sets with $M = \bigcup_i U_i$ and $U \Subset U_0$. Since each U_i is precompact in M, the closure $\overline{U_i}$ is compact and therefore admits uniform bounds on Ricci curvature, the second fundamental form of $\partial M \cap \overline{U_i}$, and a positive lower bound on the injectivity radius, so the monotonicity formula and Li–Zhou's estimates apply on each U_i. For each i choose W_i with $U_i \Subset W_i \Subset U_{i+1}$. Set $C := 2W_g(U)$ and define

$$\mathcal{V}^C(\Omega) := \big\{V \in \mathcal{V}_n(M) : \operatorname{spt}\|V\| \subset \overline{\Omega},\ \|V\|(M) \le C\big\},$$

$$\mathcal{V}^C_\infty(\Omega) := \big\{V \in \mathcal{V}^C(\Omega) : V \text{ is stationary in } \Omega \text{ with free boundary}\big\}.$$

Both sets are compact under the varifold distance function $\mathbf{F}$ (see [LZ21, Lemma 3.4]).

For each $X \in \mathfrak{X}_{\tan}(M)$, let $\{f_t\}_{t\in[0,1]}$ be the flow generated by X. The induced map $H^X : [0,1] \times \mathcal{V}_n(M) \to \mathcal{V}_n(M)$ defined by $H^X(t,V) = (f_t)_\sharp V$ is continuous with respect to $\mathbf{F}$. Similarly, the pushforward on relative cycles $H^X(t,\tau) = [(f_t)_\sharp T]$ where $T \in \tau$ gives a continuous map $H^X : [0,1] \times \mathcal{Z}_n(M, \partial M; \mathbb{Z}_2) \to \mathcal{Z}_n(M, \partial M; \mathbb{Z}_2)$.

Lemma 8.1.1 (Extension of the pull–tight map)**.** *For every i there exist a continuous map $\Phi_{U_i,W_i} : \mathcal{V}^C(M) \to \mathcal{V}^C(M)$ and decreasing sequences $\{\tau_k(U_i)\}$, $\{\varepsilon_k(U_i)\}$ converging to 0 with the following properties for all $V \in \mathcal{V}^C(M)$:*

1. $\|\Phi_{U_i,W_i}(V)\|(M) \le \|V\|(M)$;
2. *if* $\|V\|(\overline{W}_i) \le 5\,\mathbf{M}(\partial_I U)$ *then* $\Phi_{U_i,W_i}(V) = V$;

3. *if* $\|V\|(\overline{W}_i) \geq 9\,\mathbf{M}(\partial_I U)$ *and*

$$\mathbf{F}_{U_i}\big(V \llcorner Gr_n(U_i), \mathcal{V}^C_\infty(U_i)\big) \in \big[2^{-(k+1)}, 2^{-k}\big],$$

then $\|\Phi_{U_i,W_i}(V)\|(M) \leq \|V\|(M) - \varepsilon_k(U_i)$ *and* $\mathbf{F}\big(\Phi_{U_i,W_i}(V), V\big) \leq \tau_k(U_i)$.

Proof. The argument follows [LZ21, Proposition 4.17], adapted to each bounded set U_i in place of the whole manifold. The key point is that, although M is non-compact and $\mathcal{V}^C(M)$ need not be compact, each U_i is bounded, so $\mathcal{V}^C(U_i)$ and $\mathcal{V}^C_\infty(U_i)$ *are* compact under $\mathbf{F}$. The tightening construction of Li–Zhou therefore applies on U_i: one defines a continuous map $\Psi : \mathcal{V}^C(U_i) \to \mathfrak{X}_{\tan}(M)$ such that $\delta V(\Psi(V)) = 0$ if $V \in \mathcal{V}^C_\infty(U_i)$ and $\delta V(\Psi(V)) < 0$ otherwise, together with a continuous function $h : \mathcal{V}^C(U_i) \to [0,1]$ satisfying $h(V) = 0$ iff $V \in \mathcal{V}^C_\infty(U_i)$.

To ensure that the resulting map Φ_{U_i,W_i} only modifies varifolds inside W_i, we truncate the vector fields: choose a smooth cutoff $\rho_i \in C^\infty_c(W_i)$ with $\rho_i \equiv 1$ on $\overline{U}_i$ and replace each $\Psi(V)$ by $\rho_i\,\Psi(V) \in \mathfrak{X}_{\tan}(M)$. The flow generated by $\rho_i\,\Psi(V)$ is the identity outside W_i, so $\Phi_{U_i,W_i}(V)$ agrees with V on $M \setminus W_i$. The mass-decreasing property (3) is preserved because the flow still acts as the full tightening flow on U_i, while property (2) ensures that Φ_{U_i,W_i} leaves invariant any varifold with small mass in $\overline{W}_i$. □

Remark 8.1.2 (Pull-tight preserves the good sweepout structure). An important feature of the pull-tight construction is that it does not move mass from U to $\partial_I U$. This follows from two observations:

(i) By property (2), if a slice has mass $\leq 5\,\mathbf{M}(\partial_I U)$ in $\overline{W}_i$, then Φ_{U_i,W_i} acts as the identity on that slice. Since endpoints of a good sweepout have mass $\leq 5\,\mathbf{M}(\partial_I U)$ by Definition 2.2.5, the pull-tight procedure leaves the endpoints unchanged.

(ii) The flow is generated by vector fields in $\mathfrak{X}_{\tan}(M)$ supported in W_i. Since $U \Subset U_0 \Subset W_0$, the cutoff ρ_0 vanishes near $\partial_I U$, so the flow does not transport mass across $\partial_I U$.

Therefore the pulled-tight sequence $\{\mathbf{P}_i(\Gamma_t^i)\}$ remains a sequence of good sweepouts, and the non-escape property (Proposition 7.1.1) applies to the limit.

Let $\{\Gamma_t^i\}$ be a sequence of good sweepouts with $\sup_t \mathbf{M}(\Gamma_t^i) \downarrow W_g(U)$. Set

$$\mathbf{P}_i := \Phi_{U_0,W_0} \circ \Phi_{U_1,W_1} \circ \cdots \circ \Phi_{U_i,W_i}.$$

Proposition 8.1.3 (Pulled–tight minimizing sequence)**.** *After passing to a subsequence, $\{\mathbf{P}_i(\Gamma_t^i)\}$ is a minimizing sequence of good sweepouts with*

$$\lim_{i\to\infty} \sup_{t\in[0,1]} \mathbf{M}\big(\mathbf{P}_i(\Gamma_t^i)\big) = W_g(U),$$

and any min–max sequence $\mathbf{P}_i(\Gamma_{t_i}^i)$ admits a subsequence converging to a varifold in $\mathcal{V}_\infty^C(M)$, i.e. stationary in M with free boundary.

Proof. Property (3) in Lemma 8.1.1 guarantees that if the restriction of a slice to some U_j stays a definite $\mathbf{F}_{U_j}$–distance from $\mathcal{V}_\infty^C(U_j)$, then the composed map decreases its mass by at least $\varepsilon_k(U_j)$.

We now describe the diagonal argument. For each fixed j, the sequence $\{\mathbf{P}_i(\Gamma_{t_i}^i)\}_{i\geq j}$ has uniformly bounded mass (by $C = 2W_g(U)$) and is supported in $\overline{U}_j$, so by compactness of $\mathcal{V}^C(U_j)$ under $\mathbf{F}$, we can extract a subsequence converging in $\mathbf{F}_{U_j}$ to some $V_j \in \mathcal{V}^C(U_j)$. Moreover, property (3) ensures that $V_j \in \mathcal{V}_\infty^C(U_j)$, i.e., V_j is stationary in U_j with free boundary.

Applying a standard diagonal argument: first extract a subsequence $\{i_k^{(1)}\}$ converging in $\mathbf{F}_{U_1}$, then a further subsequence $\{i_k^{(2)}\}$ converging in $\mathbf{F}_{U_2}$, and so on. The diagonal sequence $\{i_k^{(k)}\}$ converges in $\mathbf{F}_{U_j}$ for every j. Since $M = \bigcup_j U_j$ and the exhaustion is nested, the limit varifold V is well-defined on all of M and satisfies $\delta V(X) = 0$ for every $X \in \mathfrak{X}_{\tan}(M)$ with $\mathrm{spt}(X) \subset U_j$ for some j. Since every compactly supported tangential

vector field is supported in some U_j, we conclude that V is stationary in M with free boundary. $\square$

8.2 almost minimizing varifolds with free boundary

We recall the notion of almost minimizing varifolds with free boundary, following Li–Zhou [LZ21, Section 4].

Definition 8.2.1 (Almost minimizing with free boundary [LZ21, Definition 4.18])**.** Let $U \subset M$ be relatively open. For $\varepsilon, \delta > 0$, define $\mathscr{A}_n(U; \varepsilon, \delta; \mathbf{F})$ to be the set of all $\tau \in \mathcal{Z}_n(M, \partial M; \mathbb{Z}_2)$ such that whenever $\tau = \tau_0, \tau_1, \ldots, \tau_m \in \mathcal{Z}_n(M, \partial M; \mathbb{Z}_2)$ is a discrete sequence with

- $\operatorname{spt}(\tau_i - \tau) \subset U$,
- $\mathbf{F}(\tau_{i+1} - \tau_i) \leq \delta$,
- $\mathbf{M}(\tau_i) \leq \mathbf{M}(\tau) + \delta$ for $i = 1, \ldots, m$,

then $\mathbf{M}(\tau_m) \geq \mathbf{M}(\tau) - \varepsilon$.

A varifold $V \in \mathcal{V}_n(M)$ is *almost minimizing in U with free boundary* if there exist sequences $\varepsilon_i \to 0$, $\delta_i \to 0$, and $\tau_i \in \mathscr{A}_n(U; \varepsilon_i, \delta_i; \mathbf{F})$ with canonical representatives $T_i \in \tau_i$ such that $\mathbf{F}(|T_i|, V) \leq \varepsilon_i$.

Definition 8.2.2 (Almost minimizing in small annuli [LZ21, Definition 5.1])**.** For $p \in M$ and $0 < s < r$, set

$$\mathcal{A}_{s,r}(p) := \begin{cases} B_r(p) \setminus \overline{B}_s(p) & \text{if } p \in \operatorname{int}(M), \\ B_r^+(p) \setminus \overline{B}_s^+(p) & \text{if } p \in \partial M, \end{cases}$$

where $B_r^+(p) := B_r(p) \cap M$. A varifold $V \in \mathcal{V}_n(M)$ is *almost minimizing in small annuli with free boundary* if for each $p \in M$ there exists $r_{\mathrm{am}}(p) > 0$ such that V is almost minimizing in $\mathcal{A}_{s,r}(p)$ with free boundary for all $0 < s < r \leq r_{\mathrm{am}}(p)$.

Proposition 8.2.3 (Stationarity [LZ21, Proposition 4.21]). *If $V \in \mathcal{V}_n(M)$ is almost minimizing in a relatively open set $U \subset M$ with free boundary, then V is stationary in U with free boundary.*

The main regularity result is the following:

Theorem 8.2.4 (Regularity of almost minimizing varifolds [LZ21, Theorem 6.2]). *Let $2 \leq n \leq 6$. Suppose $V \in \mathcal{V}_n(M)$ is stationary in M with free boundary and almost minimizing in small annuli with free boundary. Then there exist $N \in \mathbb{N}$, positive integers $m_1, \ldots, m_N$, and smooth, connected, almost properly embedded free boundary minimal hypersurfaces $(\Sigma_j, \partial\Sigma_j) \subset (M, \partial M)$ such that*

$$V = \sum_{j=1}^{N} m_j \, |\Sigma_j|.$$

Remark 8.2.5 (Dimension restriction). The restriction $2 \leq n \leq 6$ arises from the regularity theory. For $n \leq 6$, the singular set of an almost minimizing varifold with free boundary is empty by [LZ21, Theorem 6.2], which relies on Wickramasekera's regularity theory [Wic14]. For $n = 7$, the singular set is discrete, and for $n \geq 8$, the singular set has Hausdorff dimension at most $n - 7$. In those cases, the main theorem still holds with the conclusion that Σ is smooth and embedded *away from a singular set of codimension at least* 7.

8.3 LIMIT VARIFOLDS AND PROOF OF THE MAIN THEOREM

Almost minimizing selection in the non-compact setting

Let $\{\Gamma_i\}$ be the pulled-tight min–max sequence produced in Section 8.1. By Proposition 8.1.3, we may extract a subsequence (not relabelled) such that the induced varifolds converge to some $V \in \mathcal{V}_\infty^C(M)$.

In the compact setting, Li–Zhou [LZ21, Theorem 5.3] obtain the almost minimizing property from the combinatorial selection argument of

Almgren–Pitts [Pit81]. In our non-compact setting, the key observation is that the combinatorial argument is *local*: for each $p \in M$, one only needs to deform sweepouts inside a bounded neighbourhood of p. Since each element of the exhaustion U_j is bounded, the pull–tight map $\mathbf{P}_i$ restricts to a finite composition of maps Φ_{U_k,W_k} on any fixed bounded region, and the combinatorial argument of [LZ21, Theorem 5.3] applies on each U_j without modification. A diagonal argument over j then yields:

Proposition 8.3.1 (Almost minimizing selection)**.** *After passing to a further subsequence, V is almost minimizing in small annuli with free boundary in the sense of Definition 8.2.2.*

Regularity of the limit varifold

Proposition 8.3.2. *There exist finitely many connected, smooth, almost properly embedded free boundary minimal hypersurfaces $(\Sigma_j, \partial\Sigma_j) \subset (M, \partial M)$ and positive integers m_j such that*

$$V = \sum_{j=1}^{N} m_j |\Sigma_j|.$$

Proof. By Proposition 8.3.1, V is almost minimizing in small annuli with free boundary, hence stationary in M with free boundary by Proposition 8.2.3. The regularity theory of Li–Zhou (Theorem 8.2.4) then applies. We verify that the non-compactness of M causes no difficulty: the proof of Theorem 8.2.4 in [LZ21, Section 6] consists of three parts—the good replacement property, the classification of tangent cones, and the boundary regularity—all of which are *local* arguments carried out in a fixed ball $B_r(p) \cap M$. In particular:

1. The replacement construction ([LZ21, Proposition 6.3]) only modifies V inside a compact annular region $\overline{\mathcal{A}}_{s,r}(p) \cap M$, where area minimization in a bounded domain is well-posed.

2. The tangent cone analysis ([LZ21, Proposition 6.11]) uses the monotonicity formula for stationary varifolds with free boundary ([LZ21, Theorem 2.3]) and Allard-type regularity, both of which are local.

3. The boundary regularity ([LZ21, Theorem 6.2]) proceeds by blow-up at a point $p \in \operatorname{spt}\|V\| \cap \partial M$, which only sees the local geometry near p.

Therefore the conclusion of Theorem 8.2.4 holds for V in the complete, non-compact manifold M. □

The main theorem

We can now state and prove the main result of this thesis.

Theorem 8.3.3 (Main theorem)**.** *Let (M^{n+1}, g) be a complete Riemannian manifold with non-empty smooth boundary ∂M, $2 \leq n \leq 6$. Let $U \subset M$ be a* good set *(Definition 2.3.1), i.e., a bounded open set of finite perimeter satisfying*

$$\mathbf{M}(\partial_I U) \leq \frac{1}{10} W_\partial(U),$$

with $\overline{U} \cap \partial M \neq \varnothing$. Then there exists a smooth, almost properly embedded free boundary minimal hypersurface $(\Sigma, \partial\Sigma) \subset (M, \partial M)$ with $\partial\Sigma \subset \partial M$, Σ meeting ∂M orthogonally, and

$$\mathcal{H}^n(\Sigma) \leq W_\partial(U) + \mathbf{M}(\partial_I U).$$

Moreover, $\Sigma \cap \overline{U} \neq \varnothing$.

Remark 8.3.4. The hypersurface Σ may extend beyond $\overline{U}$, but the good set condition ensures $\Sigma \cap \overline{U} \neq \varnothing$ with definite area. More precisely, by Proposition 7.1.1, we have $\mathcal{H}^n(\Sigma \cap \overline{U}) \geq \varepsilon(U) > 0$ for an explicit constant $\varepsilon(U)$ depending on U.

Proof. The proof combines all preceding chapters.

Step 1: Construction of the min–max sequence. Since U is a good set, we have $W_g(U) > 0$. Indeed, the good set condition $\mathbf{M}(\partial_I U) \leq \frac{1}{10} W_\partial(U)$ implies $W_\partial(U) \geq 10\,\mathbf{M}(\partial_I U) > 0$ (since U is non-empty and has non-trivial interior boundary). By Proposition 2.2.9, $W_g(U) \geq W_\partial(U) - \mathbf{M}(\partial_I U) \geq \frac{9}{10} W_\partial(U) > 0$.

By Proposition 6.0.1, there exists a sequence of good sweepouts $\{\Gamma_t^i\}$ of U with $\sup_t \mathbf{M}(\Gamma_t^i) \downarrow W_g(U)$. Applying the pull–tight procedure of Section 8.1, we obtain a minimizing sequence $\{\mathbf{P}_i(\Gamma_t^i)\}$ whose min–max varifold limit V is stationary in M with free boundary (Proposition 8.1.3).

Step 2: Almost minimizing and regularity. By Proposition 8.3.1, V is almost minimizing in small annuli with free boundary. Proposition 8.3.2 then gives

$$V = \sum_{j=1}^{N} m_j \, |\Sigma_j|,$$

where each $(\Sigma_j, \partial\Sigma_j) \subset (M, \partial M)$ is a smooth, connected, almost properly embedded free boundary minimal hypersurface meeting ∂M orthogonally.

Step 3: Area bound and non-escape. Set $\Sigma := \bigcup_{j=1}^{N} \Sigma_j$. The area bound follows from the min–max construction:

$$\mathcal{H}^n(\Sigma) \leq \|V\|(M) = W_g(U) \leq W_\partial(U) + \mathbf{M}(\partial_I U).$$

The non-escape estimate (Proposition 7.1.1) guarantees that for every $\delta > 0$,

$$\mathcal{H}^n(\Sigma \cap N_\delta(U)) \geq \frac{\varepsilon(U)}{2} > 0,$$

so in particular $\Sigma \cap \overline{U} \neq \varnothing$.

Step 4: Non-triviality of the free boundary. It remains to show $\partial\Sigma \neq \varnothing$. The good sweepouts of U are nested sweepouts in the sense of Definition 2.2.7: each slice $\Gamma_t^i = \partial_I \Omega_t^i$ is the interior boundary of an open set Ω_t^i

with $\Omega_0^i = \varnothing$ and $\Omega_1^i = U$. Since $\overline{U} \cap \partial M \neq \varnothing$, the continuous family $\{\Omega_t^i\}$ sweeps through the boundary region, and the relative homology class $[\partial_I U] \in H_n(M, \partial M; \mathbb{Z}_2)$ is non-trivial (as U has non-empty intersection with ∂M).

Suppose for contradiction that $\partial\Sigma = \varnothing$. Then each Σ_j is a closed minimal hypersurface in $\mathrm{int}(M)$, and V is supported away from ∂M. We derive a contradiction as follows. Since $\overline{U} \cap \partial M \neq \varnothing$ and the sweepout $\{\Omega_t^i\}$ is continuous in the L^1 topology with $\Omega_0^i = \varnothing$ and $\Omega_1^i = U$, by the intermediate value theorem there exists t_i^* with $\mathcal{H}^{n+1}(\Omega_{t_i^*}^i \cap U) \in \big(\varepsilon_0, \mathcal{H}^{n+1}(U) - \varepsilon_0\big)$. The isoperimetric profile of $\overline{U}$ (cf. Proposition 7.1.1) then gives $\mathcal{H}^n(\partial_I \Omega_{t_i^*}^i \cap U) \geq 2\varepsilon(U) > 0$. In particular, the slice $\Gamma_{t_i^*}^i = \partial_I \Omega_{t_i^*}^i$ has definite mass near $\partial M \cap \overline{U}$. Passing to the varifold limit, $\|V\|(\overline{U} \cap \partial M) > 0$ by lower semicontinuity, contradicting $\mathrm{spt}\,\|V\| \cap \partial M = \varnothing$. (This argument is the free-boundary analogue of [LZ21, Lemma 5.4], adapted to the non-compact setting with $\mathbb{Z}_2$ coefficients; the key input is the non-triviality of the mod 2 relative homology class $[\partial_I U] \in H_n(M, \partial M; \mathbb{Z}_2)$.)

Therefore $\partial\Sigma \neq \varnothing$, and at least one component Σ_j is a free boundary minimal hypersurface with $\partial\Sigma_j \subset \partial M$. □

As an immediate corollary, we obtain existence under a volume growth condition.

Corollary 8.3.5. *Let (M^{n+1}, g) be a complete Riemannian manifold with non-empty smooth boundary ∂M, $2 \leq n \leq 6$, and sublinear volume growth:*

$$\liminf_{R \to \infty} \frac{\mathcal{H}^{n+1}(B_R(x) \cap M)}{R} = 0 \qquad \textit{for some } x \in M.$$

Suppose in addition that ∂M is non-compact, or more generally that there exists a good set U with $\overline{U} \cap \partial M \neq \varnothing$. Then M contains a smooth, almost properly embedded free boundary minimal hypersurface.

Proof. By Lemma 2.3.3, the sublinear growth condition guarantees the existence of a good set $U \subset M$. If $\overline{U} \cap \partial M \neq \varnothing$, the result follows from Theorem 8.3.3. $\square$

9

ALTERNATIVE REGULARITY VIA NON-EXCESSIVE SWEEPOUTS

In this chapter, we combine nested sweepouts with the non-excessive framework of Chodosh–Liokumovich–Spolaor [CLS22] and apply Wickramasekera's regularity theorem [Wic14] to obtain minimal hypersurfaces in closed Riemannian manifolds.

9.1 NON-EXCESSIVE SWEEPOUTS

An alternative approach to min–max theory, developed by Chodosh–Liokumovich–Spolaor [CLS22], is based on the notion of *non-excessive sweepouts*. The non-excessive condition provides additional structural control that simplifies the regularity analysis.

We use the notation from Section 2.2. Recall that a sweepout $\{\Omega_t\}_{t\in[0,1]}$ of M is a continuous family of Caccioppoli sets with $\Omega_0 = \emptyset$ and $\Omega_1 = M$ (Definition 2.2.1). The width is $W = W(M)$ as in Definition 2.2.2.

Definition 9.1.1 (Optimal nested volume parametrized sweepout)**.** A sweepout $\{\Phi(x) = \partial\Omega(x)\}$ is called

- *optimal* if $\mathcal{F}(\Phi) = W$;
- *nested* if $\Omega(x_1) \subset \Omega(x_2)$ for all $0 \le x_1 \le x_2 \le 1$ (cf. Definition 2.2.7);
- *volume parametrized* if $\mathrm{Vol}(\Omega(x)) = x$ for every $x \in [0,1]$.

We call such a sweepout an *optimal nested volume parametrized (ONVP) sweepout*.

Definition 9.1.2 (Critical set)**.** Given a sweepout Φ, define

$$M(x) = \limsup_{r \to 0} \{\mathbf{M}(\Phi(y)) : |y - x| < r\}.$$

If Φ is an optimal sweepout, the *critical domain of* Φ is

$$\mathfrak{m}(\Phi) = \{x \in [0,1] : M(x) = W\}.$$

We denote by $\mathfrak{m}_L(\Phi)$ and $\mathfrak{m}_R(\Phi)$ the sets of left and right critical points.

Excessive Intervals and Non-Excessive Points

We introduce the concept of excessive intervals, which identify portions of a sweepout that can be replaced by lower-mass competitors.

Definition 9.1.3 (Excessive intervals and points)**.** Suppose $\{\Phi(x) = \partial\Omega(x)\}$ is a sweepout. Given a connected interval $I \subset [0,1]$ (which may be open, closed, or half-open), we say that $\{\Phi^I(x) = \partial\Omega^I(x)\}_{x \in \bar{I}}$ is an *I-replacement family for* Φ if $\Omega^I(a) = \Omega(a)$, $\Omega^I(b) = \Omega(b)$ at the endpoints, and for all $x \in I$,

$$\limsup_{I \ni y \to x} \mathbf{M}(\Phi^I(y)) < W,$$

where $W := \mathcal{F}(\Phi)$.

We say that a connected interval I is an *excessive interval for* Φ if there exists an I-replacement family for Φ. A point x is called *left excessive for* Φ (resp. *right excessive*) if there exists an excessive interval I with $(x - \varepsilon, x] \subset I$ (resp. $[x, x + \varepsilon) \subset I$) for some $\varepsilon > 0$.

Non-Excessive Sweepouts and Stationary Varifolds

The fundamental result of Chodosh–Liokumovich–Spolaor is the existence of sweepouts where critical points are not excessive:

Theorem 9.1.4 (Existence of non-excessive min-max hypersurface [CLS22, Theorem 3.1])**.** *Let (M^{n+1}, g) be a closed Riemannian manifold with $2 \leq n \leq 6$. There exists an (ONVP) sweepout Ψ such that every $x \in \mathfrak{m}_L(\Psi)$ is not left excessive and every $x \in \mathfrak{m}_R(\Psi)$ is not right excessive.*

A sweepout satisfying the conclusion of Theorem 9.1.4 is called a *non-excessive sweepout*. We denote by $\mathcal{S}_{\mathrm{nm}}$ the collection of all such sweepouts.

Proposition 9.1.5 (Stationary varifolds from optimal sweepouts)**.** *Let (M^{n+1}, g) be a closed Riemannian manifold with $2 \leq n \leq 6$, and let Φ be an optimal sweepout with $\mathcal{F}(\Phi) = W$ (the min-max width). Then there exists a stationary n-varifold V in M with $\mathbf{M}(V) \leq W$.*

Proof. Each slice $\Phi(x) = \partial\Omega(x)$ induces a varifold $V_x = |\partial^*\Omega(x)|$. Let $\{x_i\}$ be any sequence with $\mathbf{M}(\Phi(x_i)) \to W$. By varifold compactness, some subsequence satisfies $V_{x_i} \overset{*}{\rightharpoonup} V$ with $\|V\|(M) \leq W$. We claim some such limit V is stationary.

Suppose not: every varifold limit of sequences $\{V_{x_i}\}$ with $\mathbf{M}(\Phi(x_i)) \to W$ is non-stationary. Consider $A_\eta = \{x : \mathbf{M}(\Phi(x)) \geq W - \eta\}$, which is compact. By assumption, for each $x \in A_\eta$, any limit point of $\{V_y\}_{y \to x}$ is non-stationary. The tightening flow of [CL03] strictly decreases mass near non-stationary varifolds, so by compactness of A_η, there exists $\varepsilon > 0$ (depending on η) such that tightening decreases mass by at least ε on A_η.

Choose $\eta < \varepsilon$. After tightening, slices from A_η have mass $\leq W - \varepsilon < W - \eta$, and slices outside A_η already had mass $< W - \eta$. Thus the tightened sweepout $\tilde{\Phi}$ satisfies $\sup_x \mathbf{M}(\tilde{\Phi}(x)) < W$, contradicting the definition of width. $\square$

This is the standard pull-tight argument of [CL03]. Note that the non-excessive condition is not used here; it will be essential later for establishing stability and controlling mass cancellation.

Remark 9.1.6. The varifold V obtained from Proposition 9.1.5 is stationary, but its *integrality* is not automatically guaranteed. This is the key issue that distinguishes the non-excessive framework from the classical Almgren–Pitts approach:

- In Almgren–Pitts theory, one works with integral currents $\mathbf{Z}_n(M;\mathbb{Z}_2)$ from the outset, so integrality is built into the framework.
- In the non-excessive framework, one works with varifolds, and integrality must be established separately.
- Both approaches use pull-tight to obtain stationarity, but pull-tight alone does not produce integrality.

Homotopic Minimizers and Non-Minimizing Points

To analyze the geometric properties of min-max varifolds, we introduce the notion of homotopic minimizers and points where this property fails.

Definition 9.1.7 (Homotopic minimizers)**.** Let $\Sigma = \partial^*\Omega$ be a Caccioppoli set in M, and let $U \subset M$ be an open set. We say that Σ is:

- *inner homotopic minimizing in U* if there is no Caccioppoli set $E \subset \Omega$ with $E \triangle \Omega \subset U$ and $\mathrm{per}(E, U) < \mathrm{per}(\Omega, U)$ that can be connected to Ω by a continuous family of Caccioppoli sets in U;
- *outer homotopic minimizing in U* if there is no Caccioppoli set $E \supset \Omega$ with $E \triangle \Omega \subset U$ and $\mathrm{per}(E, U) < \mathrm{per}(\Omega, U)$ that can be connected to Ω by a continuous family of Caccioppoli sets in U;
- *one-sided homotopic minimizing in U* if it is either inner or outer homotopic minimizing in U.

Definition 9.1.8 (Non-homotopic-minimizing points)**.** For a varifold V arising from a min-max procedure, we define the set of *non-homotopic-minimizing points* by

$$\mathfrak{h}_{\mathrm{nm}}(V) := \left\{ P \in \mathbf{spt}\|V\| : \begin{array}{l} \text{for all } r > 0 \text{ small, } \mathbf{spt}\|V\| \cap B_r(P) \text{ is not one-sided} \\ \text{homotopic area minimizing on either side in } B_r(P) \end{array} \right\}.$$

Intuitively, $\mathfrak{h}_{\mathrm{nm}}(V)$ consists of points where the varifold V cannot be approximated by one-sided area minimizers in any neighborhood. These notions will be used in Section 9.4 to verify the α-structural hypothesis required by Wickramasekera's regularity theorem.

The Integrality Issue

Following [CLS22], we distinguish two cases for a critical point $x_0 \in \mathfrak{m}(\Phi)$ of a non-excessive sweepout Φ:

Definition 9.1.9 (Mass cancellation)**.** Let Φ be a non-excessive (ONVP) sweepout with width W, and let $x_0 \in \mathfrak{m}(\Phi)$ be a critical point. We say that:

- *No mass cancellation* occurs at x_0 if $\mathbf{M}(\Phi(x_0)) = W$.
- *Mass cancellation* occurs at x_0 if $\mathbf{M}(\Phi(x_0)) < W$.

Without mass cancellation, the limiting varifold V equals $|\partial\Omega(x_0)|$ with no mass loss. With mass cancellation, V may have larger mass than $\mathbf{M}(\partial\Omega(x_0))$ due to sheets with opposite orientations canceling in the current while contributing to varifold mass.

In this chapter, we establish integrality and regularity without mass cancellation via Wickramasekera's theorem. The case with mass cancellation remains open.

9.2 MAIN RESULT

Theorem 9.2.1 (Existence of minimal hypersurfaces via non-excessive min-max)**.** *Let (M^{n+1}, g) be a closed Riemannian manifold with $2 \leq n \leq 6$. Then there exists a smooth closed embedded minimal hypersurface in M.*

Conjecture 9.2.2 (Integrality with mass cancellation)**.** *If the tightened stationary varifold is integral even in the presence of mass cancellation, then Wickramasekera's regularity theorem remains applicable, and the conclusion of Theorem 9.2.1 extends to this case as well.*

9.3 INTEGRALITY WITHOUT MASS CANCELLATION

Theorem 9.3.1 (Integrality without mass cancellation)**.** *Let (M^{n+1}, g) be a closed Riemannian manifold with $n \geq 2$, and let V be a stationary varifold arising from a non-excessive sweepout via the construction of Theorem 9.1.5. If $\Theta^n(V, x) = 1$ for $\mathbf{M}(V)$-almost every $x \in \mathbf{spt}(V)$, then V is integral and rectifiable.*

Case 1: No Mass Loss

The reduced boundary $\partial^*\Omega$ of a Caccioppoli set Ω induces a varifold $V = \mathcal{H}^n \llcorner \partial^*\Omega \otimes \delta_{T_x\partial^*\Omega}$.

The key tool is the following proposition from the appendix of De Lellis–Tasnady [DLT13]:

Proposition 9.3.2 (Varifolds vs Caccioppoli set limits [DLT13, Proposition A.1])**.** *Let $\{\Omega^k\}$ be a sequence of Caccioppoli sets and U an open subset of M. Assume that*

(i) $D\mathbb{1}_{\Omega^k} \to D\mathbb{1}_\Omega$ in the sense of measures in U;

(ii) $\mathrm{per}(\Omega^k, U) \to \mathrm{per}(\Omega, U)$

for some Caccioppoli set Ω. Denote by V^k and V the varifolds induced by $\partial^*\Omega^k$ and $\partial^*\Omega$ respectively, where $\partial^*\Omega$ denotes the reduced boundary. Then $V^k \to V$ in the sense of varifolds.

Proof of Theorem 9.3.1, Case 1. By Theorem 9.1.5, the varifold V arises as the weak limit of varifolds V_i associated to slices $T_i = \phi(t_i)$ with $\mathbf{M}(T_i) \to \mathcal{F}(\phi)$.

We identify each slice $T_i \in \mathbf{Z}_n(M; \mathbb{Z}_2)$ with its corresponding Caccioppoli set Ω_i (see Proposition 2.1.3). The varifold V_i is then the varifold induced by the reduced boundary $\partial^*\Omega_i$, given by

$$V_i = \mathcal{H}^n \llcorner \partial^*\Omega_i \otimes \delta_{T_x \partial^*\Omega_i}.$$

Step 1: BV convergence. By the compactness theorem for BV functions (Caccioppoli sets), the sequence $\{\Omega_i\}$ has a subsequence (still denoted $\{\Omega_i\}$) such that $D\mathbb{1}_{\Omega_i} \to D\mathbb{1}_\Omega$ weakly in the sense of measures, for some Caccioppoli set Ω.

Step 2: Perimeter convergence. We establish perimeter convergence using the no mass cancellation hypothesis.

By passing to a subsequence, we may assume $t_i \to t^*$ for some $t^* \in [0, 1]$. Since $\mathbf{M}(\phi(t_i)) \to W = \mathcal{F}(\phi)$ and the function $M(x) := \limsup_{r\to 0}\{\mathbf{M}(\phi(y)) : |y - x| < r\}$ is upper semicontinuous, we have $M(t^*) \geq W$. But $M(x) \leq W$ for all x by definition of width, so $M(t^*) = W$, i.e., $t^* \in \mathfrak{m}(\phi)$.

By the *no mass cancellation* hypothesis (Definition 9.1.9), we have

$$\mathbf{M}(\phi(t^*)) = W.$$

Since the sweepout ϕ is continuous in the flat norm topology (see Proposition 2.2.4), and flat convergence implies L^1 convergence of characteristic

functions, we have $\mathbb{1}_{\Omega_i} \to \mathbb{1}_{\Omega(t^*)}$ in $L^1(M)$. By uniqueness of BV limits, the Caccioppoli set Ω from Step 1 equals $\Omega(t^*)$.

Now we can establish perimeter convergence directly:

- **Upper bound:** By the identification of currents and Caccioppoli sets,

$$\mathrm{per}(\Omega_i, M) = \mathbf{M}(\phi(t_i)) \to W.$$

- **Lower bound:** By lower semicontinuity of perimeter under BV convergence,

$$\mathrm{per}(\Omega, M) \leq \liminf_{i\to\infty} \mathrm{per}(\Omega_i, M) = W.$$

- **Equality:** Since $\Omega = \Omega(t^*)$ and $\mathbf{M}(\phi(t^*)) = W$ by no mass cancellation,

$$\mathrm{per}(\Omega, M) = \mathrm{per}(\Omega(t^*), M) = \mathbf{M}(\phi(t^*)) = W.$$

Combining these, we obtain perimeter convergence:

$$\mathrm{per}(\Omega_i, M) \to \mathrm{per}(\Omega, M) = W.$$

Step 3: Application of Proposition 9.3.2. By Proposition 9.3.2 (De Lellis–Tasnady, Proposition A.1), since we have both BV convergence and perimeter convergence, the varifold limit V equals the varifold induced by $\partial^*\Omega$:

$$V = \mathcal{H}^n \llcorner \partial^*\Omega \otimes \delta_{T_x\partial^*\Omega}.$$

This is precisely the varifold induced by an integral current (the Caccioppoli set Ω), hence V is integral and rectifiable. $\square$

Case 2: Mass Cancellation

In this case, we assume that $\mathbf{M}(\Phi(x_0)) < W$, so that mass cancellation occurs at the critical point x_0. As noted in Conjecture 9.2.2, establishing integrality in this case remains an open problem.

Proposition 9.3.3 (Regularity assuming Conjecture 9.2.2)**.** *Suppose Conjecture 9.2.2 holds, so that the stationary varifold V arising from a non-excessive sweepout is integral even in the presence of mass cancellation. Then the same regularity argument as in Case 1 applies, and V is a smooth embedded minimal hypersurface.*

Remark 9.3.4. The fundamental obstruction to proving Conjecture 9.2.2 is that when $\mathbf{M}(V) < W$, we cannot apply the index bound $\mathrm{Index}(V) \leq 1$ from min-max, which prevents us from using Wickramasekera's regularity theory to establish integrality. This represents a genuine obstruction rather than merely a technical gap.

Proof. If V is integral, then by Proposition 9.3.2, there exists a Caccioppoli set Ω such that

$$V = \mathcal{H}^n \llcorner \partial^* \Omega \otimes \delta_{T_x \partial^* \Omega}.$$

The reduced boundary $\partial^*\Omega$ is $(n-1)$-rectifiable by De Giorgi's theorem, hence V is rectifiable and integral.

The regularity then follows from Wickramasekera's theorem (Theorem 9.4.1) once we verify the α-structural hypothesis, which can be done using the same deformation arguments as in Section 9.4 for the case without mass cancellation. □

9.4 REGULARITY VIA WICKRAMASEKERA'S THEOREM

We recall the main regularity result from Wickramasekera [Wic14]:

Theorem 9.4.1 (Wickramasekera [Wic14])**.** *Let V be a stable integral n-varifold in a closed $(n+1)$-dimensional Riemannian manifold with $2 \leq n \leq 6$. Suppose that for some $\alpha \in (0, 1/2)$, V satisfies the α-structural hypothesis (Definition 9.4.2). Then $\mathbf{sing} V = \emptyset$, i.e., $\mathbf{spt}(V)$ is a smooth embedded minimal hypersurface.*

Definition 9.4.2 (α-structural hypothesis)**.** For $\alpha \in (0, 1/2)$, a stationary integral varifold V satisfies the *α-structural hypothesis* if **no singular point** of V has a neighborhood in which V corresponds to a union of embedded $C^{1,\alpha}$ hypersurfaces-with-boundary meeting *only* along their common $C^{1,\alpha}$ boundary, and with multiplicity a constant positive integer on each of the constituent hypersurfaces-with-boundary.

Remark 9.4.3. By Wickramasekera's theorem, the following are equivalent for a stable integral n-varifold V:

(a) V satisfies the α-structural hypothesis for some $\alpha \in (0, 1/2)$.

(b) $\mathbf{sing} V = \emptyset$ if $2 \leq n \leq 6$.

(c) $\mathcal{H}^{n-1}(\mathbf{sing} V) = 0$ (vanishing $(n-1)$-dimensional Hausdorff measure of the singular set).

Thus, to prove regularity via Wickramasekera's theorem, it suffices to verify the α-structural hypothesis for some $\alpha \in (0, 1/2)$.

Verification of the α-Structural Hypothesis

Theorem 9.4.4 (Verification of α-structural hypothesis)**.** *Let V be the stationary integral varifold from Theorem 9.3.1 arising from a non-excessive sweepout without mass cancellation. Then V satisfies the α-structural hypothesis.*

Case 1: No Mass Cancellation

Assume the *no cancellation* condition:

$$\mathbf{M}(\Phi(x_0)) = W, \tag{9.1}$$

where W is the min-max width and x_0 is a critical point.

In this case, the stationary integral varifold V from Theorem 9.3.1 can be written as $V = |\Sigma| = |\partial^*\Omega|$ for some Caccioppoli set Ω.

Our proof of Theorem 9.4.4 under the no cancellation condition is similar to [CLS22, Lemma 3.4].

Remark 9.4.5 (Why the case analysis is complete). We explain why Cases 1a and 1b below are exhaustive, and why stability holds.

(1) Completeness of cases. Since $V = |\partial\Omega|$ in the no mass cancellation case, tangent cones at singular points are constrained by the Caccioppoli set structure. For stationary boundaries of Caccioppoli sets in $\mathbb{R}^{n+1}$ ($n \leq 6$), the only possible singular cones are:

- **Case 1a:** $C = |H_1| + |H_2|$ (two half-hyperplanes meeting at angle $< \pi$) — a "wedge"
- **Case 1b:** $C = 2|H|$ (single half-hyperplane with multiplicity 2) — a "fold"

Higher multiplicities ($q \geq 3$) or more complex configurations (three or more sheets meeting) would require mass cancellation, which is impossible for Caccioppoli boundaries.

(2) Stability. The min-max construction gives $\mathrm{Index}(V) \leq 1$. Combined with the deformation arguments below showing at most one unstable direction, we conclude $\mathrm{Index}(V) = 0$ (stability) or $\mathrm{Index}(V) = 1$ (one-sided minimization), both of which lead to regularity via Wickramasekera [Wic14] or Liu [Liu19].

Proof of Theorem 9.4.4, Case 1. Suppose the α-structural hypothesis fails at $Z \in \mathbf{sing}V$. Then there exists $\rho > 0$ such that

$$B_\rho(Z)\backslash \mathbf{spt}\|V\| = \left(\bigsqcup_{i=1}^{N} E_i\right) \sqcup \left(\bigsqcup_{j=1}^{N} F_j\right), \tag{9.2}$$

and we denote $\mathcal{E} := \bigsqcup_{i=1}^{N} E_i$ (resp. $\mathcal{F} := \bigsqcup_{j=1}^{N} F_j$) to represent those Caccioppoli sets within $\overline{\Omega}$ (resp. $\overline{\Omega^c}$).

Without loss of generality, we only need to show that the E_i are not homotopic minimizers. For each $i = 1, \ldots, N$, we denote by C the blow-up cone of E_i at Z. Then there are only two possibilities for the structure of the cone:

- **Case 1a:** $C = |H_1| + |H_2|$ where both H_1 and H_2 are half-hyperplanes gluing along their common boundaries with the angle between them less than π.

  We denote $C = |\partial^* \Omega_C|$ for some Caccioppoli set Ω_C of $\mathbb{R}^{n+1}$. In this case, as in [CLS22, Lemma 3.4], we can choose $C^{1,\alpha}$-coordinates on M around Z such that $E_i = \Omega_C$ in $B_\varepsilon(Z)$ so that $g_{ij}(Z) = \delta_{ij}$, which we can do since $g \in C^2$ and $\Sigma = |\partial^* E_i|$ is a $C^{1,\alpha}$-deformation of C near Z by assumption.

  Since the angle between H_1 and H_2 is less than π, by the optimal regularity theorem for one-sided minimizers from [Liu19], we can find a Caccioppoli set $E_C \subset \Omega_C$ so that $E_C \triangle \Omega_C \subset B_1 \subset \mathbb{R}^{n+1}$ and

  $$\mathrm{per}_{\mathbb{R}^{n+1}}(E_C, B_1) \leq \mathrm{per}_{\mathbb{R}^{n+1}}(\Omega_C, B_1) - \delta \tag{9.3}$$

  for some $\delta > 0$. We set

  $$E(x) := \begin{cases} (E_i \backslash B_\varepsilon(Z)) \cup (|x| E_C \cap B_\varepsilon(Z)) & x < 0 \\ E_i & x = 0. \end{cases} \tag{9.4}$$

We have that

$$\mathrm{per}_g(E(x)) - \mathrm{per}_g(\Omega) = -|x|^n \delta(1 + o(1)) \tag{9.5}$$

as $x \to 0$. This shows that Σ is not inner homotopic minimizing.

- **Case 1b:** $C = 2|H|$ where H is a half-hyperplane. In this case, we define

$$E(x) := \begin{cases} E_i \backslash B_{|x|}(Z) & x < 0 \\ E_i & x = 0. \end{cases} \tag{9.6}$$

Apparently, for any $r > 0$,

$$\mathbf{M}_{B_1}(H \backslash B_r(0)) = \mathbf{M}_{B_1}(H) - \frac{1}{2}\omega_n r^n. \tag{9.7}$$

Moreover, since E_i is also a two-valued $C^{1,\alpha}$ graph over H and $|H|$ is the blow-up cone of $|\partial^* E_i|$ at Z, we know that $\mathrm{per}_g(E_i, B_{|x|}(Z)) = o(1)|x|^n$. Thus, we can again conclude that

$$\mathrm{per}_g(E(x)) - \mathrm{per}_g(\Omega) = -|x|^n \delta(1 + o(1)), \tag{9.8}$$

where $\delta = \omega_n/2$.

We can define the corresponding $F(x)$ for F_j by changing $x < 0$ to $x > 0$. Thus, we can know that $Z \in \mathfrak{h}_{nm}(\Sigma)$, where $\mathfrak{h}_{nm}$ denotes the set of points that are not homotopic minimizing in the sense of [CLS22]:

$$\mathfrak{h}_{nm}(V) := \left\{ P \in \mathbf{spt}\|V\| : \begin{array}{l} \text{for all } r > 0 \text{ small, } \mathbf{spt}\,\|V\| \cap B_r(P) \text{ is not one-sided} \\ \text{homotopic area minimizing on either side in } B_r(P) \end{array} \right\}. \tag{9.9}$$

Since $Z \in \mathfrak{h}_{nm}(V)$, by [CLS22, Proposition 3.1], we know that there is ρ sufficiently small such that for every $X \notin B_\rho(Z)$, there exists $r > 0$ such that $\mathbf{spt}\|V\|$ is one-sided minimizing in $B_r(X)$. However, since the α-structural hypothesis fails, we can take $X \in \mathbf{sing}V$ with $X \notin B_\rho(Z)$ such

that $\mathbf{spt}\|V\| \cap B_r(X)$ has $(n-1)$-dimensional singularity, which contradicts the optimal regularity for one-sided minimizers from [Liu19].

Remark 9.4.6 (Dimension of the singular set). The proof above uses the existence of singular points $X \in \mathbf{sing}V$ outside $B_\rho(Z)$. We explain why such points exist.

If the α-structural hypothesis fails at Z, then by Cases 1a and 1b, the tangent cone at Z has an $(n-1)$-dimensional "spine" L—the common boundary of the half-hyperplanes. Since V approximates this cone in the Hausdorff sense near Z, the singular set $\mathbf{sing}V$ must contain an $(n-1)$-dimensional subset near Z, not just an isolated point. In particular, for any $\rho > 0$, there exist singular points $X \in \mathbf{sing}V \setminus B_\rho(Z)$ where the same structural analysis applies, leading to the contradiction in the proof.

Therefore, the α-structural hypothesis must hold. □

The min-max construction gives $\mathrm{index}(V) \leq 1$. This index bound holds *a priori* before regularity is established, avoiding any circular reasoning.

Thus if V is stable (i.e., $\mathrm{index}(V) = 0$), then Wickramasekera's regularity theorem [Wic14] guarantees the optimal regularity for V. And in the case when $\mathrm{index}(V) = 1$, V is one-sided area minimizing, thus we can again use the optimal regularity for one-sided minimizers from [Liu19].

Case 2: Mass Cancellation

Assume the *cancellation* condition: $\mathbf{M}(\Phi(x_0)) < W$ throughout this section.

In this case, strict mass cancellation occurs: there exists a cancellation point $q \in \mathrm{reg}\, V$ such that

$$\mathrm{per}(\Omega, B_\varepsilon(q)) < |V|(B_\varepsilon(q))$$

for all sufficiently small $\varepsilon > 0$, where $\Omega = \Omega(x_0)$ and V can be written as $V = \sum_i \kappa_i |\Sigma_i|$ for integer multiplicities κ_i.

Remark 9.4.7 (Integrality assumption). The representation $V = \sum_i \kappa_i |\Sigma_i|$ with *integer* multiplicities assumes integrality, which is precisely what Conjecture 9.2.2 seeks to establish. The analysis below should be understood as conditional on this conjecture. The fundamental obstruction to proving integrality in the mass cancellation case is that without $\mathbf{M}(V) = W$, we cannot apply the index bound to deduce regularity.

The key difference from Case 1 is that Σ need not be globally representable as $\partial^*\Omega$; instead, it may consist of multiple components with different multiplicities. Nevertheless, the cancellation condition allows us to control deformations away from the cancellation point q.

Definition 9.4.8 (Competitors in the cancellation case)**.** Given $\varepsilon, \tau > 0$ with $\Sigma = \mathbf{spt}(V) \subset \overline{\Omega}$, we call a Caccioppoli set Ω' a $(q, \varepsilon, \tau, \Sigma, \Omega)$*-competitor* if

$$(\Omega \setminus B_\tau(\Sigma)) \cup (B_\varepsilon(q) \setminus \Sigma) \subset \Omega' \subsetneq \Omega \setminus \Sigma.$$

A $(q, \varepsilon, \tau, \Sigma, \Omega)$-competitor Ω' is called a *minimizing competitor* if its perimeter is strictly less than the perimeter of any $(q, \varepsilon, \tau, \Sigma, \Omega)$-competitor Ω'' with $\Omega' \subset \Omega''$.

Intuitively, a competitor represents a Caccioppoli set that:

- Agrees with Ω outside a τ-neighborhood of Σ;
- Agrees with Ω near the cancellation point q (in $B_\varepsilon(q)$);
- Has strictly smaller perimeter than Ω in the intermediate region.

The key technical result is that such competitors cannot exist when x_0 is not excessive:

Proposition 9.4.9 (Non-existence of minimizing competitors [CLS22, Proposition 6.3])**.** *Suppose V arises from a non-excessive sweepout Φ with $x_i \nearrow x_0 \in \mathfrak{m}_L(\Phi)$ and $|\Phi(x_i)| \to V$, and suppose the cancellation condition $\mathbf{M}(\Phi(x_0)) < W$ holds. Then for every $\varepsilon > 0$, there exists $\tau > 0$ such that no minimizing $(q, \varepsilon, \tau, \Sigma, \Omega)$-competitor exists.*

Proof sketch. Suppose a minimizing competitor U exists. By the cancellation condition, we can find Caccioppoli sets Ω' with $\mathrm{per}(\Omega') \leq W + \eta - \delta(\varepsilon)$ for small η and $\delta(\varepsilon) > 0$ from the mass loss at q. The minimizing property of U implies $\mathrm{per}(U) < W$.

For τ sufficiently small, a continuity argument (as in [CLS22, Lemma 3.3]) provides a nested family $\{E(x) : x \in [0,1]\}$ with $E(0) = U$, $E(1) = \Omega(x_0)$, and $\mathrm{per}(E(x)) < W$. Combining this with the original sweepout produces a replacement family showing that x_0 is left excessive, contradicting Theorem 9.1.4. □

With the non-existence of minimizing competitors established, we can prove regularity (conditional on Conjecture 9.2.2):

Proof of Theorem 9.4.4, Case 2 (assuming integrality). By Proposition 9.4.9, there are no minimizing competitors for any $\varepsilon > 0$ and suitable $\tau > 0$. This implies:

Stability of V: If any component Σ_i had unstable regular part, one could perform a one-sided minimization in $B_\tau(\Sigma_i) \setminus B_\varepsilon(q)$ to find a minimizing competitor, contradicting Proposition 9.4.9. Thus each Σ_i has stable regular part.

$\mathfrak{h}_{\mathrm{nm}}(V) = \emptyset$: By [CLS22, Proposition 3.7], the non-existence of minimizing competitors implies that every point of V is one-sided area minimizing in a small neighborhood. Therefore, $\mathfrak{h}_{\mathrm{nm}}(V) = \emptyset$.

Verification of α-structural hypothesis: If the α-structural hypothesis fails at some point $Z \in \mathbf{sing}V$, we can construct a local area-reducing deformation as in Case 1. However, such a deformation would produce a minimizing competitor (by one-sided minimization), contradicting Proposition 9.4.9.

Therefore, V satisfies all hypotheses of Wickramasekera's theorem [Wic14], implying smoothness. □

Proof of the Main Theorem

We are now ready to prove Theorem 9.2.1.

Proof of Theorem 9.2.1. Let (M^{n+1}, g) be a closed Riemannian manifold with $2 \leq n \leq 6$. The nested sweepout framework constructed in Chapters 4–6 provides a continuous family of hypersurfaces sweeping out M. Applying the non-excessive tightening procedure of Chodosh–Liokumovich–Spolaor to this nested sweepout yields a non-excessive sweepout $\Phi \in \mathcal{S}_{\mathrm{nm}}$.

By Theorem 9.1.5, there exists a stationary varifold V with $\mathbf{M}(V) \leq \mathcal{F}(\Phi)$.

Assume there is no mass cancellation (i.e., $\Theta^n(V, x) = 1$ for $\mathbf{M}(V)$-almost every $x \in \mathbf{spt}(V)$). By Theorem 9.3.1, V is integral and rectifiable.

By Theorem 9.4.4, V satisfies the α-structural hypothesis.

The non-excessive condition ensures that V is stable. Therefore, all hypotheses of Wickramasekera's regularity theorem (Theorem 9.4.1) are satisfied. We conclude that $\mathbf{sing} V = \emptyset$, i.e., $\mathbf{spt}(V)$ is a smooth embedded minimal hypersurface. $\square$

A

APPENDIX

A.1 EQUIVALENCE OF CONTINUOUS AND DISCRETE ALMOST MINIMIZING

All (one–parameter) deformations are generated by vector fields compactly supported in the relevant relative open set and tangent to ∂M near ∂M. We always measure areas as $\mathbf{M}(\Gamma) = \mathcal{H}^n(\Gamma)$, and write $\partial_I \Omega$ for the interior boundary of a finite–perimeter set $\Omega \subset M$.

Definition A.1.1 (Family of Surfaces)**.** A family $\{\Gamma_t\}_{t\in[0,1]}$ of closed subsets of M is a **generalized smooth family of surfaces** if:

(s1) For each $t \in [0,1]$, there exists a finite set $P_t \subset M$ such that Γ_t is a smooth hypersurface with boundary in $M \setminus P_t$;

(s2) For any compact set $K \subset M \setminus P_{t_0}$, we have $\Gamma_t \to \Gamma_{t_0}$ smoothly in K as $t \to t_0$;

(s3) (No concentration of mass) For every point $x \in M$:

$$\limsup_{r\to 0} \sup_{t\in[0,1]} \mathcal{H}^n(\Gamma_t \cap B_r(x)) = 0.$$

If, in addition to (s1)-(s3) above,

(s4) The map $t \mapsto \mathcal{H}^n(\Gamma_t \cap M)$ is continuous, and $t \mapsto \Gamma_t$ is continuous in the Hausdorff topology,

we moreover say that the generalized smooth family of surfaces $\{\Gamma_t\}_{t\in[0,1]}$ is **continuous**.

Definition A.1.2 (Sweepout (Technical Version for Interpolation))**.** Let $U \subset M$ be a bounded relative open subset. A family $\{\Gamma_t\}_{t\in[0,1]}$ is a **sweepout** of U (in the sense of this appendix) if it satisfies (s1)-(s4) from Definition A.1.1 and there exists a family $\{\Omega_t\}_{t\in[0,1]}$ of relative open sets in M with finite measure such that:

(sw1) $\Gamma_t \setminus \partial\Omega_t \subset P_t$ for all t, where P_t is the finite singular set from (s1);

(sw2) $\Omega_0 \cap U = \varnothing$ and $U \subset \Omega_1$;

(sw3) $\mathcal{H}^{n+1}(\Omega_t \triangle \Omega_s) \to 0$ as $|t-s| \to 0$.

We call $\{\Omega_t\}$ the **corresponding family of open sets**. The number $\mathcal{F}(\{\Gamma_t\}) := \sup_{t\in[0,1]} \mathcal{H}^n(\Gamma_t \cap M)$ is called the **height** of the sweepout.

Definition A.1.3 (Continuous ε–a.m. in a relative open set)**.** Let $\varepsilon > 0$ and $U \subset M$ be a relative open set. A relative boundary $\Gamma = \partial_M\Omega \in Z_k(M, \partial M; G)$ is *continuously ε–almost minimizing in U with free boundary* if there is no C^0 one–parameter family $\{\Gamma_t = \partial_M\Omega_t\}_{t\in[0,1]} \subset Z_k(M, \partial M; G)$ satisfying:

(a) *(sweepout regularity)* Properties (s1), (s2), (s3), (sw1), (sw3) in Definition A.1.2;

(b) *(support in U)* $\Omega_t \setminus U = \Omega \setminus U$ for all t;

(c) *(uniform cost)* $\mathbf{M}(\Gamma_t) \leq \mathbf{M}(\Gamma) + \varepsilon/8$ for all t;

(d) *(strict gain)* $\mathbf{M}(\Gamma_1) \leq \mathbf{M}(\Gamma) - \varepsilon$.

A sequence $\{\Gamma_k\}$ is (continuously) almost minimizing in U if Γ_k is ε_k–a.m. with $\varepsilon_k \downarrow 0$.

Definition A.1.4 (Discrete (ε, δ)–a.m. in a relative open set)**.** Let $\nu \in \{\mathcal{F}, \mathbf{M}, \mathbf{F}\}$ and $\varepsilon, \delta > 0$. Write $\mathscr{A}(U; \varepsilon, \delta; \nu)$ for the set of all $\Gamma = \partial_M\Omega \in Z_k(M, \partial M; G)$ with the following property:

for every finite sequence $\Gamma^0, \Gamma^1, \ldots, \Gamma^m \in Z_k(M, \partial M; G)$ with

(i) *(support in U)* $\Omega^j \setminus U = \Omega \setminus U$ for $j = 0, \dots, m$;

(ii) *(small steps)* $\nu(\Gamma^{j+1} - \Gamma^j) \leq \delta$ for $j = 0, \dots, m-1$;

(iii) *(no overshoot)* $\mathbf{M}(\Gamma^j) \leq \mathbf{M}(\Gamma) + \delta$ for $j = 1, \dots, m$,

one necessarily has the terminal lower bound

$$\mathbf{M}(\Gamma^m) \;\geq\; \mathbf{M}(\Gamma) - \varepsilon.$$

We say that a varifold V is *discrete a.m. in U* if there are $\varepsilon_i \downarrow 0$, $\delta_i \downarrow 0$ and $\Gamma_i \in \mathscr{A}(U; \varepsilon_i, \delta_i; \mathcal{F})$ such that $\mathbf{F}(|\Gamma_i|, V) < \varepsilon_i$.

Lemma A.1.5 (Relative interpolation)**.** *Fix $L > 0$. For every $\eta > 0$ there exists $\delta > 0$ with the following property. If $\Omega_0, \Omega_1 \subset M$ are finite–perimeter sets with $\Omega_0 \subset \Omega_1$, $\mathbf{Per}(\Omega_i) \leq L$ and $\mathcal{H}^{n+1}\big((\Omega_1 \setminus \Omega_0) \cap U\big) \leq \delta$, then there exists a nested, $\mathcal{F}$–continuous family $\{\partial_M \Omega_t\}_{t \in [0,1]}$ supported in U with*

$$\mathbf{Per}(\Omega_t) \;\leq\; \max\{\mathbf{Per}(\Omega_0), \mathbf{Per}(\Omega_1)\} + \eta \qquad \textit{for all } t.$$

All flows used in the construction are tangent to ∂M near ∂M.

Idea. Cover $(\Omega_1 \setminus \Omega_0) \cap U$ by finitely many small balls whose boundary spheres meet $\partial_M \Omega_i$ transversely and have small relative area (coarea). Apply the preliminary modification 6.1.1 to add/remove these pieces one by one inside the balls, and use Lemma 5.2.2 to produce the nested homotopies with the stated perimeter bound. □

Remark A.1.6. Lemma A.1.5 is the relative version of the interpolation lemma in [LZ21, Lemma B.1], adapted to the free boundary setting.

Proposition A.1.7 (Continuous ⇒ discrete, quantitative)**.** *There exists an absolute constant $c_0 \in (0,1)$ with the following property. If $\Gamma = \partial_M \Omega$ is continuously ε–a.m. in U, then $\Gamma \in \mathscr{A}(U; \varepsilon, \delta; \mathcal{F})$ for every $\delta \leq c_0\, \varepsilon$.*

Proof. Let $\{\Gamma^j\}_{j=0}^m$ be any discrete competitor as in Definition A.1.4 with parameter $\delta > 0$ to be fixed. For each pair (Γ^j, Γ^{j+1}) we apply the relative local modification (Preliminary Modification, §6.2) to decompose their difference inside U into a finite sequence of nested elementary additions/removals supported on small balls, with the property that every intermediate slice has area bounded by

$$\max\{\mathbf{M}(\Gamma^j), \mathbf{M}(\Gamma^{j+1})\} + C\,\delta,$$

for a dimensional $C = C(n)$. Using Lemma A.1.5 on each elementary step and concatenating in time, we produce a C^0 family $\{\widehat{\Gamma}_t\}_{t\in[0,1]}$ connecting $\Gamma^0 = \Gamma$ to Γ^m, supported in U, and satisfying

$$\sup_t \mathbf{M}(\widehat{\Gamma}_t) \;\le\; \max_{0\le j\le m} \mathbf{M}(\Gamma^j) + C\,\delta \;\le\; \mathbf{M}(\Gamma) + \delta + C\,\delta.$$

Choosing $\delta \le c_0\varepsilon$ with $c_0 > 0$ sufficiently small (e.g. $c_0 \le 1/16C$), we ensure

$$\sup_t \mathbf{M}(\widehat{\Gamma}_t) \;\le\; \mathbf{M}(\Gamma) + \varepsilon/8.$$

Hence $\{\widehat{\Gamma}_t\}$ is an admissible competitor in the sense of Definition A.1.3. By the continuous ε–a.m. property we must have $\mathbf{M}(\Gamma^m) \ge \mathbf{M}(\Gamma) - \varepsilon$. This proves $\Gamma \in \mathscr{A}(U;\varepsilon,\delta;\mathcal{F})$ for all $\delta \le c_0\varepsilon$. □

Corollary A.1.8 (Discrete a.m. for the min–max limit)**.** *Let $\{\Gamma_k\}$ be a min–max sequence which is continuously ε_k–a.m. in U with $\varepsilon_k \downarrow 0$, and suppose $|\Gamma_k| \to V$ as varifolds. Then V is discrete almost minimizing in U with free boundary.*

Proof. By Proposition A.1.7, for each k we have $\Gamma_k \in \mathscr{A}(U;\varepsilon_k,\delta_k;\mathcal{F})$ with $\delta_k = c_0\varepsilon_k \downarrow 0$. Since $\mathbf{F}(|\Gamma_k|, V) \to 0$, the definition of discrete a.m. for varifolds is satisfied. □

Remark A.1.9 (Pairs of annuli). All statements above hold verbatim when U is replaced by a ball, an annulus, or a *pair* of well–separated annuli

(A_1, A_2) (in the sense of $\mathrm{CO}(\mathcal{A})$ used in the text): the deformations are supported in $A_1 \cup A_2$, and the proofs are unchanged.

A.2 FERMI COORDINATES NEAR THE BOUNDARY

In this section, we provide a self-contained exposition of *Fermi coordinates* near a boundary point of a Riemannian manifold with boundary. These coordinates are particularly well-suited for studying free-boundary minimal hypersurfaces because they naturally encode the orthogonality condition at ∂M. The presentation follows the framework developed by Li–Zhou [LZ21] and is adapted to our setting.

Definition and Basic Properties

Definition A.2.1 (Fermi coordinates)**.** Let $p \in \partial M$. Suppose $(x_1, x_2, \cdots, x_n)$ is a geodesic normal coordinate system for ∂M centered at p, defined in a small neighborhood of p in ∂M. Let $t = \mathbf{dist}_M(\cdot, \partial M)$ denote the signed distance function from ∂M, which is smooth in a sufficiently small relatively open neighborhood of p in M. Then $(x_1, x_2, \cdots, x_n, t)$ is called a *Fermi coordinate system* for $(M, \partial M)$ centered at $p \in \partial M$.

We define the *Fermi distance function from* p on a relatively open neighborhood of p in M by

$$\tilde{r} = \tilde{r}_p(q) := \sqrt{x_1^2 + \cdots + x_n^2 + t^2}. \tag{A.1}$$

The *Fermi exponential map at* p, denoted $\tilde{\exp}_p$, gives the Fermi coordinate system and is defined on a half-ball in $\mathbb{R}^{n+1}_+ \cong T_pM$ centered at the origin. Explicitly, $\tilde{\exp}_p(x_1, \cdots, x_n, t)$ is the point in M whose Fermi coordinates are $(x_1, \cdots, x_n, t)$.

The key feature of Fermi coordinates is that the t-direction is always orthogonal to ∂M, which makes them ideal for analyzing free-boundary conditions.

Lemma A.2.2 (Metric and connection in Fermi coordinates)**.** *Let g_{ab} and Γ^c_{ab} be the components of the metric and Christoffel symbols in Fermi coordinates $(x,t) = (x_1, \cdots, x_n, t)$ centered at $p \in \partial M$. Then for $i, j = 1, \cdots, n$,*

$$g_{tt} \equiv 1, \quad g_{it} \equiv 0, \quad \Gamma^t_{it} = \Gamma^i_{tt} = \Gamma^t_{tt} \equiv 0. \tag{A.2}$$

Moreover, there exist constants $r_{\text{Fermi}}, C > 0$ depending only on the geometry of M (specifically, bounds on the curvature and second fundamental form of ∂M) such that for $\tilde{r} \in [0, r_{\text{Fermi}})$,

$$\begin{aligned} |g_{ij}(x,t) - \delta_{ij}| &\leq C\tilde{r}, \\ |\Gamma^k_{ij}(x,t)| &\leq C\tilde{r}, \\ |\Gamma^j_{it}(x,t)| + |\Gamma^t_{ij}(x,t)| &\leq C. \end{aligned} \tag{A.3}$$

Proof. Since $t = \mathbf{dist}_M(\cdot, \partial M)$ is the distance function from the boundary, its gradient has unit length and is orthogonal to level sets. This immediately gives $g_{tt} \equiv 1$ and $g_{it} \equiv 0$. The vanishing of the Christoffel symbols Γ^t_{it}, Γ^i_{tt}, and Γ^t_{tt} follows from the fact that $\frac{\partial}{\partial t}$ has constant length and is orthogonal to $\frac{\partial}{\partial x_i}$.

For the quantitative estimates, observe that $\frac{\partial}{\partial x_i}$ are Jacobi fields along the geodesics $\gamma_x(t) = (x,t)$ emanating orthogonally from ∂M. They satisfy the Jacobi equation

$$\nabla_{\frac{\partial}{\partial t}} \nabla_{\frac{\partial}{\partial t}} \frac{\partial}{\partial x_i} = -R\left(\frac{\partial}{\partial t}, \frac{\partial}{\partial x_i}\right)\frac{\partial}{\partial t},$$

where R is the Riemann curvature tensor of M.

By standard estimates for geodesic normal coordinates on ∂M, for $|x| < r_{\text{Fermi}}$ we have

$$|g_{ij}(x,0) - \delta_{ij}| \leq C|x|, \tag{A.4}$$

as ∂M has bounded sectional curvature. Along the t-direction, we have $\partial_t g_{ij}(x,0) = h_{ij}(x)$, where h_{ij} are the components of the second fundamental form of ∂M in the coordinates $(x_1, \cdots, x_n)$. Using the Jacobi equation, one derives the Riccati-type equation (see [Mar17, Lemma 2.2])

$$\partial_t^2 g_{ij} = -2R_{titj} + 2g^{rs}(\partial_t g_{ir})(\partial_t g_{sj}).$$

Since the curvature is bounded, integrating this equation yields for $t < r_{\text{Fermi}}$,

$$|g_{ij}(x,t) - g_{ij}(x,0)| \leq Ct. \tag{A.5}$$

Combining (A.4) and (A.5) gives $|g_{ij}(x,t) - \delta_{ij}| \leq C(|x| + t) \leq C\tilde{r}$. The estimates for the Christoffel symbols follow from similar arguments using bounds on the derivatives of the curvature tensor. □

Geometric Properties of the Fermi Distance Function

Lemma A.2.3 (Gradient and Hessian of the Fermi distance)**.** *The Fermi distance function $\tilde{r}$ is smooth (where it is defined) and there exist constants $r_{\text{Fermi}}, C > 0$ depending only on the geometry of M such that for $\tilde{r} \in [0, r_{\text{Fermi}})$,*

(i) $\left\|\nabla\tilde{r} - \frac{\partial}{\partial\tilde{r}}\right\|_g \leq C\tilde{r}$,

(ii) $\left\|\mathbf{Hess}\tilde{r} - \frac{1}{\tilde{r}}g_{\tilde{r}}\right\|_g \leq C$,

where $g_{\tilde{r}}$ denotes the induced metric on the sphere of radius $\tilde{r}$ in Euclidean space (i.e., the round metric scaled by $\tilde{r}$).

Proof. The gradient of $\tilde{r}$ in Fermi coordinates is given by

$$\nabla \tilde{r} = \sum_{i,j=1}^{n} g^{ij}(x,t) \frac{x_i}{\tilde{r}} \frac{\partial}{\partial x_j} + \frac{t}{\tilde{r}} \frac{\partial}{\partial t}. \tag{A.6}$$

Using $g^{ij} = \delta_{ij} + O(\tilde{r})$ from Lemma A.2.2, we obtain

$$\nabla \tilde{r} = \frac{1}{\tilde{r}} \left(\sum_{i=1}^{n} x_i \frac{\partial}{\partial x_i} + t \frac{\partial}{\partial t} \right) + O(\tilde{r}) = \frac{\partial}{\partial \tilde{r}} + O(\tilde{r}),$$

proving (i).

For (ii), direct computation in Fermi coordinates yields

$$\mathbf{Hess}\tilde{r} \left(\frac{\partial}{\partial t}, \frac{\partial}{\partial t} \right) = \frac{\tilde{r}^2 - t^2}{\tilde{r}^3}, \quad \left| \mathbf{Hess}\tilde{r} \left(\frac{\partial}{\partial x_i}, \frac{\partial}{\partial t} \right) + \frac{x_i t}{\tilde{r}^3} \right| \leq C,$$

$$\left| \mathbf{Hess}\tilde{r} \left(\frac{\partial}{\partial x_i}, \frac{\partial}{\partial x_j} \right) - \frac{\delta_{ij}\tilde{r}^2 - x_i x_j}{\tilde{r}^3} \right| \leq C.$$

These formulas show that $\mathbf{Hess}\tilde{r}$ agrees with the Hessian of the Euclidean distance function up to bounded error, which is precisely the statement (ii). □

Fermi Balls and Convexity

Definition A.2.4 (Fermi balls and spheres)**.** For each $p \in \partial M$, we define the *Fermi half-ball* and *Fermi half-sphere* of radius r centered at p respectively by

$$\tilde{B}_r^+(p) := \{q \in M : \tilde{r}_p(q) < r\}, \qquad \tilde{S}_r^+(p) := \{q \in M : \tilde{r}_p(q) = r\}.$$

The following lemma summarizes the key geometric properties of Fermi half-balls that are most relevant for our applications.

Lemma A.2.5 (Convexity and comparison with Euclidean balls)**.** *There exists a constant $r_{\text{Fermi}} > 0$, depending only on the geometry of M, such that for all $0 < r < r_{\text{Fermi}}$ and $p \in \partial M$,*

(a) $\tilde{S}_r^+(p)$ is a smooth hypersurface meeting ∂M orthogonally,

(b) $\tilde{B}_r^+(p)$ is a relatively convex domain in M,

(c) $B_{r/2}(p) \cap M \subset \tilde{B}_r^+(p) \subset B_{2r}(p) \cap M$,

where $B_s(p)$ denotes the Euclidean ball of radius s centered at p in the ambient space $\mathbb{R}^L$.

Proof. Properties (a) and (c) follow directly from the estimates in Lemmas A.2.2 and A.2.3. For (b), the relative convexity of $\tilde{B}_r^+(p)$ follows from the estimate (ii) in Lemma A.2.3: the Hessian of $\tilde{r}$ is positive definite (up to a bounded perturbation from the Euclidean case), which implies that sublevel sets of $\tilde{r}$ are convex for sufficiently small r. □

Remark A.2.6 (Applications to free-boundary theory). Fermi coordinates are particularly useful in the regularity theory for free-boundary minimal hypersurfaces. The orthogonality condition $\Gamma \perp \partial M$ translates to the statement that Γ is a graph over a domain in $\{t = 0\}$ near ∂M, and the free-boundary condition becomes a Neumann boundary condition for the graphing function. This perspective is exploited extensively in the work of Li–Zhou [LZ21] and in our application of their regularity results in Chapter 8.

A.3 WICKRAMASEKERA'S REGULARITY THEORY FOR STABLE VARIFOLDS

In this section, we provide a detailed exposition of Wickramasekera's regularity theory for stable codimension 1 integral varifolds [Wic14], which plays a crucial role in establishing the smoothness of minimal hypersurfaces arising from non-excessive min-max procedures in Chapter 9.

Stationary and Stable Varifolds

We begin by recalling the key definitions. Let V be an integral n-varifold on a smooth $(n+1)$-dimensional Riemannian manifold N.

Definition A.3.1 (Stationary varifold)**.** V is *stationary* if it has zero first variation with respect to area, i.e., for any compactly supported C^1 vector field ψ on N,

$$\int_{N \times G_n} \operatorname{div}_S \psi(X)\, dV(X,S) = 0,$$

where G_n denotes the space of unoriented n-planes in the tangent bundle of N, and div_S denotes the tangential divergence on the plane S.

For a stationary varifold V, the *regular part* $\operatorname{reg} V$ is the set of points where the support of V is a smooth embedded hypersurface, and the *singular set* $\operatorname{sing} V$ is its complement in $\operatorname{spt}\|V\|$. By Allard's regularity theorem [All72], $\operatorname{reg} V$ is always a dense, open subset of $\operatorname{spt}\|V\|$ when V is stationary.

Definition A.3.2 (Stable varifold)**.** A stationary integral n-varifold V is *stable* if its regular part satisfies the *stability inequality*: for every open set $\Omega \subset N$ with sufficiently small singular set (e.g., $\operatorname{sing} V \cap \Omega = \emptyset$ when $n \leq 6$, or $\mathcal{H}^{n-7+\gamma}(\operatorname{sing} V \cap \Omega) = 0$ for all $\gamma > 0$ when $n \geq 7$),

$$\int_{\operatorname{reg} V \cap \Omega} |A|^2 \zeta^2 \, d\mathcal{H}^n \leq \int_{\operatorname{reg} V \cap \Omega} |\nabla \zeta|^2 \, d\mathcal{H}^n \quad \forall\, \zeta \in C^1_c(\operatorname{reg} V \cap \Omega), \tag{A.7}$$

where A denotes the second fundamental form of $\operatorname{reg} V$, $|A|$ its norm, and ∇ the gradient on $\operatorname{reg} V$.

Equivalently, stability means that V has non-negative second variation of area under normal deformations supported away from the singular set. The stability inequality (A.7) is a quantitative expression of this variational property.

The α-Structural Hypothesis

The key geometric hypothesis in Wickramasekera's theorem is the α-Structural Hypothesis (Definition 9.4.2 in Chapter 9), which we recall here for convenience:

> For $\alpha \in (0, 1/2)$, a stationary integral varifold V satisfies the *α-Structural Hypothesis* if no singular point of V has a neighborhood in which V corresponds to a union of embedded $C^{1,\alpha}$ hypersurfaces-with-boundary meeting only along their common $C^{1,\alpha}$ boundary, with multiplicity a constant positive integer on each constituent hypersurface.

In other words, the α-Structural Hypothesis rules out the possibility that the varifold locally looks like three or more sheets meeting along a common $(n-1)$-dimensional junction.

Remark A.3.3. The α-Structural Hypothesis is implied by either of the following conditions:

(a) V corresponds to an absolutely area-minimizing rectifiable current with no boundary.

(b) The singular set has vanishing $(n-1)$-dimensional Hausdorff measure: $\mathcal{H}^{n-1}(\operatorname{sing} V) = 0$.

Condition (b) is the weakest size hypothesis on the singular set that guarantees the regularity conclusions below. No hypothesis in terms of Hausdorff measure of dimension $> n-1$ can suffice, as shown by examples of transversely intersecting hyperplanes.

Wickramasekera's Main Theorem

We can now state the complete version of Wickramasekera's regularity and compactness theorem:

Theorem A.3.4 (Wickramasekera's Regularity and Compactness Theorem [Wic14]). *Let V be a stable integral n-varifold on a smooth $(n+1)$-dimensional Riemannian manifold N. Suppose V satisfies the α-Structural Hypothesis for some $\alpha \in (0, 1/2)$. Then:*

(i) ***Regularity:***

- *If $2 \leq n \leq 6$: $\operatorname{sing} V = \emptyset$ (complete regularity).*
- *If $n = 7$: $\operatorname{sing} V$ is discrete.*
- *If $n \geq 8$: $\mathcal{H}^{n-7+\gamma}(\operatorname{sing} V) = 0$ for all $\gamma > 0$.*

In all cases, V corresponds to a smooth embedded hypersurface away from the singular set, with multiplicity a constant positive integer on each connected component.

(ii) ***Compactness:*** *For any fixed $\alpha \in (0, 1/2)$, each mass-bounded subset of the class of stable codimension 1 integral varifolds satisfying the α-Structural Hypothesis is compact in the varifold convergence topology.*

The dimension estimates on the singular set in part (i) are optimal, as demonstrated by well-known examples of stable minimal cones (e.g., the cone over $\mathbf{S}^3(1/\sqrt{2}) \times \mathbf{S}^3(1/\sqrt{2}) \subset \mathbf{R}^8$ has an isolated singularity at the origin).

Application to Non-Excessive Min-Max Theory

The relevance of Wickramasekera's theorem to our work in Chapter 9 is as follows. The Chodosh–Liokumovich–Spolaor theory [CLS22] produces a stationary varifold from a non-excessive sweepout, but does not automatically guarantee integrality. Our contribution is to show that in the multiplicity-one case, no mass cancellation occurs in the limit, which implies:

(a) The limiting varifold is integral and rectifiable.

(b) The singular set has vanishing $(n-1)$-dimensional Hausdorff measure.

(c) The α-Structural Hypothesis is satisfied.

Given these properties, together with the stability implied by the non-excessive condition, Theorem A.3.4 immediately yields smoothness of the resulting minimal hypersurface when $n \leq 6$.

Key Ideas in the Proof

While we do not reproduce the full proof of Theorem A.3.4 (which occupies a substantial monograph), we outline the main ideas:

1. **Sheeting Theorem:** Wickramasekera proves that if a stable varifold satisfying the α-Structural Hypothesis is weakly close to a hyperplane of multiplicity q, it must locally decompose into q disjoint smooth graphs ("sheets") over the hyperplane.

2. **Minimum Distance Theorem:** No such varifold can be close to a cone consisting of three or more half-hyperplanes meeting along a common axis.

3. **Inductive argument:** The proof proceeds by induction on the multiplicity, using blow-up analysis and refined excess-decay estimates to control the geometry near potential singularities.

4. **Use of stability:** The stability inequality (A.7) is used crucially throughout, particularly in regions where the Hausdorff dimension of the singular set is sufficiently small.

A key technical point is that Wickramasekera's proof does *not* assume a priori that the singular set has finite $(n-2)$-dimensional Hausdorff measure (as in the earlier work of Schoen–Simon [SS81b]). The α-Structural

Hypothesis is a purely geometric condition that does not directly imply any bounds on the size of the singular set, yet it suffices to conclude optimal regularity.

Remark A.3.5 (Comparison with area-minimizing theory). For absolutely area-minimizing codimension 1 rectifiable currents, the same regularity conclusions were established through the combined work of De Giorgi [DG61], Reifenberg, Fleming, Almgren, Simons [Sim68], and Federer. Wickramasekera's theorem subsumes this classical theory: if T is an area-minimizing current with no boundary, it automatically satisfies the α-Structural Hypothesis (since no union of three or more sheets can be area-minimizing).

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